

YESPARENCY: A WEB-BASED BIDDING SYSTEM

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Chapter 1

INTRODUCTION

In an era where digital transformation is reshaping the way institutions operate, governments and organizations worldwide are increasingly adopting technology-driven solutions. The trends toward digital transformation evolved based on the concept that integrating technology for different public service operations and procedures can lead to improved service quality, effectiveness, and efficiency (Mergel et al., 2023). Public procurement, in particular, has been one of the most critical areas where digitalization has made a significant impact, with institutions moving away from paper-based bidding processes toward electronic systems that streamline document handling and promote greater transparency in the awarding of contracts (OECD, 2025).

In the Philippines, the government has taken concrete steps toward transparent and efficient public procurement through the enactment of Republic Act No. 9184 (2003), also known as the Government Procurement Reform Act, which mandates transparency, competitiveness, and accountability in all government procurement activities. The Philippine Government Electronic Procurement System, commonly known as PhilGEPS, further reflects the national government's commitment to modernizing procurement practices by providing a centralized online portal for procurement transactions across government agencies and institutions (PhilGEPS, n.d.).

Despite these national efforts, many local government institutions still operate procurement processes that have not fully embraced digital transformation (Bacarisas et al., 2026; Kalupe et al., 2025). One such organization is Southern Luzon State University, where the

Procurement Office still uses traditional, mainly manual methods for its bidding process. In order to streamline Southern Luzon State University's procurement process in accordance with public procurement law requirements for transparency, integrity, and accountability, the capstones research proposes to design and develop YesParency, a web-based bidding system.

Background of the Study

Public procurement plays a vital role in the operations of government institutions, as it governs the acquisition of goods, services, and infrastructure necessary for the delivery of public functions (Kumar, 2022; OECD, 2021). The integrity of this process directly affects how public funds are managed and how accountable institutions are to the communities they serve (Sikanyiti & Chibomba, 2025). As digital technology continues to advance, the digitization of procurement processes has become increasingly important in ensuring that bidding procedures are not only efficient but also secure, transparent, and resistant to manipulation (Celestin, 2021; Fourie & Malan, 2021). A web-based bidding system that incorporates secure document handling and verified authentication mechanisms serves as a critical tool in upholding these standards, reducing human error, minimizing opportunities for corruption, and creating a verifiable digital record of every transaction within the procurement cycle.

Southern Luzon State University (SLSU), as a government funded institution of higher learning, is governed by the Commission on Higher Education (CHED) and is subject to the procurement regulations set forth by the Government Procurement Policy Board (GPPB). The university regularly conducts bidding and procurement activities through its Bids and Awards Committee (BAC), which is tasked with managing the acquisition of goods, services, and

infrastructure projects in accordance with Republic Act No. 9184 (2003). These proceedings are required to be conducted transparently and must be made accessible to all qualified observers and interested parties. The Procurement Office of SLSU serves as the administrative support unit of the BAC and is responsible for the coordination, documentation, and dissemination of all procurement-related activities within the university.

Currently, the Procurement Office of SLSU follows the traditional two-envelope bidding process as prescribed by Republic Act No. 9184 (2003). Under this process, bidders are required to physically submit two sealed envelopes, the first containing their eligibility and legal documents proving their legitimacy as a supplier, and the second containing their financial bid offer. These envelopes are not opened until the scheduled bid opening date, during which the BAC conducts the opening proceedings and broadcasts the event live through Zoom as a measure of public transparency. While this process complies with procurement regulations, it still has potential for improvement and presents several operational vulnerabilities that undermine its efficiency and security. According to the head of the procurement office, it would be more efficient for both the office and the bidders to adopt an electronic submission system rather than relying on physical document submission, as this exposes bid envelopes to risks of tampering, loss, or damage prior to the opening. The use of Zoom, a third-party platform not owned by the university, also raises concerns regarding data privacy, platform dependency, and the lack of institutional control over the broadcast. Furthermore, the absence of a digital audit trail means there is no verifiable record of who accessed the documents, when they were opened, and under whose authority, making it difficult to investigate irregularities should they arise. Additionally, the manual nature of the process places a significant administrative burden on

procurement personnel, particularly in the organization, storage, and retrieval of physical bid documents.

These conditions collectively point to the need for a more secure, efficient, and institution-controlled procurement system. A digital solution that preserves the integrity of the two-envelope process while eliminating the vulnerabilities of its physical implementation would significantly improve the operations of the SLSU Procurement Office. Specifically, a system that allows bidders to submit their documents online, encrypts those submissions to prevent premature access, and requires authorized procurement officers to authenticate through digital signatures before any bid document can be opened would directly address the gaps identified in the current process. The integration of an institution-owned live streaming feature would further eliminate the university's dependence on third-party platforms while maintaining the public transparency that the bid opening process demands. Furthermore, the development of YesParency is aligned with the United Nations Sustainable Development Goals, particularly SDG 8 (Decent Work and Economic Growth), by promoting fair and efficient procurement practices that support economic participation among local suppliers and businesses; SDG 9 (Industry, Innovation, and Infrastructure), by leveraging digital innovation to modernize institutional procurement infrastructure; and SDG 16 (Peace, Justice, and Strong Institutions), by strengthening the transparency, accountability, and integrity of public procurement processes in a government institution. It is within this context that YesParency, a web-based bidding system, was conceptualized and developed for the procurement office of Southern Luzon State University.

Objectives of the Study

The general objective of this study is to design, develop, evaluate, and deploy YesParenity, a web-based bidding system for Southern Luzon State University.

To achieve the general objective, the project will pursue the following specific objectives:

1. Design and develop a Web-Based Bidding System called YesParenity capable of:
 - a. allowing registered bidders to securely submit their eligibility documents and financial bid offers online through a digitized two-envelope submission process;
 - b. allowing the encryption of submitted bid envelopes and the authentication of authorized procurement officers through digital signatures before granting access to submitted bid documents during bid opening;
 - c. allowing real-time live streaming of bid opening proceedings conducted by the Bids and Awards Committee (BAC);
 - d. allowing the maintenance of an audit trail that records all document access, signature authentication events, and system activities;
 - e. allowing the management of user authentication for bidders and authorized procurement administrators while also allowing public viewing access to live bid opening broadcasts and the publication of procurement-related announcements, including bidding schedules, submission deadlines, and bid opening reminders.
2. Evaluate the YesParenity system based on ISO/IEC 25010:2023 Software Quality Standards in terms of:

- Functional Suitability
 - Performance Efficiency
 - Compatibility
 - Usability
 - Reliability
 - Security
 - Maintainability
 - Flexibility
 - Safety
3. To deploy YesParenCy at Southern Luzon State University, including system installation and transition from the current manual two-envelope bidding process to a digitized, signature-based document access controlled bidding system.

Significance of the Study

The development of YesParenCy is significant, as it provides a modern, secure, and transparent solution to the current procurement process of Southern Luzon State University. By integrating digital bid submission, encryption, digital signatures, live streaming, and audit trails functionalities, the system addresses the limitations of manual processes while ensuring

compliance with the transparency and accountability requirements of Republic Act No. 9184. Specifically, the system is expected to benefit the following stakeholders:

Southern Luzon State University (SLSU) stands to benefit from the system, which provides the university with an institution-owned digital platform that improves procurement transparency, strengthens compliance with government regulations, and enhances overall operational efficiency. It also supports better record management through centralized and traceable digital data.

Bids and Awards Committee (BAC) benefits from YesParency by providing a structured system for managing bid submissions, conducting live bid openings, and securely accessing documents through digital signature authentication. It also ensures that all actions are recorded through an audit trail, supporting accountability and easier verification during procurement processes.

Procurement Office benefits from the system, which reduces reliance on manual document handling and third-party platforms by offering a centralized digital solution. It simplifies the organization, storage, and retrieval of procurement documents while minimizing risks such as document loss, tampering, and inefficiencies in administrative tasks.

Bidders and suppliers benefit from a more convenient and secure way of submitting documents through an online platform. The digitized two-envelope system ensures fairness by protecting the confidentiality of submissions while providing equal access to procurement opportunities.

Public and stakeholders benefit from the integration of live streaming, which allows them to observe bidding processes in real time, promoting transparency and trust. It ensures that procurement activities are visible and accessible even to those who cannot be physically present.

Other State Universities and Colleges (SUCs) may benefit from the proposed system as it can serve as a reference for institutions seeking to adopt digital procurement solutions. It demonstrates how technology can be used to enhance transparency, efficiency, and compliance with national procurement policies.

Researchers benefit from the study by gaining practical experience in designing and developing a real-world web-based system. It enhances their knowledge in areas such as system security, digital governance, and software development while contributing to their academic and professional growth.

Future researchers may benefit from this study as it contributes to the field of digital procurement and transparency systems. It may serve as a basis for further research, including system improvements, integration with national procurement platforms, or the development of more advanced features such as automation and enhanced security mechanisms.

Scope and Limitation

This study focuses on the design, development, evaluation, and deployment of YesParency, a web-based procurement bidding system for the Procurement Office of Southern Luzon State University. The system aims to improve the old manual bidding process by providing a secure, organized, and digital platform where bidders can submit their eligibility

documents and financial proposals online. YesParency replicates the traditional two-envelope bidding process in a digital format. Registered bidders can submit their eligibility documents and financial bid proposals through the platform, which stores these files as two separate encrypted digital envelopes. Submitted documents remain inaccessible until the scheduled bid opening, ensuring confidentiality and compliance with standard procurement procedures.

The system requires authorized procurement officers to authenticate using their registered digital signatures before accessing any submitted bid documents. This ensures that only authorized personnel can open and review the bids during the official bid opening session. The system's core feature is the electronic submission of bidding documents, allowing registered bidders to securely upload their eligibility documents and financial proposals online.

Moreover, the YesParency web-based bidding system will support three classifications of users: super administrators, procurement administrators, registered bidders, and public observers. The super administrator, typically the head of the Procurement Office, will have full access to all modules and system functions, including managing procurement administrators, monitoring system activities, and overseeing bid openings. The permission for deleting submitted documents will be restricted to super administrators, while they may grant or reject access or editing permissions for procurement administrators as needed. Procurement administrators will be responsible for verifying bidder accounts, managing submitted eligibility and financial documents, conducting bid opening sessions, and generating downloadable reports. They can also monitor system logs to ensure transparency, integrity, and accountability during the procurement process. Registered bidders, as non-administrative users, will be able to securely submit eligibility and financial documents online, view submission status, and access live streaming of bid openings. Public observers or watchers will be provided access to view the live

streaming of the bid opening proceedings, enabling stakeholders and members of the public to observe the process and support transparency. To maintain integrity and prevent unauthorized access, the system requires bidder authentication during registration, including submission of personal information such as name, organization, email address, and contact details. The platform automatically records the date and time of each bid submission to display bidders in the order of submission, assisting procurement administrators in tracking submissions accurately.

This digital submission process improves the integrity of the procurement procedure by making sure that all documents are safely stored, encrypted, and inaccessible until the scheduled bid opening. To assist in monitoring submissions, the platform also records the exact date and time of each bid and will display bidders in the order they submitted their documents, allowing procurement officers to efficiently track the sequence of submissions and maintain an organized and transparent record.

To further promote transparency and accountability, YesParency includes a built-in live streaming feature, enabling stakeholders and observers to watch the bid opening proceedings in real time. The system also maintains a detailed audit trail that logs document access, digital signature authentications, and other system activities, providing an accurate record of all actions within the platform.

The study encompasses the development of the system interface, database management, encryption, access control, and live streaming functionalities. Furthermore, YesParency will be evaluated based on the ISO/IEC 25010 Software Quality Standards, considering functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, flexibility, and safety. The deployment of the system at Southern Luzon State University forms

part of the transition from the current manual bidding process to a fully digitized platform with signature-based document access control. The system emphasizes improving operational efficiency and minimizing manual errors associated with the current bidding process. Through the centralization of submission, authentication, and review within a single digital platform, the system reduces administrative delays, prevents document tampering, and provides an accessible record for both procurement officers and observers. This design aligns with the university's goal of enhancing transparency, auditability, and compliance with procurement regulations. The system ensures that all bidding activities are traceable, facilitating the management, monitoring, and review of submissions by the Bids and Awards Committee (BAC).

Despite the system's features, several limitations are acknowledged. YesParency is designed specifically for the procurement procedures of Southern Luzon State University; its structure, workflow, and functionalities reflect the operational requirements of the university's procurement office. Implementation in other institutions may require modifications to meet different policies and processes. The system focuses primarily on the bid submission and bid opening stages, leaving other procurement activities such as project planning, supplier evaluation, contract management, and project monitoring outside its scope. Its operation is dependent on stable internet connectivity; poor connectivity may affect document uploads, live streaming quality, and overall system performance. Digital signature authentication only works for signatures registered within the system and does not integrate with external certification authorities or national procurement platforms such as the Philippine Government Electronic Procurement System. Although encryption and access control protect submitted documents, advanced cybersecurity measures like blockchain verification or third-party auditing are not included. The quality of live streaming may also be affected by the institution's server capacity

and internet speed, which could reduce video quality during peak usage. The system does not provide full financial management for submitted bids; it only displays the bid amounts during the live bid opening for transparency purposes. Another limitation is that the in-order display of submissions relies on accurate timestamps from the system. If the server time is incorrect or technical issues occur, the display order could be affected, although safeguards are in place to minimize such risks.

Definition of Terms

The following terminologies are defined both technically and operationally to help readers understand the study.

Audit Trail is a record that tracks financial or system activities back to their source. It helps verify transactions, detect errors or discrepancies, and ensure transparency and compliance with regulations (A. Hayes, 2025).

Application Programming Interfaces (APIs) core component used in many applications, whether on mobile devices, computers, or websites. It allows different software systems or services, often created by different organizations, to communicate with each other and share data or functionality.

Bidder is an individual, company, or organization that submits a proposal or bid to provide goods, services, or infrastructure to a government institution during a procurement process.

Bid Opening is the official procedure during which submitted bid documents are opened and reviewed by the Bids and Awards Committee to determine compliance with procurement requirements.

Bids and Awards Committee (BAC) is the body within Southern Luzon State University that is responsible for managing and overseeing all procurement and bidding activities of the university in accordance with the provisions set forth by Republic Act No. 9184.

Compatibility is the ability of a system to operate effectively with other software systems, devices, or platforms without conflicts or performance issues.

Digital Signature a digital signature is a cryptographic technique used to validate the authenticity and integrity of digital documents (Hashemi-Pour, Gillis, & Lutkevich, 2024).

Encryption is a security technique that converts readable data into coded information to prevent unauthorized access to sensitive documents and system data.

Flexibility is the capability of a system to adapt to changes in requirements, user needs, or operational environments without requiring major modifications.

Functional Suitability is the degree to which a system provides functions that meet specified requirements and user needs when used under defined conditions.

ISO/IEC 25010:2023 is a family of an international standard that defines a comprehensive quality model for software product evaluation in terms of functional suitability, performance efficiency, usability, reliability, security, maintainability, flexibility, and safety.

Live Streaming/Broadcasting is the real-time transmission of video and audio content over the internet that allows viewers to witness events as they occur without the need for physical presence at the location where the event is being conducted.

Maintainability is the ease with which a system can be modified, updated, corrected, or improved by developers to adapt to changes or fix issues.

Procurement is the process through which a government institution acquires goods, services, and infrastructure projects through a structured and regulated bidding process in accordance with applicable laws and policies.

Performance Efficiency is the capability of a system to perform its functions effectively while utilizing appropriate amounts of system resources such as processing time, memory, and network capacity.

Reliability is the capability of a system to maintain consistent performance and operate correctly without failure over a specified period of time.

Republic Act No. 9184 (RA 9184). It is the Government Procurement Reform Act of the Philippines that mandates all government agencies, including State Universities and Colleges, to conduct their procurement activities in a manner that is open, competitive, transparent, and accessible to the public.

Safety is the degree to which a system operates without causing harm, data loss, or unintended consequences that could negatively affect users or system operations.

Security (Software Quality) is the ability of a system to protect data and system resources from unauthorized access, ensuring confidentiality, integrity, and availability of information.

State Universities and Colleges (SUCs) are government-funded institutions of higher learning in the Philippines that are governed by the Commission on Higher Education and are required to comply with public procurement regulations set by the Government Procurement Policy Board.

Transparency is the principle of conducting institutional activities in an open and accessible manner that allows stakeholders such as students, faculty, suppliers, and the general public to observe and verify how public funds and government processes are being managed.

Two-Envelope Bidding system is a procurement method where bidders submit two separate envelopes: one with technical details and another with financial proposals. The technical envelope is evaluated first, and only those that meet requirements have their financial bids opened, ensuring fairness and transparency.

Usability is the extent to which a system can be easily learned, understood, and used by users to achieve their intended tasks efficiently and satisfactorily.

Web is an information system that can enable documents and other web resources to be accessed through the internet. The documents and downloadable media are made available to the network through web servers and can be accessed by programs such as web browsers software.

Web-Based System is a system that runs in a web browser. It can also be used to describe applications that have a very small component of the solution loaded on the client PC. The host server for a web-based system could be a local server, or it could be accessed via the internet.

YesParency is a web-based bidding system developed for Southern Luzon State University that enables digitized bid submission, signature-based document access control, live bid opening broadcasts, and audit trail monitoring to improve transparency, efficiency, and security in the procurement process.

Zoom is a commercial cloud-based video conferencing platform that enables users to conduct virtual meetings, webinars, and live sessions over the internet.

Chapter II

REVIEW OF LITERATURE AND STUDIES

Bidding

The process of competitive bidding is very significant to ensure fair and transparent business practices. However, there are certain firms that tend to influence bidding practices through practices such as bid-rigging, where firms tend to influence the outcome of contracts through bidding. According to Carbone et al. (2024), firms tend to ensure that they secure contracts while maintaining the essence of fair competition by colluding and influencing bidding practice through similar bids, temporary association and strategic participation. On the other hand, having more bidders in the bidding process may assist in ensuring that fairness and efficiency are achieved. The more bidders that are involved in the process, the higher the chances of organizations will be able to get the best deals. As explained by Tátrai et al. (2024), by increasing the level of competition in public procurement enhances transparency and encourages suppliers to provide better proposals during the bidding process.

Transparency and proper regulation are essential in making the procurement process honest, fair, and accountable. When procurement policies and bidding procedures are clear and open, suppliers are given a fair chance to compete, and the risk of irregularities is reduced. According to the Organisation for Economic Co-operation and Development (2024), transparent procurement systems help organizations and governments manage resources more effectively while building trust among stakeholders. When the bidding process is transparent, everyone involved can clearly understand how proposals are submitted, evaluated, and selected. This makes the process more dependable and simpler to monitor. Transparency helps stakeholders to

track all stages of the bidding process. This is important in ensuring that decisions are based on objective criteria rather than bias. Because of this, many organizations are encouraged to adopt electronic procurement systems, which can further improve both transparency and efficiency.

According to Taufiq et al. (2023), making procurement information available online improves transparency and helps for better oversight of the government procurement process. The transparency of the bidding process is also improved by the availability as offered by the digital platforms. This is because online platforms make it easier for stakeholders to access important information regarding the bids. Also, they are in a position to monitor the way the contracts are awarded, and check whether procurement procedures are properly followed. When this information is made accessible to the public, organizations can reduce the chances of irregularities and ensure that the entire process remains fair and transparent. Thus, through ensuring that the information used in the bidding process is accessible to the public, these organizations are able to effectively prevent irregularities in the process.

In relation to these findings, the development of the Yesparency system can also be seen as being in line with the necessity of having a transparent and regulated process in the bidding system. This is where the integration of digital tools that support open access of bid information, contract tracking, and procedural monitoring, Yesparency aims to strengthen accountability and reduce opportunities for irregularities. The system is intended to promote transparency, ensure fair competition between suppliers, and provide an objective evaluation in the bidding process. This is where YesParency contributes to improving efficiency, integrity, and trust in procurement activities.

E-Bidding Systems

According to Perante et al. (2025), the implementation of a cloud-based Purchase Order (PO) and Inspection and Acceptance Report (IAR) management system has substantially improved procurement operations through three key benefits which are improved workflow processes and better data accuracy and higher system accessibility. The system's computerized data entry and validation reduced errors which occur in manual processes because it implemented role-based functionality together with enhanced security measures to ensure reliable operational performance. Users reported that the intuitive interface and instant access to historical and current records helped them complete tasks efficiently while promoting transparency and organizational accountability. The study shows how public sector procurement requires digital systems because Republic Act No. 9184 forces organizations to follow strict rules which mandate specific document formats and multiple system processes for official approvals. The system achieves process improvement through automation which creates regulatory compliance while proving that technology maintains both precise results and system adherence and business productivity across public sector and private sector purchasing environments.

Extending the discussion on digital procurement systems, Tran et al. (2025) examined the construction industry's adoption of electronic construction bidding, established through the 2023 Bidding Law as both a mandatory requirement and a method to assess construction company operations. The study shows that companies differ in their preparedness for operations because they have different levels of technological resources and human resources and internal business procedures and organizational values. The recommended actions include creating capacity assessments which will lead to establishing e-bidding support centers in various regions and

building an online digital learning resource and working together with enterprises and industry associations and government organizations. The measures work to improve readiness while supporting knowledge sharing and enforcing mandatory e-bidding requirements which will start in 2025. The research provides a base for conceptual and methodological development but the author requests large quantitative studies to test the E-Readiness framework because it will help create policies that enhance enterprise capacity and boost public support effectiveness for Vietnam's construction industry.

Similarly, Venant (2022) found that the implementation of an e-bidding system in public procurement processes within Rwandan public institutions leads to improved operational results because it enables better scheduling, increased control capabilities, decreased expenses, enhanced operational visibility, and improved capacity to meet customer needs. The research findings demonstrate that E-Noticing improves planning and control processes while E-Submission delivers cost savings for procurement operations and E-Evaluation supports transparent evaluation processes and E-Access establishes faster communication methods with stakeholders. The research findings indicate strong agreement among respondents, confirming that the electronic flow of documents streamlines supply chain communication, clarifies requirements, and allows better assessment of market conditions. The public institutions should establish complete e-procurement systems according to the research findings while they should invest in training their employees and develop infrastructure to enable effective system operation. The implementation of electronic procurement systems leads to faster business processes which enhance regulatory compliance while simultaneously creating higher levels of operational visibility and decreasing corrupt practices and fostering stakeholder cooperation with current technological advancements.

The studies collectively demonstrate that digital procurement systems improve operational efficiency, reduce manual errors, and enhance regulatory compliance and transparency across diverse sectors and countries. Perante et al. (2025) Tran et al. (2025) and Venant (2022) all highlight how automation and electronic workflows facilitate accurate, timely and accountable procurement processes. The findings establish a connection between YesParency's features and its digitized two-envelope bidding system which enables encrypted document access and creates an audit trail to maintain secure operations with reduced errors and complete transparency. YesParency implements evidence-based functions to assist organizations in achieving efficient procurement processes while they comply with regulations and track their operational activities.

Digital Signature

In the study of Gowda et al. (2023), digital signatures provide more benefits than traditional handwritten signatures because they create anonymity protection and untraceable verification which includes evidence of tampering and non-repudiation and improved data integrity for all system requirements. The survey of existing literature presents practical solutions which use digital signatures together with watermarking and two-factor authentication and QR codes and blockchain technology to create enhanced security protection. The studies show multiple limitations which include inadequate performance testing and insufficient security testing and restricted functionality to specific situations and dependence on specific types of hardware and software. The literature has major gaps because it lacks detailed evaluations of current advanced digital signature systems and it only partially studies how different parameters

like key sizes and hash functions and elliptic curves affect system performance and it only supports specific new technology environments which include Internet of Things and blockchain and cloud computing and it pays insufficient attention to how users adopt systems and how usable they are and it only partially addresses legal requirements and regulatory issues and post-quantum security threats. The survey demonstrates how digital signatures create transformative possibilities while requiring additional studies to improve their security and scalability and their ability to work across different situations.

Complementing this view, Sum and Nyaboga (2024) stated that digital signatures function as an essential security method which improves electronic transactions by making them more secure and efficient while establishing their legal validity for multiple industry sectors. The adoption of digital signatures faces multiple obstacles because their benefits are countered by problems with standardization and secure key management and inconsistent legal recognition and technical problems that prevent small businesses from using them. Digital signatures establish superior advantages which exceed those of handwritten signatures because they provide better protection against forgery through their secure document execution process and less expensive operations together with their ability to create traceable audit records. E-commerce financial services, government and healthcare all use their practical applications because they protect the authenticity and integrity of digital records and agreements. The study shows that businesses need to adopt digital signatures because their electronic transactions without paper documents become more common. This adoption leads to faster commerce processes while building trust and legitimacy together with data security for digital business interactions.

Further expanding on the technical mechanisms, Pattnaik et al. (2023) explained that digital signature privacy systems use existing cryptographic methods together with new

technological developments to protect the three main elements of digital documents which are authenticity and integrity and confidentiality. The study explains that digital signatures are generated using a sender's private key combined with a hash of the document, while the recipient verifies authenticity using the sender's public key. The digitally signed documents maintain document integrity yet they fail to protect document privacy because all users can access the document even when they cannot alter its content. To solve this problem cryptographic algorithms which include RSA and AES and Diffie-Hellman and blockchain technology provide solutions to maintain privacy. RSA secures data and signatures through asymmetric encryption, AES provides symmetric block cipher encryption for data privacy, Diffie-Hellman enables secure key exchange, and blockchain authenticates transactions while ensuring tamper-proof records. The paper proves that these methods strengthen privacy and security protection, and the research team will test RSA-AES systems to measure their ability to safeguard digital signature systems.

While the technology is technically robust, Geels (2024) highlighted that digital signatures provide online security protection through their ability to let users confirm their identity while showing their digital presence commitment. The technology remains underused because regular people do not understand how it works despite their knowledge of its basic functions which allow them to show their identity through "It is Me" and express their agreement through "I Agree". The study's interviews which followed those with users discovered four main barriers which prevent people from using the service because they think digital signatures work only in formal situations which require high security or because they doubt the signatures ability to protect against security threats or because they value the situation more than the ability.

The research works demonstrate that digital signatures function as essential tools which protect document authenticity and document integrity and document non-repudiation during digital transactions. The research of Gowda et al. (2023) together with Pattnaik et al. (2023) demonstrates that digital signatures together with cryptographic algorithms such as RSA and AES and blockchain technology and other supporting systems create strong security measures which protect user privacy. Sum and Nyaboga (2024) demonstrate that digital signatures establish legal recognition together with traceable audit documentation and protection against forgery which organizations require to conduct business across different industries. The research of Geels (2024) demonstrates that organizations face difficulties when trying to implement new systems because they require both systems to be usable and for users to receive proper training. The research findings support YesParency's encrypted document access system which creates audit trails through digital signatures that protect sensitive bidding materials while maintaining record integrity and authenticity and establishing traceable document histories and fulfilling all technical and legal security standards.

Document Encryption

Data security remains a central concern in the digitalization of organizational processes. Based on the study of Sudershan et al. (2023), file encryption operates as an essential method which secures sensitive digital data through the combination of symmetric and asymmetric cryptographic methods. The system starts processing when users provide a file through their input of JPG or MP4 or PDF or DOCX or other supported file formats. The system uses the RSA algorithm to create asymmetric keys which require the recipient to establish a public and private

key pair while sharing the public key with the sender. The system creates a random 256-bit symmetric key which the AES algorithm uses to encrypt the file thereby providing secure and efficient data protection. The system encrypts the symmetric key through the RSA public key before sending it to the recipient. The recipient first uses the RSA private key to decrypt the symmetric key during decryption which he uses to restore the original file. The hybrid encryption method provides enhanced data protection through its ability to secure confidential data and protected communications and digital resources.

Complementing these results, Isewon et al. (2022) established that technological advancements and digital data expansions create more obstacles for secure data transmission throughout diverse industries. Organizations that implement digital systems need to develop effective ways to protect their confidential data when they store and transmit information. The study presents the Hyde system which functions as an encryption solution that improves data integrity while delivering extra protection for personal and organizational information. The system uses encryption to protect documents and texts which stops unauthorized users from accessing them despite physical or remote system breaches. The system protects data from further attacks while building stronger overall data security. The study proposes that future encryption development will enable direct operating system integration of encryption tools which will let users encrypt and decrypt files through built-in shortcut menu features that will make encryption technologies more accessible and boost their usage.

Similarly, Daalen (2023) highlighted that governmental restrictions on encryption technologies create major threats to privacy rights and data protection rights and freedom of speech rights because these measures increase the danger of unauthorized access to confidential information. The analysis shows that actual encryption systems apply policies which include

using weaker encryption algorithms and storing keys with external parties and performing client-side scanning and allowing secret government access to communications. The measures create new security weaknesses which both law enforcement agencies and foreign intelligence outfits and cybercriminals can use to exploit systems. The assessment of encryption-related regulations requires organizations to consider the likelihood of unauthorized access as their primary element. The study shows that courts and policymakers must assess security risks before imposing encryption restrictions which need strong evidence to prove their necessity because these restrictions will have serious effects on personal privacy and data security.

Expanding on this perspective, Oladoyinbo et al. (2024) stated that data encryption algorithms protect digital information through their ability to convert plaintext into ciphertext by using mathematical algorithms together with cryptographic keys. The system protects sensitive data because it restricts access to authorized users who hold the correct decryption key throughout the process of data transmission and storage and actual use of the information. The literature demonstrates that encryption improves various data security elements which include authentication and privacy protection as well as regulatory compliance and data integrity. Encryption acts as a fundamental protection measure because it prevents hackers and service providers and other unauthorized parties from accessing protected data which helps organizations defend against ransomware and man-in-the-middle attacks. Encryption algorithms help organizations meet compliance requirements for regulatory systems which include HIPAA and PCI-DSS and GDPR. The research shows that strong encryption systems create fundamental security protection which secures confidential information while helping organizations build trust and meet legal requirements and defend against new cybersecurity risks.

The literature collectively demonstrates that e-procurement systems enhance organizational transparency and operational efficiency and financial performance across different business environments. Kumar et al. (2023) shows that e-procurement reduces monopolistic power, corruption risks, and enhances public trust, while Ntshingila (2024) emphasizes that SMEs benefit from automation, error reduction, and competitive advantages through digital procurement processes. Singh and Vij (2022) explain that successful technology adoption requires organizations to develop their technological capabilities and organizational resources and executive leaders to handle both internal and external challenges. Empirical findings from Sum and Nyaboga (2024) confirm that e-procurement improves operational results, cost savings, supplier management, and accountability, and Aung et al. (2023) demonstrate that it strengthens supplier relations and enhances overall procurement performance. The studies demonstrate that digital procurement platforms such as YesParency enable organizations to achieve their desired results through structured transparent bidding processes which maintain operational efficiency and prevent manual errors and ensure proper competition between bidders while enabling proper accountability processes.

Procurement

Public procurement is a significant role in any government agency or public organizations that ensures that public procurement activities are carried out in a transparent, efficient, and sustainable manner (Hochstetter et al., 2023). Transparency in public procurement is important as it enhances the credibility of an institution and increases public trust in the handling of public resources. Transparency in public procurement also supports good governance

within public institutions, including universities. This will help in ensuring that there is accountability in the use of public funds and that stakeholders are able to monitor the activities involved in public procurement more effectively . Additionally, public access to procurement data promotes public discussions about the matter and helps in the fight against corruption in the procurement process. For this reason, organizations are encouraged to implement digital monitoring systems and accountability mechanisms to enhance transparency and ensure responsible contract planning (Mebrate & Shumet, 2024).

According to Felizzola et al. (2024), open data has a significant role to play in ensuring transparency in public procurement. It gives detailed information about the different stages involved in the procurement process, from planning, bidding, awarding, and executing contracts. This information is important to stakeholders since it helps them understand how public funds are being used. However, the availability of procurement data alone does not guarantee that it will be used effectively, which highlights the need for proper monitoring and analysis to maximize the benefits of open procurement data. Another approach that improves procurement processes is the implementation of e-procurement systems. The system aims to enhance the efficiency of procurement activities procedures and enable procurement committees to report their activities to higher authorities and the public more efficiently. Despite these benefits, public procurement systems are not without limitations. It is also possible that some of the suppliers may still engage in dishonest practices, which indicates that technology must go hand-in-hand with ethical standards and the enforcement of laws for fair and honest public procurement (Novitaningrum, 2024).

The studies show that transparency is important in public procurement. Hochstetter et al. (2023) explain that transparency is essential for building trust in government institutions and for

good governance by making the use of public funds more accountable. Mebrate and Shumet (2024) also emphasizes that there is a need for digital monitoring systems for the enhancement of transparency and ensuring that procurement activities are properly managed. According to Felizzola et al. (2024) note that with the availability of open data for public procurement processes, stakeholders are able to understand the use of public funds. However, the data should be monitored for its effectiveness. On the other hand, Novitaningrum (2024) emphasizes that technology alone is not enough to prevent unfair practices, ethical standards and proper enforcement of rules are still necessary to ensure honest procurement processes. These findings relate closely to the features of YesParency. The system's digital two-envelope bidding system, encrypted document access, real-time live streaming of bid openings, and audit trail logging all support transparency and accountability in procurement. Just like the studies suggest, YesParency helps make procurement activities more visible, organized, and traceable. To keep procurement processes transparent, secure, and reliable, the system's monitoring and record-keeping features support proper oversight and provide accessible information.

E-Procurement

Kumar et al. (2023) states that the implementation of e-procurement by Pakistan's government procurement agencies functions as a vital tool for combating both systemic corruption and operational inefficiencies. The study shows that suppliers who hold monopoly power over markets create obstacles to proper competition while preventing successful implementation of e-procurement systems because these companies will not accept any technological changes that threaten their current market dominance. The procurement process

becomes more difficult to understand because different parties involved start to gather information about the process at different times which results in biased treatment of certain individuals through corrupt activities. The structural problems of the system demonstrate that organizations need to establish transparency and accountability as essential components which they should use to achieve their procurement reform objectives. E-procurement enables organizations to access procurement information while following uniform processes which leads to decreased monopolistic power and decreased corruption risks and improved public trust. The study demonstrates that e-procurement functions as a complete system transformation which enables public procurement processes to achieve better operational performance and improved governance and restored public trust.

Similarly, research shows that e-procurement provides operational benefits for smaller enterprises. Ntshingila (2024) states that e-procurement systems help small and medium enterprises (SMEs) in Gauteng province South Africa to improve their business operations. The study shows that three elements which include data quality management and e-procurement practices and competitive advantage together help businesses to achieve better results while competitive advantage gives the biggest impact. The results demonstrate that system automation together with digital procurement processes enables organizations to achieve better procurement visibility and operational efficiency through error reduction. The e-procurement system integration enables SMEs to share information while decreasing risks and achieving better strategic advantages. The research shows that organizations use technological tools for procurement because these technologies help them to create competitive advantages which drive their business success. The research demonstrates that automated e-procurement systems provide SMEs with an effective solution which enables them to improve their procurement operations

and boost their business results despite their research limitations in geographic reach and participant variety.

Expanding on this perspective, the adoption and success of e-procurement systems depend on technological, organizational, and contextual factors. Based on the study of Singh & Vij (2022), e-procurement emerged as a result of information and communication technology advances which have fundamentally changed public and private procurement methods. The literature shows that e-procurement exists as a potential solution which various operational environments and their different interpretations prevent from achieving common understanding. The studies show that organizations need to comprehend three main themes which include adoption drivers and barriers and critical success factors that determine e-procurement system performance. The structural political and technological limitations of developing countries create obstacles for adoption but organizations still pursue implementation because it will bring better transparency and efficiency and lower costs. The Indian Government's GeM platform demonstrates how organizations can achieve digital procurement through technology while facing challenges that come from needing to create accountability systems against institutional pushback. The literature shows that organizations need strong leadership and proper technological infrastructure and organizational readiness to succeed in e-procurement system implementation and sustainability.

In addition, empirical studies show the measurable benefits of e-procurement on organizational performance and financial outcomes. Sum and Nyaboga (2024) states that Uasin Gishu County in Kenya has improved its procurement results through e-procurement implementation because it now achieves better operational results and cost savings and increases its transparency. The study shows that e-payments and e-ordering and e-sourcing all lead to

positive results which help organizations achieve better financial outcomes and decrease their operational tasks and improve their supplier management processes. E-payments decrease both transaction fees and procurement costs while e-ordering streamlines procurement operations through automatic processes that decrease mistakes and boost order precision. E-sourcing enables organizations to enhance their supplier selection process and their contract management system as well as their ability to compare prices which results in reduced expenses and increased revenue. The implementation of e-procurement systems boosts organizational performance through its ability to develop efficient operations which encourage suppliers to maintain fair competition while ensuring procurement activities remain accountable and traceable. The results show that digital procurement platform implementation requires more than standard procedures since it creates fundamental changes which lead to actual increases in performance and financial discipline and administrative transparency throughout county government operations.

Similarly, international contexts demonstrate the link between e-procurement practices, supplier relationships, and operational efficiency. According to Aung et al. (2023), international non-governmental organizations (INGOs) in Myanmar achieve better procurement results and supplier relations through their use of e-procurement practices. The study shows that online purchase specification and potential supplier identification activities improve operational efficiency while enhancing communication between buyers and suppliers. The Pearson correlation analysis shows that e-procurement establishes positive links to procurement performance ($r = 0.341$, $p < 0.042$) and supplier relations ($r = 0.567$, $p < 0.01$) while supplier relations directly drive procurement results ($r = 0.338$, $p = 0.044$). The study shows that effective e-procurement systems enable organizations to optimize their processes while minimizing expenses and building better supplier relationships which lead to higher organizational

performance. The study shows that organizations need to establish organized methods for managing supplier communication because this process enables them to achieve better procurement results while e-procurement provides organizations with both technological capabilities and strategic advantages for their operations.

The literature shows that e-procurement functions as a revolutionary system which improves public and private procurement processes through better transparency and operational efficiency and greater accountability. The research demonstrates that this system helps organizations of all sizes through its three main benefits which include lower corruption rates and better supplier relations and increased operational efficiency. The successful implementation of a project requires organizations to establish the necessary technological systems and executive leadership and to prepare their teams to deal with both institutional and political obstacles. The complete e-procurement system provides organizations with an efficient system to achieve better governance and financial performance and competitive benefits in all operational environments. The research results from these studies demonstrate that YesParency and other digital procurement platforms can achieve equal outcomes because they create structured bidding processes which provide transparent bidding procedures that enable traceability. The bidding process allows multiple parties to compete while maintaining system integrity through effective error reduction and efficiency improvements. The system maintains security through methods which ensure all activities remain visible for the purpose of conducting audits.

Live Streaming

According to Mahmoud and Abozariba (2024), WebRTC demonstrates considerable versatility and potential as a real-time communication tool across multiple sectors, including telemedicine, online gaming, education, and live streaming. Its peer-to-peer architecture and low-latency capabilities, coupled with robust security protocols, facilitate high-quality data transmission and enable diverse applications beyond conventional video conferencing. However, the review identifies several limitations and research gaps: the lack of large-scale deployment data, constrained language scope, and the rapidly evolving technological landscape may affect the generalizability and timeliness of findings. Moreover, improvements in scalability, security, and handling of non-video data are needed, alongside exploration of integration with emerging technologies such as machine learning and edge computing. Overall, while WebRTC holds substantial promise for shaping future real-time communication applications, targeted research is essential to address existing challenges and expand its practical and interdisciplinary applications.

Building on the potential of real-time technologies, Huang et al. (2023) stated that live streaming has changed social commerce because it creates new methods for online user interaction which affect how people use the internet and how companies develop their marketing plans. The primary elements which determine user engagement and motivation and ongoing participation include social interaction and content accessibility and advertisement interactivity and humor and entertainment value. The research demonstrates that people who use live streaming extensively develop addictive tendencies which self-efficacy and self-esteem and information-seeking attributes control. Streamers and platform managers use their content creation and user interaction and virtual gifting activities to drive user engagement and build

trust with users while they sell products because their activities produce immediate results and their audience base generates extra benefits. Live streaming enables users to see its benefits while decreasing their risk assessments which leads to increased social commerce participation. Platform designers need to combine Human-Computer Interaction design principles with participatory design methods so that user behavior and motivations match their needed service results.

Similarly, Zhu (2022) noted that live streaming creates an essential change in all business and social and political areas because it modifies how people connect with each other and how they communicate and how they adapt to new situations. The research shows that live streaming enables companies to interact with customers through live video, which helps businesses and social media marketers develop new methods for their marketing and branding and customer relationship management. Live streaming supports commercial use, but it also drives cultural and social communication evolution because it enables people to connect beyond standard communication methods while creating minor changes in how they interact with others. Organizations need to understand live streaming because it helps them build audience connections and create better user experiences and develop their businesses through new technologies. The study establishes live streaming as a fundamental element that drives both social change and economic development, which demonstrates how live streaming affects human communication and social organization and business operations.

The research findings demonstrate that real-time communication technologies which include WebRTC live streaming capabilities lead to increased user engagement and better user interactions and improved access to information in various fields. Mahmoud and Abozariba (2024) show that WebRTC offers strong technical protection and secure systems while Huang et

al. (2023) and Zhu (2022) show its effects on social and business and institutional environments. The evidence supports YesParency's live streaming system which delivers secure low-latency bid opening broadcasts that enhance transparency and stakeholder engagement and enable real-time interactions. YesParency uses these proven advantages to maintain technical reliability together with social accessibility for live procurement activities which follow digital communication standards and organizational transparency requirements.

Audit Trail

Afadzinu et al. (2024) stated that technological developments create major changes which affect audit processes by introducing new methods that guarantee accuracy and complete disclosure of information to auditors in the digital era. The combination of data analytics and artificial intelligence and automated systems improves audit transparency through better documentation methods and immediate reporting capabilities and creation of complete audit records which stakeholders can access. The system increases audit objectivity because it uses data analysis methods which can work with extensive datasets to produce reliable results and maintain audit standards. The implementation of these technologies faces several obstacles which include difficulties in execution and limited resources of small businesses and issues related to data protection and potential biases present in algorithms. The study shows that auditors must develop specific abilities and regulatory systems must adapt to manage emerging threats. The research shows that technology enables significant audit quality improvements but organizations need to handle its implementation together with ethical supervision to establish dependable auditing methods which succeed in present and upcoming challenges.

Complementing this perspective, Bani et al. (2025), in their systematic review examining 84 articles, found that digital innovations including AI, machine learning, big data analytics, blockchain technology, cloud computing, and robotic process automation have created major changes in public and private sector audit operations. The technologies assess the study's aim to examine digital audit developments and their effects because they improve audit work through nine different quality aspects which include effectiveness and efficiency and transparency and accountability and timely performance. The organization faces three main obstacles which stem from technical difficulties and human resource restrictions and regulatory limitations because businesses experience expensive implementation requirements and their systems become difficult to integrate and they encounter cybersecurity threats while their employees lack digital skills and refuse to adapt because they require additional training and there are no proper standards in place to address ethical matters and maintain public trust. The research demonstrates that organizations need to resolve systematic issues while developing institutional capabilities to achieve successful implementation of digital auditing systems. The research establishes a basis for policymakers, practitioners, and academics through its global evidence synthesis which connects adoption to specific regulatory contexts while upcoming research should validate frameworks through empirical testing and assess different national approaches and investigate how digital technology has changed auditors' roles into strategic partners for governance.

Similarly, Angeles et al. (2023) examined the benefits and challenges of digitalization in the auditing profession, particularly within the Big Four firms in the Philippines. The research found multiple benefits which included better auditor skills and improved operational flexibility and efficiency and advanced data processing methods and new audit assurance services and

better audit report quality and reliability and increased operational effectiveness and better auditor reputation. The main difficulties which were identified in this research included inadequate personnel resources and training gaps and the new auditor requirements and resistance from stakeholders and the new auditing environment and issues related to data protection. The study identifies future development needs which include research on new digital technologies such as IoT and augmented reality and ERP systems and drones and machine learning and research expansion into various international research contexts. The study stresses that auditors must acquire essential knowledge and skills while the research recommends using face-to-face interviews with different interview subjects to achieve better results during the digital auditing transition.

The research findings demonstrate that digital technologies provide major benefits to audit operations because they enhance both audit quality and transparency and audit reliability. Afadzinu et al. (2024), Bani et al. (2025), and Angeles et al. (2023) emphasize the need for secure, comprehensive, and accessible digital records to support accountability and regulatory compliance. YesParenity's audit trail feature proves its value through these findings because it records all procurement activities in a secure format that cannot be changed and which auditors can review easily. By incorporating these evidence-based practices, YesParenity strengthens stakeholder trust and enhances the reliability and governance of digital procurement processes.

Role-Based Access Control

In the study of Sodikova & Abduvaliyeva (2022), role-based access control systems enable organizations to strengthen their security through their ability to manage authentication

and authorization and policy enforcement which controls access to resources. The study demonstrates that traditional authentication methods still receive broad usage yet their unencrypted transmission and static password and email verification weaknesses create security vulnerabilities which require organizations to adopt dynamic encrypted authentication systems. The implementation of role mining and structured policy mapping functions as a vital method for organizations to decrease role redundancy while establishing system access that corresponds with their hierarchical organization structures. The evaluation of open-source RBAC solutions against commercial RBAC solutions demonstrates the trade-off between system flexibility and user-friendly design. The research results demonstrate that organizations need to implement RBAC through specific role definitions and flexible access policies and security systems which include VPN access controls to protect their complete resource inventory.

Further emphasizing the security benefits of structured access control mechanisms, Mehra (2024) stated that RBAC establishes structured access controls which designate particular access rights to specific system functions and protected information. The study shows that RBAC protects organizations from insider threats and ransomware attacks and malware attacks through its access rights restrictions which decrease potential attack vectors and its security incident recovery controls. RBAC enables organizations to achieve regulatory compliance with GDPR and HIPAA because it controls which users can access data based on their required organizational duties and existing legal obligations. The use of storage device technology enables organizations to implement effective access control systems which provide better auditing and monitoring capabilities while supporting their system growth requirements. Case studies from financial institutions and healthcare organizations demonstrate that RBAC enhances data security

and enables faster recovery operations and builds organizational strength which establishes its value as a key security method for contemporary data protection systems.

Similarly highlighting the impact of RBAC on system security and user experience, Cardos et al. (2025) stated that the system security and user satisfaction within the SEAIT OJT Evaluation and Feedback System show improvement through Role-Based Access Control (RBAC) implementation together with data privacy protection measures, which users evaluate positively across all system features including layout and responsiveness and access accuracy and secure data management. The findings show that access permission updates together with stable system operation, which enabled users to access their feedback, built their trust in the system through its RBAC implementation, which serves secure digital feedback environments. The study results show that users face a communication problem because they need security guidelines beyond technical functions to understand system design. The study demonstrates that organizations must combine advanced access security methods with user-friendly design processes to create institutional feedback systems, which protect sensitive data while building user trust and system visibility and operational efficiency.

The studies demonstrate that Role-Based Access Control (RBAC) enhances system security by implementing controlled methods for authentication and authorization and policy enforcement, which secures access to protected resources. RBAC reduces vulnerabilities associated with traditional authentication methods while protecting systems from insider threats and ransomware and malware attacks through controlled access rights. The literature shows that RBAC helps organizations meet regulatory standards while its monitoring and auditing capabilities improve system reliability and disaster recovery processes. The implementation of

RBAC leads to better user trust and satisfaction when organizations maintain system stability and provide transparent security procedures.

Public Procurement

Public procurement plays an essential role in government agencies and public organizations as it helps ensure that procurement activities are conducted in a transparent, efficient, and sustainable manner (Hochstetter et al., 2023). Maintaining transparency in procurement processes is particularly important because it strengthens institutional credibility and builds public trust in how government resources are managed. Transparency in public procurement also supports good governance within public institutions, including universities. Providing public access to procurement information promotes accountability and enables stakeholders to monitor procurement activities more effectively. In addition, open access to procurement data encourages informed public discussions and contributes to efforts aimed at addressing corruption in procurement practices. For this reason, organizations are encouraged to implement digital monitoring systems and accountability mechanisms to enhance transparency and ensure responsible contract planning (Mebrate & Shumet, 2024).

According to Felizzola et al. (2024), open data plays an important role in promoting transparency in public procurement. It provides detailed information about the different stages of the procurement process, including planning, bidding, awarding, and contract execution. Access to this information allows stakeholders to track how public funds are used, identify potential irregularities, and support better policy decisions. However, the availability of procurement data

alone does not guarantee that it will be used effectively, which highlights the need for proper monitoring and analysis to maximize the benefits of open procurement data.

Another approach that improves procurement processes is the implementation of e-procurement systems. The system aims to enhance the efficiency of procurement activities procedures and enable procurement committees to report their activities to higher authorities and the public more efficiently. Despite these benefits, public procurement systems are not without limitations. Some suppliers may still engage in dishonest practices, which suggests that technological solutions must be supported by strong ethical standards and strict law enforcement to ensure fairness and integrity in public procurement (Novitaningrum, 2024).

In line with these efforts to improve transparency, several researchers emphasize that public procurement serves as an important instrument for promoting transparency and accountability in public institutions. Transparent procurement practices encourage informed public discussions and allow stakeholders to monitor how public funds are utilized, identify potential irregularities, and support better policy decisions. However, studies also indicate that the availability of procurement data alone does not guarantee its effective use, and dishonest practices may still occur within procurement processes. This suggests that technological solutions and open procurement systems must be supported by strong ethical standards, proper monitoring, and strict law enforcement to ensure fairness and integrity in public procurement. These findings highlight the importance of implementing transparent and accountable public procurement systems that allow stakeholders to effectively monitor procurement activities.

Document Management System

Document Management system (DMS) is a software designed to reduce the use of paper by storing, tracking and managing electronic documents . It supports managing digital information and has been widely adopted by government agencies (Gani et al., 2024) . It helps organizations to improve document storage and accessibility (Jain, 2021). This system replaces the traditional manual way of managing and processing documents and can be used in different industries that require a large number of documents to be managed. In addition, a document management system improves document organization through features such as search functions, and version control. These features allow users to easily locate, update and share documents within the organization, improving efficiency and ensuring that important records are properly maintained.

Furthermore, Tkachenko and Denisoca (2022) concluded that document management systems can classify documents using information system technologies, improving the retrieval of digital records. The system enables users to manage extensive digital document collections because it organizes documents according to the specific content and characteristics. Users can find their documents through automatic classification because it encourages them to search without needing to examine all available files. Document management systems deliver document organization benefits while they enhance document retrieval speed and enable organizations to handle their digital records more effectively.

The implementation of a document management system (DMS) offers many advantages compared to conventional systems. It significantly helps save time, cost and papers. With digital management, documents can be shared and accessed quickly without having to search

through a stack of paper, enabling faster document retrieval and improving productivity across departments. This not only saves employees' time but also reduces the storage costs (Sari et al., 2024). Making it more efficient and reliable in terms of handling large volumes of documents.

The reviewed studies show that document management systems have become essential for organizations which need to handle extensive document collections. The systems enable organizations to lessen paper use because it allows electronic document storage and tracking and management which leads to time and cost savings. Organizations can achieve better operational results by implementing document management systems that replace their existing paper based document handling procedures. The document management systems enable employees to work more efficiently because they provide fast digital solutions for document organization and storage and retrieval. The search functions and document indexing and version control features enable users to find essential files while maintaining correct document updates. Digital document management enables employees to work together better because they can easily access and share documents throughout the organization. Overall, the implementation of document management systems enhances record management, reduces administrative workload, and increases productivity in document-related processes.

Visual Studio

Balić (2026) noted that developer satisfaction and autonomy in tools are important aspects in IDE usage, especially before the emergence and widespread use of AI-based tools. The study indicates that developer preference is an environment that provides clarity, control, and minimal dependency issues, which contribute to a more reliable workflow. Visual Studio

Code, as well as the development tools, indicate that the usability, extension and integration of the tools improve developer's understanding of the project. This shows that tools such as Visual Studio Code enhance transparency in web development where developers can clearly monitor their code behavior, manage configurations, integrate extensions, and maintain structured workflows.

Rodriguez et al. (2026) studied the core Visual Studio Code (VS Code) environment by looking at user experience issues, also referred to as UX smells, through large language models. From the results it is clear that most of the concerns in VS Code are not about the performance of the system, but about how clear, informative, and easy the system is for the user. This means that improvements are more needed in how the tool communicates with users and supports their thinking process, rather than in its basic structure. The study also explains that in complex development tools, usability problems usually appear in areas where users interact with the system most often. The researchers provide a clear foundation upon which is to measure how effectively VS code performs as a development tool. The findings emphasize how improvements to VS Code should prioritize how effectively it can communicate to developers through its interface.

In addition to usability and interface, system-level integration within IDEs also contributes to transparency and functionality. Ernst et al. (2021) propose a way of integrating symbolic execution tools with existing IDEs using the Debug Adapter Protocol. Their implementation for the SecC verifier was described as straightforward and easy to apply. However, the approach has a limitation related to exponential path growth when program branches are not merged, which can affect performance in complex cases. Some potential improvements to this method have also been identified, such as being able to target specific

branches during debugging, as well as other implementation strategies such as Zippers. This study has shown that verification tools can be integrated into an existing IDE in a flexible manner, which can make program verification more interactive, thus bridging the gap between development and verification.

The reviewed studies collectively emphasize that transparency in development environments is enhanced through clear interface design, user-centered system communication, as well as integrated debugging tools. In addition, Balić (2026) emphasizes the importance of tool autonomy as well as workflow, while Rodriguez et al. (2026) similarly stated the need for the improvement of the interaction with the system in order to increase usability as well as understanding for the developers. Ernst et al. (2021) further demonstrate that embedding verification and debugging mechanisms directly into IDEs enhances visibility of program behavior and supports structured code analysis. In relation to YesParency, these findings support the system's digitized two-envelope mechanism, encrypted access to documents using digital signatures, and audit trails, all of which enhances process, activity, and system transparency within a bidding system.

XAMPP

Ray and Vashisht (2024) states that XAMPP serves as an essential tool for developing and testing web-based rental platforms because it enables developers to create and improve their applications at a secure local server environment which they can use before launching their work to the public. The research demonstrates that developers can create interactive websites which users find easy to use by combining PHP to handle server-side operations with HTML which

provides website structure and CSS which delivers visual design. XAMP provides website developers with the ability to test their website components by using htdocs directory access to localhost which enables them to work without needing internet access or external servers. Developers can create their applications through local testing because it enables them to test their work at multiple points which speeds up their development process while maintaining system safety through access control. The organization of XAMP establishes a systematic development framework which enables developers to create web applications with functional elements that include user-friendly navigation and property display based on geographical information.

Similarly, other studies have examined the performance of XAMPP when compared with alternative local development environments. According to the study of Chandra and Setyaningsih (2025), XAMPP serves as the most efficient and scalable testing environment through their comparison of local development server setups which they tested under different levels of concurrent user activity that included MAMP and Laragon as their competitors. The results show that XAMPP achieves better response times when users access the system because it uses its resources effectively to manage multiple requests during peak traffic periods. The response time of Laragon shows competitive results but its extra layers of abstraction make it less efficient. When the number of requests increases, MAMP is not able to adjust to higher pressure. Its resources become scarce and we also see inflexibility in system design. The study shows that XAMPP achieves faster processing times through its lightweight modular design which allows complete Apache MySQL and PHP system integration. XAMPP displays the highest reliability among local development environments because it delivers stable performance throughout extensive testing on demanding projects.

In addition to these findings, literature also highlights the core features and accessibility of XAMPP as a development platform. XAMPP is described as a complete web server solution which enables users to develop and test web applications through its easy-to-use local web server system. Apache Friends offers the software as free open-source software which includes essential components such as the Apache HTTP Server and MariaDB database and PHP and Perl interpreters to create a complete development environment. The study shows that developers can use XAMPP to create local web servers through its lightweight and simple installation process which requires no advanced setup work. The software maintains its usefulness because it runs smoothly on Linux and Mac and Windows operating systems. The software uses the same main components which web servers use in real-world operations to help users transition from local testing to actual server implementation. The platform serves as an essential web development tool because it provides developers with an efficient environment to construct and evaluate their complete web application packages before deployment. (Gordon College – College of Computer Studies, 2022).

The literature highlights XAMPP as an essential tool for developing and testing web-based applications in a secure local environment. Ray and Vashisht (2024) demonstrate how the platform enables users to create interactive websites through its integrated support for PHP HTML and CSS while conducting complete testing in their local environment. Chandra and Setyaningsih (2025) demonstrate that XAMPP outperforms alternative local servers in response time, scalability, and reliability due to its lightweight, modular design. The system achieves practical efficiency for web application development because its open-source components and cross-platform compatibility enable Gordon College users to build complete web applications which they can test before deployment to create YesParenity systems.

HTML5

According to Jnrid & Issn (2023), HTML5 is the fifth major version of the Hypertext Markup Language, the formal completion of which was accomplished in October 2014. HTML5 provides various features that eliminate the limitations of its predecessors and meet the needs of modern web-based applications which are constantly changing. The other new elements of HTML5 like `<header>`, `<footer>`, `<article>`, and `<section>` are aimed at providing a more accessible and meaningful way of content organization while enhancing the semantic structure. In addition, HTML5 with `<audio>` and `<video>` elements has greatly improved native multimedia support thus it is easier to integrate multimedia, performance has been boosted, and a more consistent experience across devices is assured. Moreover, the new Application Programming Interfaces brought in by HTML5 provide the opportunity for advanced functionality without the use of third-party plugins. Because of these improvements, HTML5 has become a key technology in developing modern and dynamic web applications.

Similarly, Cherukuri et al. (2022) emphasized that HTML5 has tremendously developed the functionality and management of input forms in real use. For example, new input types including date, time, email, Url and number have in-built validation, and the correct input methods that are compatible with mobile devices. Many techniques were used to offer input hints, while the released placeholder attribute provides a native and efficient solution. Validation of form data can now be done to a certain extent using HTML5 attributes such as required, pattern and min/max allowing for the exclusion of javascript validation in most cases. Moreover, some new items as datalist for auto complete provide new capability in form interaction for a

user experience. All of these enhancements are not only for the ease of development but also for enhancing the usability of interfaces for the end-users.

In relation to the overall development of modern web systems, Ashtari (2022) stated that HTML5 has made a number of innovations that greatly facilitate the process of developing modern web-based systems. One of the major attractions of HTML5 is the introduction of the new elements in markup language that greatly enhance its semantic structure, such as `<header>`, `<main>`, `<section>`, and `<footer>` which give content structure that is clear, thus, improving even more the accessibility and the search engine optimization, which are very important for the dynamic online platforms. The other improvements that HTML5 brings include eliminating the use of plugins for common rich media elements like `<video>` and `<audio>`, that makes multimedia integration become simple, and it also guarantees better performance and security. Moreover, HTML5's XML compatibility and separation of design from content lead to cleaner code and easier maintenance which are necessary for scalable web systems. In addition, HTML5 has a responsive design and accessibility support through semantic tags and ARIA, thus ensuring operability across different devices and assistive technologies. Another aspect contributing to the ease of development and the increase of interactivity is the introduction of new Application Programming Interfaces (APIs) like web storage, geolocation, and drag-and-drop among others. The ability to perform local storage in a persistent way is an advantage in data management and the possibility of working offline, therefore, web-based systems become more efficient, secure, and user-centered.

The literature highlights HTML5 as a foundational technology for modern, dynamic web applications. Inrid and Issn (2023) demonstrate that their semantic elements and native multimedia capabilities and APIs enable advanced functions without the need for third-party

extensions. Cherukuri et al. (2022) demonstrate that HTML5 enhances form management and validation and mobile support, which leads to better user experience and faster development processes. Ashtari (2022) identifies HTML5 as a technology that improves web systems through its support for responsive design and accessibility and offline functionality and development of maintainable code. The research studies demonstrate that YesParency should use HTML5 to develop its secure platform which includes two-envelope bidding and real-time streaming and signature-based access and audit trail features that users can access across multiple devices.

CSS

According to Kuparinen (2023), the process of website performance optimization requires developers to first understand CSS limitations, which become more critical when visitors create larger DOM structures and more unoptimized elements on their web pages. The research demonstrates that specific methods which include shadow DOM and style scoping and style concatenation and CSS containment, lead to performance improvements which work across all web browsers. The researchers found that main-thread activities could be reduced through two methods, which involved deleting unneeded CSS and creating easier-to-use selectors and better layout practices. Users face challenges when they try to measure performance because their environments differ from each other, which leads to the need for customized benchmarks and Real-User Monitoring and Core Web Vitals as feasible assessment methods. The research shows that CSS optimization needs to be studied in different contexts because current knowledge on this topic remains insufficient and needs further research until browser development brings new changes.

Expanding on the technical and performance-related role of CSS, Jain et al. (2024) stated that the CSS system functions as the main element which generates the complete visual experience and operational performance and user interactive capabilities of present-day websites by turning basic HTML elements into interactive website experiences. CSS extends its functions beyond basic styling by establishing standardized branding which controls all elements of design from layout to typography and color schemes to create a consistent professional appearance that remains intact across different platforms. The system uses animations together with transitions to create multiple responsive elements which display visual information to help users navigate through dynamic website content. CSS functions as an accessibility tool for website users because it enhances search engine optimization through its support of semantic HTML elements which make content easier to understand and locate. The combined functions of CSS elements demonstrate that this system functions as a vital design instrument which helps website developers create better user experiences while maintaining consistent interface designs and usable design systems which can easily grow across all digital platforms.

Further highlighting the broader impact of CSS beyond technical implementation, Wilson et al. (2022) explained that CSS has created a fundamental change in web development because it connects design to programming which has changed how professionals work and how web designers view their work. CSS created a system which enabled designers to develop their work because it transformed visual elements into specific design elements and built design systems which required front-end developers to work with the new system. The CSS Zen Garden initiative showed how CSS functioned as a communication tool which built community ties while it helped designers create their work according to industry standards of responsive design. The development of Bootstrap and Tailwind frameworks demonstrates a trend which moves

towards standardized development of components, which leads to problems because modern web interfaces lose their unique design elements. The research demonstrates how CSS functions as both a technical system and a cultural element, which drives team work and job development and creates standardized procedures for current web design methods.

The studies demonstrate that CSS functions as an essential element which enhances website performance and visual design and accessibility and site responsiveness. The optimization techniques which include shadow DOM and style scoping and unnecessary CSS removal, help decrease processing requirements while enhancing browser performance. CSS enables standardized branding and interactive navigation and development of scalable interfaces which function across different platforms. CSS serves as a technical styling solution which enables designers to develop contemporary web development standards through its use as a design element.

Javascript

According to Sharov et al. (2024), JavaScript provides interaction of the webpage with the user and it processes the data. The researchers emphasized that JS is an essential tool for web development, which encourages the programmers to learn JS. In addition to that, they stated that JS is a cross platform language so it can run in different operating platforms. In addition, Sharov et al., (2024) highlighted that it can also be used to create browser extensions, applications for mobile OS, server applications and desktop applications and is also easy to use and productive since it can quickly process requests without signaling the server. This shows the capabilities of JavaScript as a tool for creating web applications.

Building on the cross-platform and application capabilities of JavaScript, Kholmatov (2023) noted that JavaScript is a versatile programming language that powers the web. The author stated that JavaScript's popularity as a front-end and server-side language led to the rise of libraries and frameworks, each providing unique advantages to the developers. In the study, with libraries like React, jQuery, and Node.js, JavaScript is highlighted as flexible and powerful. React's components and virtual DOM promotes optimized performance, dynamic, and reusable user interfaces. As for jQuery, The author described it as still relevant as it simplifies DOM manipulation and event handling, especially in medium or smaller projects. Node.js brings JavaScript to the server side, allowing developers to create full-stack applications with its non-blocking I/O system and a vast npm library that supports a repository of packages. This demonstrates the flexibility of libraries making it adaptable and relevant for the project needs.

Similarly, Sayed and Patel (2023) emphasized the transformative role of JavaScript in web development. They stated that it changed the way users interact with online platforms and it empowered the developers to create dynamic and responsive web pages that incorporate server-side functionalities, allowing real-time updates and better experience for the users. As a result, JavaScript has become an essential tool for modern web developers. Thus, JavaScript still remains as the main component in web application projects with its features.

Extending these observations, Pathak et al. (2024) highlighted that JavaScript executes on the client side to make the execution fast. JavaScript is compatible with modern browsers, languages and benefits from company resources, as well as it is deployable anywhere on the web. It allows developers to create dynamic and interactive applications, on any device. This emphasizes the versatility of JavaScript across multiple platforms.

The studies demonstrate that JavaScript possesses three primary qualities which allow developers to use it across different platforms and to create both client-side and server-side applications. The research shows that JavaScript enables the creation of dynamic, responsive, and interactive web applications. The system achieves rapid processing with instant updates through its dual execution on client and server environments while libraries such as React and jQuery and Node.js enhance system performance and code reusability and development efficiency. JavaScript has become a fundamental instrument for contemporary web development because it enables developers to create adaptable and high-performing applications.

PHP

According to Syaifudin et al. (2025), web systems have become immensely popular and widely used in various fields. It is estimated that 75.9% of websites using server-side languages are using PHP. The researchers explained that PHP remains popular in web development because it supports database connectivity and allows developers to create dynamic and interactive websites. It is also noted that PHP can easily be integrated with HTML and databases such as MySQL. As an open-source language, it has a large and active community of developers who contribute to its development and create useful tools and libraries. Toal (2022) also stated that developers can easily integrate PHP code with other technologies because it is also known for its compatibility with other development tools that can be used to create a more ideal and dynamic web system.

PHP is commonly used in developing web-based systems because it is efficient and flexible. Pure PHP, CodeIgniter, and Laravel were used to compare the development of the

academic information system, according to a study by Niarman et al. (2023). The study revealed that pure PHP uses less memory because it does not have the framework's built-in components, enabling the programmer to choose the functions used. However, CodeIgniter had a slightly better performance in terms of overall performance because its system design is more structured. Despite this, the performance gap between using pure PHP and CodeIgniter was small, and this demonstrates that PHP can be relied upon for the development of web applications.

In addition, Kho, Fababeir, and Torres (2024) stated in their study that PHP demonstrates strong memory efficiency and performs well when handling multiple requests at the same time. Because of this, PHP remains a reliable choice for server side web development, particularly for applications that require stable and consistent web functionalities. The study also explained that PHP frameworks such as Laravel provide various tools and features that assist developers in building and scalable web systems. These frameworks simplify the development process by offering built in libraries, and support for database integration.

The reviewed studies highlight that PHP continues to play an important role in web system development. It was said that PHP remains widely used in modern websites because of its strong compatibility with databases such as MySQL and its ability to integrate easily with other technologies used in web development. Studies also emphasize that PHP allows developers to create interactive and dynamic web applications due to its capability to efficiently process multiple server requests. In addition, studies comparing PHP with frameworks such as CodeIgniter and Laravel shows that PHP can demonstrate efficient memory usage while still maintaining reliable performance in web systems. Although frameworks may provide more structured architectures and additional development tools, pure PHP still performs competitively in terms of system efficiency and functionality. Furthermore, the availability of various PHP

frameworks, built-in libraries, and support from a large community of developers has contributed to the constant improvement and adaptability of the language in modern development environments. These factors indicate that PHP provides both flexibility and stability for developers who build web-based systems. Based on the reviewed literature, it appears that PHP remains a practical and effective programming language for server-side web development and continues to support the creation of scalable, dynamic, and reliable web applications.

MySQL

According to Wahyudi et al. (2022), MySQL was developed by Michael "Monty" Wildenus, who also created the web-based UNIREG database, which is incompatible with the dynamic web. MySQL is developed using the UNIREG and mSQL open source code to develop a dynamic database. Therefore, MySQL became known as a relational database management system and open source. MySQL AB produces free software, which is created by the community at large and whose authors hold the rights to the source code. In addition to being used to query databases in large quantities to facilitate dataset access, MySQL's non-transaction mode has a weakness that implies it cannot ensure the accuracy of the data it stores.

Building on this foundational description of MySQL's development and limitations, further details on its structure and functionality are provided. Based on Erickson (2024), MySQL is an open source RDBMS that uses SQL, which stands for Structured Query Language, a programming language that's used to retrieve, update, delete, and otherwise manipulate data in relational databases. MySQL stores data in tables of rows and columns organized into schemas. A schema defines how data is organized and stored and describes the relationship among various

tables. With this format, developers can easily store, retrieve, and analyze many data types, including simple text, numbers, dates, times, and, more recently, JSON and vectors.

In addition to its structural capabilities, the practical advantages of MySQL further explain its widespread adoption. Annala (2022) stated that MySQL has a number of advantages that makes it popular among developers and organizations. First of all, MySQL is cost-effective to use because it is open-source and it is free of charge to use, thus allowing the businesses to use it without going through licensing costs. At the same time, MySQL is assisted by a large and active community of developers, users, and contributors who provide help, share knowledge, and take part in the improvement of the database. MySQL also gives fast performance and high capacity in concurrent users, so it can be used for a wide range of applications ranging from small websites to big enterprise systems. Moreover, MySQL is very flexible in terms of programming languages, frameworks, and platforms which makes it very easy to integrate with existing systems and technologies.

The reviewed studies establish MySQL as a flexible open-source relational database management system which stores and retrieves structured data for different applications. The system supports YesParency's requirement for structured data storage through its schema-based design and ability to handle multiple data types needed for audit trail logging and digitized two-envelope bidding system operations. The system has a restriction about non-transaction mode which affects data accuracy thus requiring extra protective measures according to YesParency's practice of using encrypted document access and digital signatures to safeguard data integrity. MySQL demonstrates both scalability and performance capabilities which enable the system to manage multiple bidding processes together with user interactions at the same time.

Software Development Life Cycle

The Software Development Life Cycle (SDLC) is a structured framework used to guide the development of software systems through phases such as planning, design, development, testing, and maintenance. It ensures that systems are developed systematically, efficiently, and with high quality. One of the key strengths of SDLC is its ability to integrate risk management throughout the development process. According to Adriani et al. (2024), SDLC incorporates risk management across all phases, emphasizing that elements such as time, cost, security, process, and data are highly vulnerable during software development. These risks must be continuously identified and mitigated to ensure project success; however, challenges such as project complexity, communication gaps, limited resources, and organizational culture may hinder effective risk management.

In addition, SDLC has evolved over time to address the limitations of traditional software development approaches. Dávila-Campos et al. (2024) highlight that modern SDLC frameworks now incorporate Agile, Scrum, and hybrid methodologies to better respond to changing project requirements. While traditional models still provide a structured approach, there is an increasing shift toward flexible and adaptive frameworks that can accommodate dynamic environments. Their study also indicates that there is still limited research on specialized SDLC applications, suggesting the need for further development of more adaptable and widely applicable frameworks.

Furthermore, recent developments show that SDLC practices have significantly transformed, particularly in response to the post-COVID-19 environment. Wiguna et al. (2024) explain that organizations increasingly adopt Agile and Scrum methodologies due to their ability

to enhance collaboration, flexibility, and rapid system development. These methodologies are widely used across different sectors such as business, education, and government, as they allow organizations to respond effectively to continuous changes and evolving user needs compared to traditional rigid models.

The reviewed literature highlights that SDLC has evolved from a rigid and linear process into a more flexible and adaptive framework that integrates both traditional and modern methodologies while maintaining strong risk management practices. In relation to the development of the YesParenity web-based bidding system, applying SDLC ensures a systematic and organized approach to system development. Integrating risk management across all phases helps minimize potential issues such as system errors, security vulnerabilities, and performance limitations. Moreover, adopting modern approaches such as Agile and Scrum supports continuous improvement, flexibility, and responsiveness to user needs. Overall, SDLC serves as a fundamental framework that ensures the successful implementation and long-term sustainability of the YesParenity system.

Agile Scrum Methodology

Agile Scrum is a widely used software development methodology that emphasizes flexibility, collaboration, and iterative progress. It organizes development into short cycles called sprints, allowing continuous feedback and improvement throughout the process. According to Aleryani et al. (2024), Scrum serves as a structured Agile framework that enables teams to deliver incremental outputs while adapting to changing requirements. Their study highlights that successful Scrum implementation depends on critical factors such as management commitment,

team collaboration, customer involvement, and adaptability. However, challenges such as scope creep and misalignment between stakeholders must be carefully managed to ensure project success.

In addition, Agile methodologies, including Scrum, have been shown to significantly enhance project performance and adaptability. Rahmah et al. (2024) emphasize that Agile practices improve productivity, customer satisfaction, and responsiveness to change. Their findings further reveal that factors such as organizational culture, team knowledge, communication, and management support greatly influence the effectiveness of Agile Project Management (APM). However, organizations may face difficulties in adopting Agile methods due to resistance to change, lack of trust within teams, and challenges in communication, especially in diverse or distributed environments.

Furthermore, Scrum continues to evolve as a modern development approach integrated with other practices such as DevOps. Pastrana et al. (2025) highlight that Scrum promotes incremental delivery, continuous integration, and transparency, allowing stakeholders to monitor progress and provide feedback regularly. Their study also shows that Scrum practices, such as stakeholder involvement, continuous communication, and short delivery cycles, contribute to maintaining software quality throughout the development process. However, despite its advantages, there is still a lack of clear implementation guidelines, particularly for organizations of different sizes, making adoption a trial-and-error process in some cases.

The reviewed literature indicates that Agile Scrum is an effective and adaptive methodology that supports continuous improvement, collaboration, and high-quality software development. In relation to the development of the YesParenity web-based bidding system, the

use of Scrum methodology enables iterative development, allowing the system to be continuously improved based on user feedback and evolving requirements. The emphasis on collaboration, transparency, and stakeholder involvement ensures that the system aligns with user needs and expectations. Additionally, practices such as continuous integration and regular feedback help maintain system quality and performance. Overall, Agile Scrum provides a flexible and efficient approach that supports the successful implementation and sustainability of the YesParenity system.

Hostinger

According to Hoseini (2022), Hostinger was launched in Lithuania in 2004 as Hosting Media and was renamed Hostinger in 2011. Hostinger has gained over 24 million active users since its foundation. The 17 largest data centers in this collection exist across Brazil, the Netherlands, Lithuania, the United States, and Singapore. The platform provides multiple services which include VPS hosting and domain registration. Hostinger provides its customers with multiple benefits that include high-performing host servers and affordable registration plans with complete features and extensive website building capabilities, worldwide data center options at eight locations, free SSL certificates for domains and hosting, an easy-to-use control panel, various payment methods, WordPress speed optimization, secure SSH access, simple website administration, and unlimited bandwidth.

Expanding on the general features and services of Hostinger, user experience and accessibility are further emphasized in practical applications. Hostinger is basically a commercial web server that hosts websites permanently. It offers a wide range of resources at quite affordable

prices and at the same time, presents a user-friendly and simple interface. Companies like the 17th of May Association that are already acquainted with Hostinger can say that managing a website will be easier for them since they have a pre-existing account (Pina, 2022).

Further supporting these observations, additional research highlights the factors that contribute to Hostinger's widespread adoption and reliability. Based on Hadian (2024), affordability is ranked as the major factor that made Hostinger a success. The company's low-cost hosting offers are tailored to customers who are short of money. A user-friendly and straightforward interface of the site is suitable for novice users, and a few packages include a free domain, which is an idea to help users keep their money. Besides that, Hostinger gives out free SSL certificates, thus making the connections to websites secure. To ensure data security, the platform commits to daily backups and weekly full backups of the website information. On top of that, 24/7 customer support through live chat is available to respond to customers' issues quickly. The shared hosting plans offered by Hostinger, which are simple to maintain and use, are what people usually choose rather than paying more and facing the complexity of managing a Dedicated Private Server (DPS).

The reviewed studies highlight that Hostinger provides secure hosting features such as free SSL certificates and regular backups and secure access which support YesParency's requirement for encrypted document access via digital signatures. The platform maintains system availability through its high-performing servers which operate from worldwide data centers to deliver continuous access to records and audit trail logging. The literature shows that YesParency achieves easy deployment and management through its user-friendly interface and affordable pricing. The system can handle multiple users and transactions because of its scalable infrastructure and complete service capabilities.

Functional Suitability

According to Trendowicz et al. (2023), the aspect of functional suitability appeared most frequently in the articles they reviewed and that it is often described in conjunction with the quality of the data. They coded some of these aspects using the ISO/IEC 25012 data quality model, which has a series of dimensions, including currentness, credibility, and correctness. Issues in any of these aspects have a negative effect on the user's experience because their quality directly impacts how users perceive the performance and reliability of the system. The same may be true of poor efficiency, reliability impacted by instability, and some cases of compatibility between components, all of which will have an effect on the communication between the components of the system. These findings suggest that functional suitability is closely connected with other quality attributes that contribute to the overall effectiveness of a system.

Similarly, as stated by Silva et al. (2022), the demand for software applications that are suitable across contexts and for multiple users continues to grow, making the evaluation of functional suitability paramount. Functional suitability is the degree to which software functionalities are complete and executed correctly in specific environments. This concept is often misinterpreted or overlooked due to the difficulties and costs associated with evaluating it. Among the sub-characteristics, completeness and correctness are often treated as the most important; functional appropriateness is usually evaluated more indirectly through user acceptance of functionalities delivered in the software. Because of these challenges, the authors suggest that a more systematic approach to measuring functional suitability in software projects is needed.

In line with this explanation, functional suitability refers to the ability of software to deliver functions that satisfy both explicit and implicit user needs under specific conditions. It has three sub-characteristics: functional completeness; functional correctness; and functional appropriateness. Functional completeness evaluates whether the software provides all functions necessary for users to accomplish their goals; functional correctness assesses the results as to whether they are accurate; and functional appropriateness measures how well functions help users accomplish their goals. These dimensions ascertain the software is performing its intended purpose and produces output that users expect (Sarwosri et al., 2023).

Furthermore, based on the study by Ariyani et al. (2021), functional suitability is an important software quality indicator investigating whether a product's functions meet user needs in practice. The authors maintained that part of testing this quality indicator is to verify if intended functionality aligns with enacted functionality. The authors also explained that usability testing extends from functional suitability methods which evaluate how interactive systems meet user goals and tasks while reducing negative user effects. The two indicators together provide a broader understanding of software quality from both technical and user perspectives.

The literature consistently shows that functional suitability represents a vital software quality component which directly determines system performance and customer satisfaction and system dependability. The researchers Trendowicz et al. (2023) found that functional suitability relates to data quality dimensions which include currentness and credibility and correctness whereas deficiencies in these dimensions create negative effects on both user experience and system performance. The researchers Silva et al. (2022) and Sarwosri et al. (2023) established that functional suitability includes three elements which are functional completeness and correctness and appropriateness to make sure software operates all essential tasks in ways that

support user objectives. The researchers Ariyani et al. (2021) established functional suitability through their study which studies actual usability testing to show that organizations must verify functionality execution results through technical assessments and user experience testing. The research indicates that systems which use precise functionality to manage digitized bidding and encrypted document access and audit trail tracking must establish functional suitability standards for their operations to function correctly and deliver accurate results which meet user expectations.

Performance Efficiency

According to Amazon Web Services, Inc. (2024), achieving and maintaining performance efficiency in a system requires a data-driven approach that carefully evaluates access patterns and trade-offs to optimize overall performance. Analyzing and testing with benchmarks helps in picking the right resource types and settings. Moreover, when infrastructure is treated as code, the living and dying of architecture can be rapid and safe, and this is through the help of data-driven decision-making. To make sure the efficiency remains constant, it is a must to use both active and passive monitoring because this way performance decline can be realized and stopped before it happens.

In accordance with Shethiya's (2025) article, performance optimization is very important in web application development as it makes applications fast, responsive, and efficient. Poor performance can lead to user frustration, higher bounce rates, and lost revenue opportunities. The enhancement of performance requires a combination of multiple strategies such as caching, database optimization, API optimization, content delivery networks (CDNs), and load balancing.

Caching stores the most often accessed data to enhance caching efficiency which will also result in the minimization of database queries and reduced latency. The methods used in database optimization like indexing, query tuning, and denormalization are all aimed at improving data retrieval and system responsiveness. API optimization is mainly concerned with the transfer of data that is efficient and reliable. CDNs are excellent when it comes to reducing the delay time because of their ability to deliver content from different places around the world while load balancing distributes traffic by not only counting but also sharing it among the servers in a manner that none of the servers becomes overloaded and the system keeps working. These optimization practices, altogether, lead to the development of web applications that are capable of delivering high performance, reliability, and scalability.

Based on the study of Klinchova et al. (2025), sustainability during the software development process is an important factor that greatly contributes to the achievement of high efficiency, particularly when the world is experiencing high energy prices and energy shortages. Energy-efficient applications are becoming a necessity more and more, particularly for mobile devices where the performance and consumption of resources directly affect the user's experience. Researchers conducted an experiment where websites employing adaptive, responsive and mobile-first design approaches were created to evaluate their sustainability in terms of user interfaces (UIs). The experiments measured load times and overall performance for each method and they uncovered the differences in energy efficiency and responsiveness. The results pointed out that UI design for sustainability should offer optimized layouts, efficient resource loading, and adaptive elements which will reduce energy consumption but not affect the user experience.

The studies show that web systems achieve their maximum performance levels through three essential components which include data monitoring systems and optimization methods and sustainable design approaches. Performance management requires organizations to analyze their access patterns together with caching systems and database performance and API efficiency and load distribution and content delivery networks. The combination of energy-efficient designs and interactive UI elements creates better system performance especially in mobile applications. The research findings demonstrate that organizations need to combine technical performance improvements with effective resource management to keep their systems running at optimal capacity.

Compatibility

According to Haindl et al. (2024), the evaluation of software quality can be done using ISO/IEC 25010 standard, which includes different aspects of software such as reliability, security, portability, maintainability, and compatibility. Their study on a software platform for human AI collaboration in smart manufacturing examined how different stakeholders prioritize these quality characteristics when evaluating system performance. From the results, it was clear that compatibility, just like portability and maintainability is one of the less critical attributes in this particular use case, but it is important in enabling software systems to operate and integrate with other technologies in a shared environment.

Based on the study by Purbaratri et al. (2023), the quality of the software can be assessed by the ISO/IEC 25010 standard, which helps to identify important software quality characteristics such as functional suitability, performance efficiency, compatibility, usability,

reliability, security, maintainability, and portability. In their proposed framework for smart government systems, compatibility was described as the capability of a software system to interchange information and execute functions with other systems or components within a shared environment, which is significant in providing integrated digital government services. Similarly, the arc42 Quality Model (2023) described that compatibility, as defined in the ISO/IEC 25010 standard, means that the software has the ability to function properly together with other systems and share information in one environment. Since modern software must often interact effectively with other applications and technologies in order to function properly, this highlights the significance of interoperability and coexistence.

Based on the reviewed literature, compatibility under ISO/IEC 25010 refers to a system's ability to function in a shared environment in which it can interact effectively with other application systems through interoperability and coexistence. Literature studies conducted by Haindl et al. (2024) and Purbaratri et al. (2023), along with the Arc42 Quality Model (2023), explained that compatibility allows a system to exchange information as well as function smoothly with other applications systems. Compatibility may not always be considered a very important quality attribute, but it is necessary to allow effective integration of a system in today's digital world, as today's application systems often communicate and exchange information, making it necessary to avoid technical conflicts between them. In relation to YesParency, high compatibility ensures that there is stability in the exchange of data, easier verification, and efficient information flow between various stakeholders. This promotes transparency and accountability in the system.

Usability

According to Thamilarasan et al. (2023), usability essentially means how usable something is, which means how easy-to-use the system is as well as the overall satisfaction of the user with the system. To measure usability, one can say usability is just an attempt to measure the level of friendliness of the system, which can also mean measuring specific end-user attributes, which can consist of user skill, time taken to be able to learn the system, productivity from using the system, and overall interaction with software. In the world of software web development, usability is an important aspect, regardless of company type, to ensure the users of the system are able to interact with the system efficiently - which means you can save time by retaining users' productivity on the system. When usability for web interfaces has increased, the user engagement, effectiveness and satisfaction has increased as a result too, which is one contributor to using a web interface effectively to improve performance and provide an overall positive experience for users. ISO 9241-11 defines usability as "the extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use." Usability can be evaluated using three criteria: effectiveness, efficiency, and satisfaction. Similarly, ISO 9126 describes usability as "the capability of the software to be understood, learned, used, and liked by the user when used under specified conditions," and states five criteria: understandability, learnability, operability, attractiveness, and compliance with usability standards.

Expanding on the assessment of usability, Giri et al. (2023) stated that the user experience of a system can be evaluated through usability testing. Usability is an important quality characteristic of user experience when using software, websites, mobile applications, or other digital products. Usability testing will yield better insight into real user interactions, in

comparison to evaluation methods that are not based on observing what users actually do with a system. In web development, it is especially important to be able to evaluate usability to understand the ease or efficiency of user experience. Usability evaluation methods, for example cognitive walkthroughs, performance measures, and the System Usability Scale (SUS), can capture qualitative and quantitative data effectively. Cognitive walkthroughs and performance measures provide contextualized data about user behavior and task completion, while the System Usability Scale (SUS) evaluates satisfaction with and perception of the system.

Further emphasizing the cognitive dimension of usability, Darejeh et al. (2024) argued that the evaluation of usability requires a consideration of cognitive load variables in terms of user interaction with different types of interfaces. It provides a substantive analysis of cognitive load measurement strategies for the assessment of usability in many Human-Computer Interaction (HCI) contexts. While assessing the nature of how users interact with various information and components of the system, the authors propose guidelines and a framework for the evaluation of interface usability, which supports the work of designers and developers of software web-based applications. In this context, the application of cognitive load theory can enable the designer and developer to produce interfaces that diminish unnecessary cognitive load, support performance of tasks, and promote overall interaction capacity. Usability is defined, to an extent, by both effectiveness and efficiency of the interface, but also the degree of intuitive use that users can experience without cognitive load.

The research results demonstrate that usability exists as a complex quality that comprises five elements which include user ease of use, user contentment, system effectiveness and

efficiency, and management of cognitive load. The usability evaluation process uses cognitive walkthroughs and performance assessment methods and the System Usability Scale to deliver both qualitative and quantitative data about how users interact with the system. The study shows that better usability leads to improved user engagement and satisfaction while it decreases cognitive demands and boosts operational effectiveness. Therefore, usability stands as an essential element which designers and evaluators must prioritize when they create and assess web-based software applications.

Reliability

According to Pacific Certifications (2025), reliability is the quality of a system that performs consistently and predictably while making sure its operation is effective even under certain conditions. It consists of four major sub-characteristics: the first one is maturity which is an indication of the reliability of the software in terms of meeting requirements under normal operating conditions; the second one is availability which makes sure that the system is always there when it is required; the third one is fault tolerance which allows the system to keep on working even in the presence of errors; and the fourth one is recoverability which gives the system the capacity to get back its operations and data after a failure. Reliability is, for instance, of utmost importance in the healthcare, aviation, and security sectors where unpredictable performances and long downtimes are no-go areas for safety and trust.

Expanding on the fault tolerance aspect of reliability, Noor et al. (2021) stated that fault tolerance is imperatively important in the area of managing faults, especially in distributed database systems and other critical data environments. The paper presents the various fault

detection and recovery methods, comparing them in terms of their pros and cons, and thus showcasing the importance of these methods in the process of keeping the system reliable and stable. However, it also points out that the present fault-tolerance models are facing a number of challenges and that these issues need to be dealt with in the future. The trend of increasing complexity in systems will also mean that the fault-tolerance mechanisms need to be advanced, so that they can handle the design-related faults and still keep the operation error-free during the time when the faults are present.

Complementing the discussion of system reliability, Gavtade (2022) noted that most web browsers have a wide range of security vulnerabilities like unblocked pop-ups, unsafe plugins, weak antivirus protection, and unsecured password storage. Despite the fact that the result of the research mentions that a great majority of people trust browsers for daily activities, still a considerable number of users regard ads, pop-ups, and saved passwords as top threats. Moreover, users are very much divided on the subject of cookies, as they see both their advantages and disadvantages. The research indicates that there is still a significant proportion of the population who are not aware of the growing number of cyber threats associated with browsers, making them more susceptible to attacks. To mitigate these issues, the study comes up with a list of recommendations which include; updating browsers frequently, using HTTPS for secure connections, creating strong and unique passwords, turning off auto-complete features, blocking pop-ups, and handling cookies in a responsible manner. Moreover, it points out that the improvement of browser security does not only rely on the implementation of technical measures but also on user education, because most cyber crimes do not attack software flaws but human errors.

The studies demonstrate that modern software systems need to achieve their essential software properties of reliability and security as their most important requirements. The concept of reliability includes four components which are maturity availability and fault tolerance and recoverability to guarantee stable and predictable system performance. The distributed systems and critical systems require fault tolerance to sustain system functionality when unexpected problems arise. The existence of security vulnerabilities together with human factors creates major threats which demonstrate the necessity for implementing both technical security measures and educating users about security. The research results show that systems need both strong system design and user knowledge to achieve their desired level of robustness and reliability and security.

Security

According to Chiarelli (2022), web application security must consist of protective measures to prevent external attacks such as data loss, denial of service and privacy violation. Since web applications are available to everyone once deployed, developers should assume exposure to potential security risks. The author highlighted three (3) ideas to help in understanding web application security. Threats refers to possible harmful events that could target the system, while vulnerabilities are the weaknesses whether in design or the code that an attacker might take advantage of. And risks which are the potential harm or damage that can happen when a threat exploits a vulnerability in the system such as information of clients. These concepts highlight the importance of understanding the different elements that contribute to web

application security. The study emphasizes the importance of designing a web application with security, recognizing that all users both authorized and unauthorized can pose a threat.

Similarly, based on Pant et al. (2022), website security is a crucial component of web development since it creates trust between the client and the server. The authors added that a secure website can be achieved by implementing appropriate methods and tools to apply authentication and authorization within the system. In their study it said that to ensure authorization, user sessions play an important role in storing login information. It utilizes session IDs to verify and grant user access. In addition, users are commonly categorized according to their access rights such as administrator and as users who can perform CRUD (Create, Read, Update and Delete). The users are also limited to viewing content, and to perform an action a user permission is verified through authorization process. Through these mechanisms, systems are able to control user access and protect sensitive data. The authors concluded that authentication and authorization was researched to keep user data safe from hackers, and it is a must to provide authentication and authorization to the users of the website.

In relation to authentication practices, Albeshier (2023) examined authentication procedures among 20 popular websites, and found that 11 of them offered the option of signing up using other accounts or existing accounts such as Google or Facebook. It shows that integrating the feature reduces cognitive load to the users by reducing the number of passwords to memorize and the need to follow new password policies. As a result, this improves the user experience as well as security effectiveness relying on protections from established providers that employ multi-factor authentication such as Google.

The existing research demonstrates that web application security protects sensitive data while establishing user confidence. Chiarelli (2022) identifies threats, vulnerabilities, and risks as critical elements which authorized and unauthorized users can use to endanger system security. Pant et al. (2022) demonstrate that strong authentication and authorization serve as essential requirements which organizations must implement to protect their data through role-based access control and session management. Albeshir (2023) found that security providers who offer Google and Facebook authentication through multi-factor authentication create a safer user experience. The research demonstrates that organizations need secure access controls and authentication systems and authorization mechanisms to protect their data from unauthorized access while achieving dependable system performance. The findings support YesParency because its encrypted document access together with signature-based authentication enables secure digital bidding processes which require strong security measures to create trust and efficiency.

Maintainability

It has been considered that the maintainability of software is one of the external quality attributes, and hence it can't be measured directly from the source code. Instead, it has to be estimated indirectly by some measurable internal attributes. These internal attributes include code complexity, system size, documentation quality, modularity, and the presence of comments within the code. Systems that are difficult to maintain will require more time, effort, and financial resources for debugging, updating, improving, and adapting to new requirements. These concepts are consistent with the findings of Heričko and Šumak (2023), who explained

that different versions of the Maintainability Index may give different numerical values when used on the same software system.

Maintainability is a significant quality attribute in modern software systems, especially in systems that need to be constantly updated and improved. Özdemir and Buzluca (2024) stated that maintainability can be evaluated through key factors such as testability and modifiability. Their study introduced a quality assessment framework that analyzes code metrics and software features using ISO/IEC quality standards to determine the maintainability level of microservices. To examine the source code and calculate the maintainability scores, the framework applies clustering algorithms allowing developers to better understand how easily a system can be modified and tested. The study indicates that this method can help developers gain useful insights to identify areas in the system that need improvement.

According Gu (2025), various factors can be considered that might impact the maintainability of high-performance web applications, like coding standards of different programming teams, dependency issues of component based systems, and the need to balance performance with security requirements.

The study shows that the above challenges can be overcome by ensuring the implementation of common coding standards, code review processes, and the design of a proper modular structure with clear system levels. These methods will ensure that the system is more maintainable by simplifying the structure, improving the efficiency of the development and keeping the system stable. The web application needs architectural design and development standards as essential components for achieving its maintenance goals. The reviewed studies emphasize the importance of maintainability as a quality aspect in software systems, especially

in web-based applications that need to be updated and improved continuously (Heričko & Šumak, 2023). It is commonly measured through internal factors in the software such as complexity of the code, documentation, modularity, system structure, testability, and coding standards, which help assessing the maintainability of the system and the ease with which it can be maintained over time (Ozdemir & Buzluca, 2024; Gu, 2025). These factors help in ensuring that the system is updated and improved in a smooth manner in the future. In relation to the YesParenity system, the importance of maintainability lies in the smooth upgrades, long-term improvements of the system.

Flexibility

According to Adaptable (2024), website flexibility is the feature of a digital platform that can smoothly adjust to new technological developments, changes in the industry, and the expectations of users. The flexible website not only enables but also encourages the developers and companies to be very quick in reacting to customer feedback, making improvements, and adding new functions. This very quality gives rise to an agile development method which allows for constant updating, thus quicker delivery of better digital experiences. Website flexibility is like a cushion that supports continuous iterations, innovations, and faster deployments of new functionalities to the market which keep the site relevant and user-centered even after a long time. Because of these advantages, flexibility has become an important consideration in modern web development.

Similarly, as Washington (2023) stated, responsive site design and web accessibility work as essential components that provide the best online user experience. Web accessibility and

responsive design create more viewing and interaction opportunities which enable a larger audience to access content. Modern responsive web design techniques enable users to access websites through various devices, which include large television displays and small smartwatches. Accessible design benefits disabled users because it provides them with necessary accommodations, while all other users gain benefits from the same accessibility features. The process of creating accessible and responsive designs presents developers with multiple challenges because their attempt to implement accessibility features will result in an extended negative impact on user experience. The design process requires designers to create institutional websites which enable all users to access the site through proper design methods.

In relation to responsive web development, Velan (2025) explained that both adaptive and responsive web design have their own specific advantages, although their goals different methods which they work and use their functions. Responsive Web Design (RWD) uses an all-inclusive design method which creates flexible designs that work well on all devices while maintaining uniformity throughout the entire website. Adaptive Web Design (AWD) uses dynamic layout systems which adjust to different devices by delivering content according to the particular device being used, thus achieving better performance results and improved user experience. The most effective approach demands that both methods be combined starting with a complete responsive system which needs adaptive methods for tracking speed and performance and user experience enhancements. The combined method creates an online experience which achieves both strong performance and user-friendly design.

The literature establishes that website adaptability and user interface responsiveness together with website accessibility functionality serve as fundamental requirements needed to develop digital platforms which focus on user needs. Adaptable (2024) demonstrates that flexible

websites enable organizations to implement rapid updates which result from technological progress and user input and industry changes while maintaining their operational efficiency. Washington (2023) proves that designers who create responsive and accessible websites for their products enable users to access their content from different devices while providing support for users with disabilities although they face difficulties during their development process. Velan (2025) shows that organizations can achieve better user experience results by using both responsive and adaptive web design methods which provide stable system operation together with tailored performance for different devices. The insights develop YesParency through their explanation which proves that the project requires a flexible and responsive accessible interface to ensure delivery of SLSU's digitized two-envelope bidding and real-time bid streaming and signature-based access and audit trail functions which must remain efficient and user-friendly and adaptable to new technological and institutional needs.

Safety

According to Purbaratri et al. (2024), organizations must assess web system security because this assessment protects confidential data and maintains online service trustworthiness. The researchers performed a security evaluation of the government whistleblower system according to the quality standards which ISO/IEC 25010 establishes. The results demonstrated that ISO 25010 security criteria application enables organizations to discover security weaknesses while it improves their defenses against unauthorized access. The research demonstrated three essential elements which security evaluation assesses: confidentiality, integrity, and authentication which help protect sensitive data. Organizations can improve the

overall safety and protection of their web platforms that manage critical data by implementing these security mechanisms.

Kurniasari and Rochimah (2024) also focuses on the importance of evaluating website systems to ensure that web applications operate efficiently and reliably. In this research, the researchers were able to develop a framework for testing a website based on the quality characteristics of the ISO/IEC 25010 standard, particularly on performance efficiency. In addition, to identify the technical problems that may cause instability and failure in the system, developers must assess the performance of the system. The developer is able to minimize the possible risks that may affect the functionality of the website by carrying out the right testing and ensuring the system runs smoothly. The stable performance of the system also helps in the improvement of the safety of the website platforms, particularly in the protection of the user data and ensuring that the services run effectively for the users.

Ukor (2025) explained that aspect-oriented programming enhances web system security by separating security elements which include authentication and authorization and encryption and input validation from the rest of the system functions. Security components can be handled by developers through this separation method which enables them to maintain system performance without any interruptions. The security development process requires developers to establish component separation between various system elements because it helps decrease security vulnerabilities while boosting system protection. To establish necessary protection measures that contribute to the system's overall safety is by ensuring sensitive data and user interactions are protected by implementing structured security mechanisms during the development process.

The reviewed studies highlight the significance of safety practices implementation and system evaluation in ensuring safety within web systems. Purbartri et al. (2024) state that ISO/IEC 25010 can be applied for security evaluation, which can lead to the identification of vulnerabilities, especially by considering critical aspects such as confidentiality, integrity and authentication. Similarly, Kurniasari and Rochimah (2024) emphasize that performance testing can identify technical problems that may lead to instability, thus increasing reliability and security. Ukor (2025) also further emphasizes the importance of integrating the security mechanism during the development stage, including the separation of authentication, authorization, and encryption from core system functions to reduce vulnerabilities. In relation to this, the YesParenity system applies structured testing, development practices, and security evaluation to ensure stable, reliable, secure and safe operation throughout the bidding process.

Organizational Readiness and Change Management

The organization's readiness in adopting and using the system plays a vital role in making sure it works properly. The level of preparedness indicates the extent to which the organization's members are prepared to adopt and utilize a particular system, especially in the face of the changes from their usual way of doing things. Organizations should also understand and apply the importance of being prepared for change, as this helps guide people in adapting to new systems and processes (Kraus et al., 2022). As stated by Teerikangas et al. (2026), organizational readiness is related to resistance and user participation, which suggests that organizational readiness and participation in the process play a significant role in determining whether the change is successful or not. Lack of user readiness and resistance to changes lead to

unsuccessful implementation and utilization of the system. User readiness and participation in the process have positive effects on the changes' success (Hanelt et al., 2021).

Following this, implementing appropriate change management techniques will be critical to facilitate a smooth transition from manual to workflow systems. As Errida and Lotfi (2021) suggest, organizational change depends on factors such as strong leadership support, clear communication, employee involvement, and proper training. These factors help reduce resistance and guide employees as they move from traditional manual processes to a more efficient digital system. In addition, individual readiness also plays a key role in how employees respond to this transition. According to Vakola (2022), it has been noted that users are likely to show support for digitalization where they have an understanding of the benefits it brings them and improve their work. When users see the advantages of moving from manual to digital processes, they become more willing to adjust, which helps minimize resistance and improves overall acceptance of the new system.

The reviewed literature highlights that both organizational readiness and effective change management are essential for the successful implementation of the YesParenity web-based bidding system. Transitioning from manual to digital procurement procedures implies not only an adaptation of the required technology but also a readiness of the personnel to use it. If the personnel is ready and involved, it would reduce the potential resistance and increase the chances for successful usage of the system. Furthermore, using the appropriate change management techniques, such as good leadership, effective communication, users' involvement, and training, would facilitate the transition to digital workflow. Highlighting the advantages of the YesParenity solution, especially those related to increased efficiency, transparency, and convenience of document submission and live bidding, could also contribute to user acceptance. In general, the

readiness of the organization and its personnel, along with change management, would be an essential factor in implementing the YesParenthood system successfully and sustainably.

System Interoperability and Installation Technicalities

The development of digital systems shows the importance of having strong and well-connected digital infrastructure to support efficient operations. Ciasullo et al. (2026) explained that digital transformation involves moving from traditional processes to connected digital systems that improve coordination, data sharing, and overall efficiency. In other words, the success of the system is not only dependent on the application itself, but also on the support of the technical environment. The infrastructure can form a solid basis for the functioning of the system and make it easily accessible and able to perform all required activities in time. On the other hand, according to Flores (2025), it was found that State Universities and Colleges are frequently faced with issues concerning budget and resource allocation. In this way, insufficient resources will result in outdated devices, insufficient system updates, and fewer technical personnel. This could delay the implementation process and affect the performance of the systems. Furthermore, system interoperability and proper data management are essential in making sure that various parts of the system are able to interact effectively. It is much harder to maintain consistency in situations when there are several databases involved. Abbasi et al. (2026) explained that it is important to have an integrated and efficient data management system for facilitating communication and ensuring the efficiency of the system operation. This shows the need for proper database migration and system integration when deploying the system.

Based on the reviewed studies, system interoperability and installation technicalities are important for the successful deployment of digital systems. A strong and well-prepared technical setup helps the system work properly and run smoothly in its environment. However, problems such as limited budget, outdated equipment, and lack of technical support can affect how the system is installed and used. In addition, when multiple databases and systems are involved, it becomes harder to maintain consistent and accurate data. This shows the importance of proper database migration, system integration, and server setup to make sure all parts of the system can work together. Therefore, for the YesParenity web-based bidding system, making sure that it is compatible with the existing setup and properly installed is important to achieve reliable performance and successful system deployment.

User Acceptance and Usability

User acceptance and usability are important in determining if a system will be successfully used after implementation. In Technology Acceptance Model (TAM), users are more likely to use a system if they find it useful and easy to use. According to Bustaman et al. (2023), these two factors affect how users feel about the system and if they plan to use it. This means that if users think the system helps them do their tasks better and is not hard to use, they will likely accept it. Similarly, Nuralam et al. (2024) explained that when a system is easy to use and useful, it also increases user satisfaction, which encourages continued use. It is evident that users will not only adopt the technology but also continue to use it. The perceived usefulness and perceived ease of use significantly influence users' intention to adopt digital systems (Nguyen & Nguyen, 2023). In addition, Addotey-Delove et al. (2024) explained that user acceptance of digital systems is strongly influenced by how useful and easy the system is to use. Users feel satisfied using a system when they find that it improves their efficiency and

requires no great effort on their part to learn how to use it. It shows that usefulness and ease of use both become crucial factors for determining the success of the adoption of a system. For YesParency, making the system easy to use and useful in enhancing the process of bidding becomes a key factor towards achieving user satisfaction and acceptance.

The reviewed studies show that user acceptance and usability are important in making sure a system is successfully used after it is implemented. Many studies agree that perceived usefulness and perceived ease of use affect how users feel about a system and whether they will continue using it. If users think the system helps them do their tasks better and is easy to use, they are more likely to accept it and keep using it. However, if the system is hard to use or does not give clear benefits, users may feel dissatisfied and choose not to use it. The Technology Acceptance Model (TAM) helps explain why users in SLSU may accept or reject the YesParency system. According to TAM, users will adopt the system if they believe it makes their work easier, faster, and more organized (usefulness), and if it is simple to understand and use (ease of use). On the other hand, if users find the system complicated or do not see its benefits compared to the manual process, they may resist using it. Therefore, making the YesParency system easy to use and clearly helpful is important to ensure that the Bids and Awards Committee will accept and continuously use it.

Research Paradigm

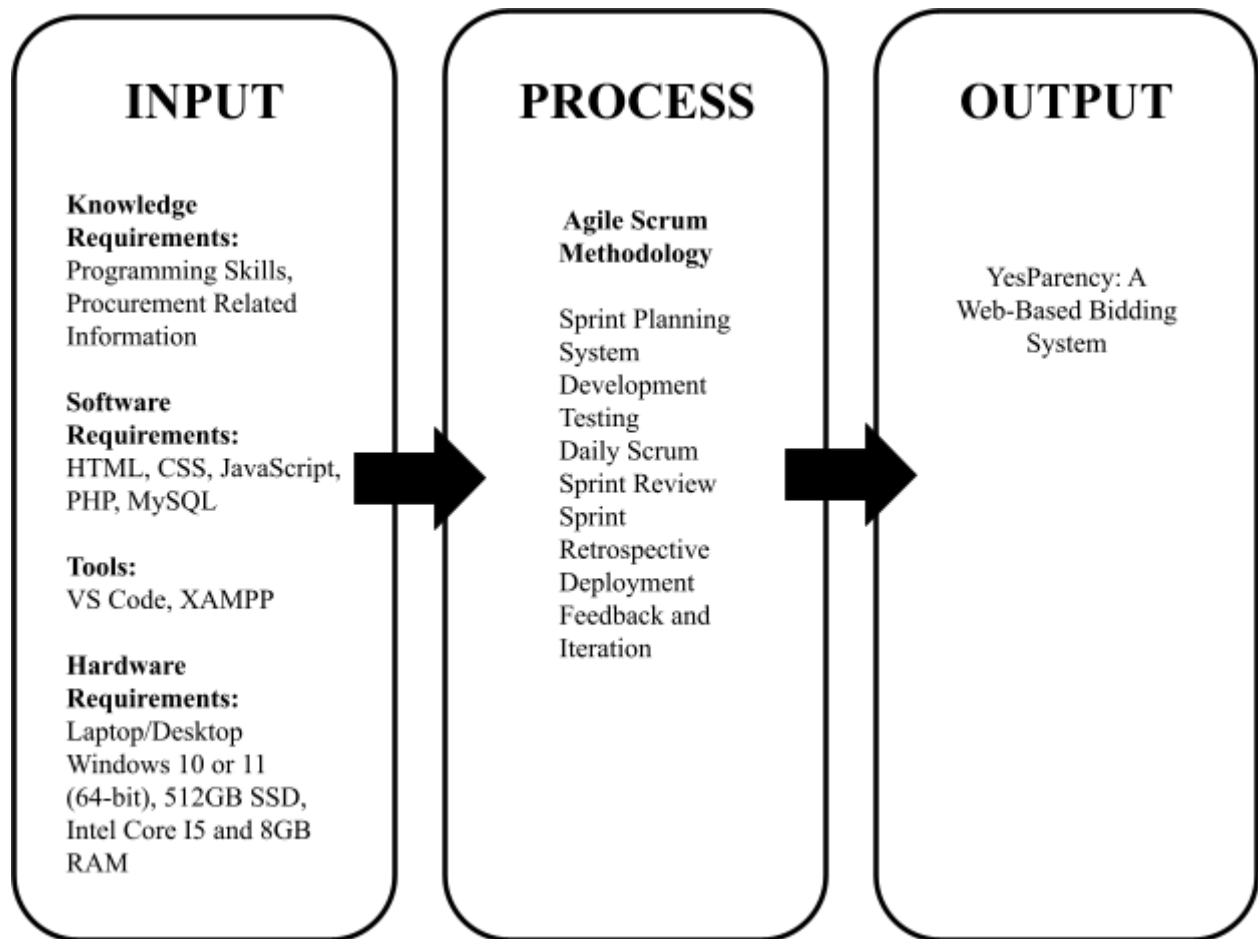


Figure 1. Input- Process – Output (IPO) model of YesParenCy: A Web-Based Bidding System

Figure 1 illustrates the input, process, and output of the YesParenCy: A Web-Based Bidding System with Digitized Bid Submission and Signature-Based Document Access Control for Southern Luzon State University. The input consists of knowledge, software, and hardware requirements. Knowledge requirements include programming skills and procurement-related information such as bidding procedures, document management, digital signature authentication, and public procurement regulations under Republic Act 9184. Software requirements include PHP, HTML, CSS, JavaScript, and MySQL, which will be implemented using software

development tools such as VS Code and XAMPP. For hardware requirements, the developer suggests a laptop or personal computer with Windows 10 or 11 (64-bit), a 512 GB SSD, an Intel Core i5 processor, and 8 GB RAM.

The process follows Agile Scrum methodology phases, including sprint planning, system development, testing, daily scrum, sprint review and retrospective, deployment, and feedback and iteration.

The output is the developed system of YesParency: A Web-Based Bidding System with Digitized Bid Submission and Signature-Based Document Access Control for Southern Luzon State University

Chapter III

METHODOLOGY

This chapter presents the methods and procedures used by the researchers in conducting the study. This includes research locale, research design, research instrument, data gathering collection, and statistical treatment of the data.

Research Locale

This study will be conducted in the municipality of Lucban, located in the province of Quezon, in Region IV-A (Calabarzon), Philippines. The municipality of Lucban is considered a developing municipality and hosts educational institutions and has a growing student population, which is ideal for an academic research study.

The research will be specifically carried out at Southern Luzon State University, a recognized public higher education institution in the region. The university serves as an appropriate research setting due to the presence of administrative offices and personnel directly involved in bidding processes and document management. Furthermore, the university was selected as the research setting due to its current need to improve administrative processes, particularly in terms of online bidding procedures and the digitalization of document submission. Currently, these processes might involve less efficient systems, which may cause delays in processing, lack of transparency, and difficulty in managing records. Through this study, the proposed system aims to address these problems by providing a more efficient, organized, and transparent digital solution.

The locale was chosen because of its accessibility as well as availability of relevant data from offices that take part in bidding processes and document submission. This will allow the researchers to collect accurate and significant information needed to achieve the objectives of the study.

Research Design

In this study, the developmental research method was used for designing, developing, and evaluating an innovative web-based system aimed at addressing specific real-world problems in public procurement, as the researchers intended to create a system that would help the Procurement Office of Southern Luzon State University in managing its bidding activities more securely, efficiently, and transparently.

The researchers will develop YesParency, a web-based bidding system where all procurement-related activities, including bid submission, document access control, bid opening proceedings, and audit trail monitoring, can be managed within a single institution-controlled digital platform. It will also investigate changes over a period of time and be accountable for improvements and adjustments identified through iterative development cycles guided by the Agile Scrum Methodology. The researchers will utilize this method to properly obtain necessary information about the procurement operations and requirements of the Procurement Office and the Bids and Awards Committee of Southern Luzon State University in Lucban, Quezon, ensuring that the developed system is aligned with the provisions of Republic Act No. 9184 and the actual operational needs of its users.

Requirements Analysis

This requirements analysis includes the software and hardware requirements necessary for the development and deployment of YesParenity. It serves as a roadmap for the entire development process, guaranteeing that the system is built on a solid technical foundation and that all components are properly identified and prepared before implementation begins. The following section will also contain the system's data flow diagram, which illustrates how data moves through the different modules and user roles within the YesParenity platform.

Software Requirements

Visual Studio Code (VsCode). Was used as the primary code editor in the development of the system prototype. It enabled the developers to efficiently design and structure the front-end component of the system because it provided a flexible and user-friendly environment. The software offers essential features such as syntax highlighting, intelligent code completion, debugging support, and a wide range of extensions that enhance productivity and streamline the development process.

Git and GitHub. Were also utilized as tools for storing and backing up the source code of the project while developing the system prototype. The researchers used these tools for ensuring that the code was securely stored and protected from accidental loss or deletion, rather than using advanced version control features. Through GitHub, team members were able to easily access the repository, download the project files in compressed format, and share the code with one another. This method was helpful in enabling basic collaboration and ensuring that all team members had a copy of the latest version of the prototype.

Google Chrome. was used as the main web browser for testing the system prototype. It allowed the researchers to view the interface in real time and check how the layout and design actually looked while running. Through this, they were able to spot small issues like alignment problems and needed adjustments. The browser's developer tools were also used to inspect elements and help fix minor errors during development. This made the testing process faster and more convenient. It also helped the researchers ensure that the prototype displayed correctly across different views during testing.

Hardware Requirements

Developers Workstation. refers to the tools used by the researchers or developers in creating, developing, testing, and maintaining the Yesparency system. This workstation must be capable of running the development tools, local servers, and database management tools properly. The minimum specifications for a workstation to operate properly are an Intel Core i5 processor (10th Gen or higher) or AMD Ryzen 5 equivalent, 8 GB RAM, and at least 256 GB SSD storage. For better performance, at least 512 GB SSD storage is recommended. The workstation must run on the Windows 10/11 operating system. To support online resources, updates, and deployment processes, the device must use integrated graphics and a stable internet connection of at least 10Mbps.

Procurement Office Devices. 64-bit Microsoft Windows 10 or newer operating system; Intel Core i3, 8th generation or higher, or AMD equivalent processor; minimum of 4 GB RAM and also 8 GB recommended for smoother multitasking, at least 128 GB storage; display resolution of 1366×768 or higher; capable of running web-based applications efficiently while

handling multiple tasks such as procurement, bidding operations, and transaction management; stable internet connection via Wi-Fi or Ethernet with a minimum speed of 5–10 Mbps.

Bidder Devices. The system operates with current online platforms when users access the system through recent versions of Google Chrome and Microsoft Edge and Mozilla Firefox. The system requires Intel Core i3 processors or AMD compatible processors together with 4 GB RAM and 128 GB storage for computer users. The system needs mobile devices to have at least 64 GB of storage. The system enables users to submit bids and upload documents while accessing procurement features without any interruptions. The system needs a stable internet connection which can use Wi-Fi or mobile data but requires speeds between 5 and 10 Mbps as a minimum requirement.

Requirements Documentation

This requirements documentation involves user and system requirements. It is used as an overview throughout the development procedure, ensuring that the finished system corresponds to the demands of stakeholders and the specified standards.

Functional Requirement

Bid Submissions. For this requirement, bidders should have a registered account to use in logging in to the system, securely stored in the system's database. Through their individual accounts, bidders can access the bid submission module. The system allows bidders to upload and submit all required documents for participation in the bidding process. The system also lets the bidders review their submitted documents before finalizing their submission. The system should notify bidders upon successful submission of their bids. Additionally, bidders should be able to view the status and details of their submitted bids through their designated accounts.

Audit Trail. For this requirement, the system should include an audit trail module that records all activities performed within the platform. The system should track actions such as document submissions, updates, deletions, and user logins. All recorded data should be securely stored in the system's database for future reference. Each record should include relevant details such as the user, type of action, date, and time of the activity. The administrators are authorized to access the audit trail through their designated accounts. This will allow administrators to monitor user behavior and ensure accountability within the system.

Live Bidding. For this requirement, the system must include a live bidding module accessible to all users. The system will allow the users to join the live online bidding using their accounts. During the live bidding process, the users will be able to view the entire process in real time. This will include viewing the opening of the submitted documents and the evaluation process by the administrators. On the administrator side, they will be in charge of facilitating the live bidding process. The admins should be able to open submitted documents, review them, and verify if bidders meet the required qualifications. The YesParenity system will be able to allow smooth and uninterrupted live bidding processes.

Announcements. The announcement module allows the procurement office to issue updates regarding bidding results, schedules for the next bidding activities, and other important system updates through the system. Using a dedicated interface, admins can create announcements and they can specify the message content and relevant details. These announcements can also be organized and managed for easier access and reference. Users receive the updates through the system in real time. The system ensures that important information is properly delivered to all users to maintain transparency and timely communication.

Non-Functional Requirement

Functional Suitability. The system should provide all functions necessary to support the complete procurement bidding process of Southern Luzon State University, including bid submission, document encryption, digital signature authentication, live streaming of bid opening proceedings, and audit trail monitoring. All features should operate as intended across all user

roles including super administrator, procurement administrator, registered bidder, and public observer, without any broken or incomplete feature present in the deployed system.

Performance Efficiency. The system should be able to handle multiple simultaneous users, including concurrent bid submissions and live stream viewers during bid opening sessions, without experiencing significant delays or degradation in response time. This includes optimizing server response times, database query performance, and streaming bandwidth to maintain stable operation even during peak usage periods.

Compatibility. The system should be accessible through major web browsers and operate seamlessly across different devices such as smartphones, laptops, and desktop computers without conflicts or loss of functionality. This ensures that bidders and public observers can access YesParency regardless of the device or browser they are using.

Usability. The system should provide a clean, intuitive, and responsive user interface that allows all types of users to navigate and accomplish their tasks efficiently with minimal training. This includes clear navigation menus, informative feedback messages, and role-appropriate dashboards that simplify procurement workflows for both technical and non-technical users.

Reliability. The system should maintain consistent and stable operation at all times, particularly during critical procurement activities such as bid submission deadlines and live bid opening sessions. This involves implementing proper server infrastructure, automated backups, and error handling mechanisms to minimize downtime and data loss.

Security. The system should protect all submitted bid documents, user credentials, and procurement data from unauthorized access through encryption, role-based access control, and digital signature authentication. This includes safeguards against premature document access, failed authentication logging, and compliance with data protection standards to ensure the integrity and confidentiality of all procurement records.

Maintainability. The system should be structured in a way that allows developers to easily identify, correct, and improve its components without disrupting ongoing procurement operations. This includes maintaining clean and well-documented code, modular system architecture, and regular data backups to prevent loss of procurement records in the event of technical issues.

Flexibility. The system should be capable of accommodating future changes in procurement requirements, user roles, or institutional policies without requiring a complete system overhaul. This includes the ability to add new procurement projects, update announcement content, adjust user permissions, and expand system capacity as the needs of the Procurement Office evolve.

Safety. The system should operate without causing unintended harm to data, users, or procurement operations. This includes preventing accidental deletion of submitted bid documents by unauthorized users, ensuring that encryption and access control mechanisms do not inadvertently lock out authorized personnel, and maintaining data integrity throughout the entire bidding process.

User Requirement

Super Admin/ Admin Account. The Super Admin account is assigned to the Bids and Awards Committee (BAC) in the procurement office, who serves as the highest authority within the YesParenity system. This account is responsible for the overall management of the system, including the creating, modifying and removing user accounts, particularly admin-level accounts. The Super Admin also has full access to all systems, modules, and functionalities, ensuring that all procurement-related activities are monitored and controlled appropriately.

In addition, the Super Admin is responsible for overseeing the entire procurement process within the system, which includes the submission of required documents and online bidding processes. They are also responsible for managing system configurations and maintaining proper access control to prevent unauthorized actions.

On the other hand, Admin accounts are responsible for managing and reviewing the documents submitted by bidders. They ensure that all submitted requirements are complete, accurate and properly formatted before they are processed for procurement evaluation. Admins also monitor bidding activities and assist in maintaining the proper flow of procurement transactions within the system.

Bidders Account. Is intended for suppliers or participants who wish to join procurement activities within the system. Before they are allowed to submit bids, bidders must first upload and submit their legal requirements as proof of eligibility to participate. The documents that are uploaded and submitted will then be processed and validated by the admin. Once approved, the bidders will be notified through the system that they are qualified to participate in the bidding process.

Before proceeding with any bid, the bidders are required to purchase the official bid documents from the SLSU Procurement Office. After purchase, they will be given a receipt, which they will encode in the system as part of the verification process. Only after successful validation of the receipt number will bidders be granted access to proceed with the online bidding process.

Additionally, the Bidders account allows users to view procurement opportunities, monitor their submitted bids, and monitor the status of their participation within the system.

Design of Software, Systems, Product and/or Processes

This part includes the different diagrams of the system such as Software Design Methodology, Use Case Diagram, Activity Diagram, Sequence Diagram, Class Diagram, Database Diagram, and the Deployment Diagram.

Software Design Methodology

The Software Development Life Cycle (SDLC) used in this study follows the Agile Scrum methodology, an iterative approach that promotes collaboration, adaptability, and continuous improvement allowing the system to be refined as new requirements and feedback arise (Ekechi et al., 2024) . This contains Product Backlog, Sprint Planning, Sprint Development and Sprint Review. Agile Scrum is characterized by a dynamic development phase when all tasks are monitored and repeated .

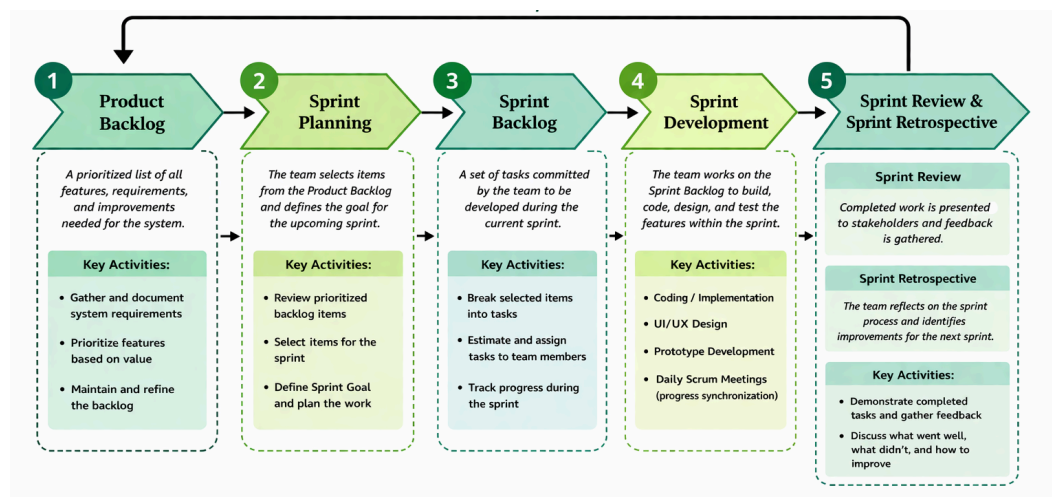


Figure 2. Agile Scrum Software Development Lifecycle of the YesParenity System

In figure 2, the Agile Scrum Software Development Lifecycle (SDLC) of the YesParenity system is illustrated, showing the step-by-step process followed by the researchers during system development. Each sprint produces incremental improvements based on feedback, ensuring that the system evolves in alignment with user and project requirements. The process starts with the Product Backlog, which contains all the system, features, and possible improvements.

During Sprint Planning, selected tasks from the product backlog are placed into the Sprint Backlog, which serves as the guide for what will be done in a specific sprint. This is followed by the Sprint Development phase, where the team works on coding, designing the user interface, and building system features within a set time. After the sprint, a Sprint Review is conducted to check the completed work and collect feedback. Then, a Sprint Retrospective is done to reflect on the process and identify ways to improve.

Use Case Diagram

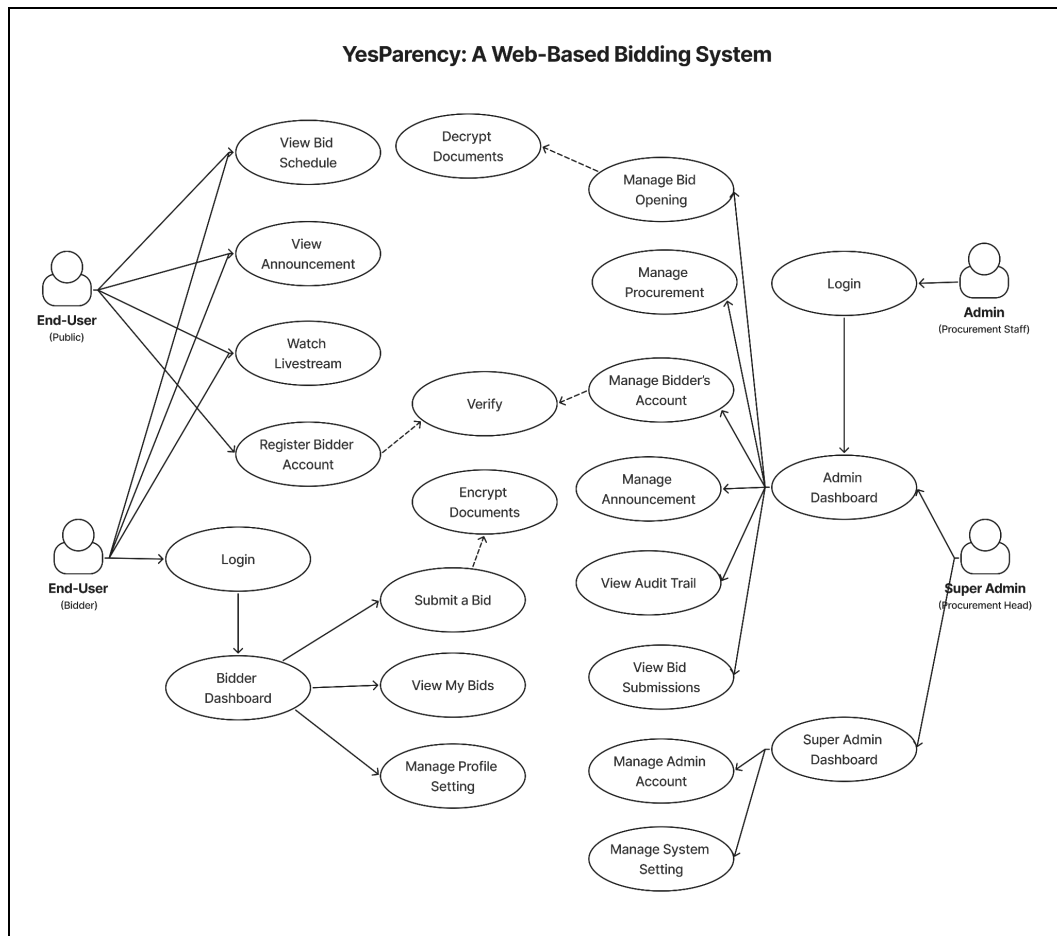


Figure 3. Use Case Diagram of “YesParency: A Web-Based Bidding System.”

The researchers used a use case diagram to illustrate the relationship among users, procurement administrators, and the system. This diagram represents the system's features and the accessibility levels of the Super Administrator (SAdm), Administrator (Adm), and End-Users (Public and Bidder). As shown in the figure above, the End-User (Public) can access publicly available features without logging in, including viewing bid schedules, announcements, and watching the live stream of bid opening proceedings. The End-User (Bidder) may register a bidder account and, upon logging in, gains access to the Bidder Dashboard where they can submit a bid, view their submitted bids, and manage their profile settings. When a bid is

submitted, the system automatically encrypts the documents, and the bidder account undergoes a verification process before being granted full participation in the bidding activity.

The Admin (Procurement Staff) logs into their account to access the Admin Dashboard, which provides access to procurement management functions such as managing bid openings, managing bidder accounts, managing announcements, viewing the audit trail, and viewing bid submissions. During the bid opening phase, the Admin is responsible for decrypting submitted documents, which is performed through the Manage Bid Opening use case and requires proper authentication before access is granted. The SAdm (Procurement Head) holds full control over the system and shares the same permissions as the Admin through the Admin Dashboard. In addition, the SAdm has exclusive access to the Super Admin Dashboard, which allows them to manage administrator accounts and configure system settings, giving the Procurement Head complete oversight and authority over both system operations and user management.

Activity Diagram

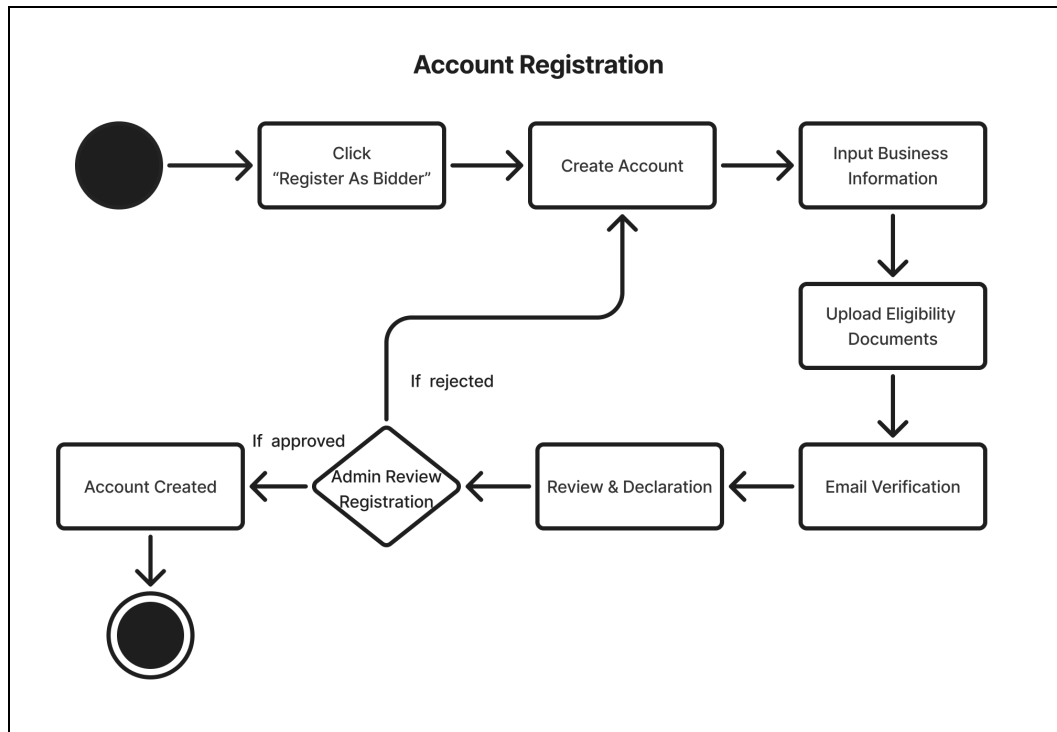


Figure 4. Account Registration Activity Diagram

Figure 4 shows how a user can register an account as a bidder within the system. The process begins when the user clicks the "Register As Bidder" button, which redirects them to the account creation page. The user then inputs the required business information and uploads the necessary eligibility documents. After uploading, the user proceeds to email verification, followed by a review and declaration of the submitted details. Once completed, the Admin reviews the registration. If the registration is approved, the user is granted access leading to the end of the process. If the registration is rejected, the system redirects the user back to the account creation step to resubmit the required information.

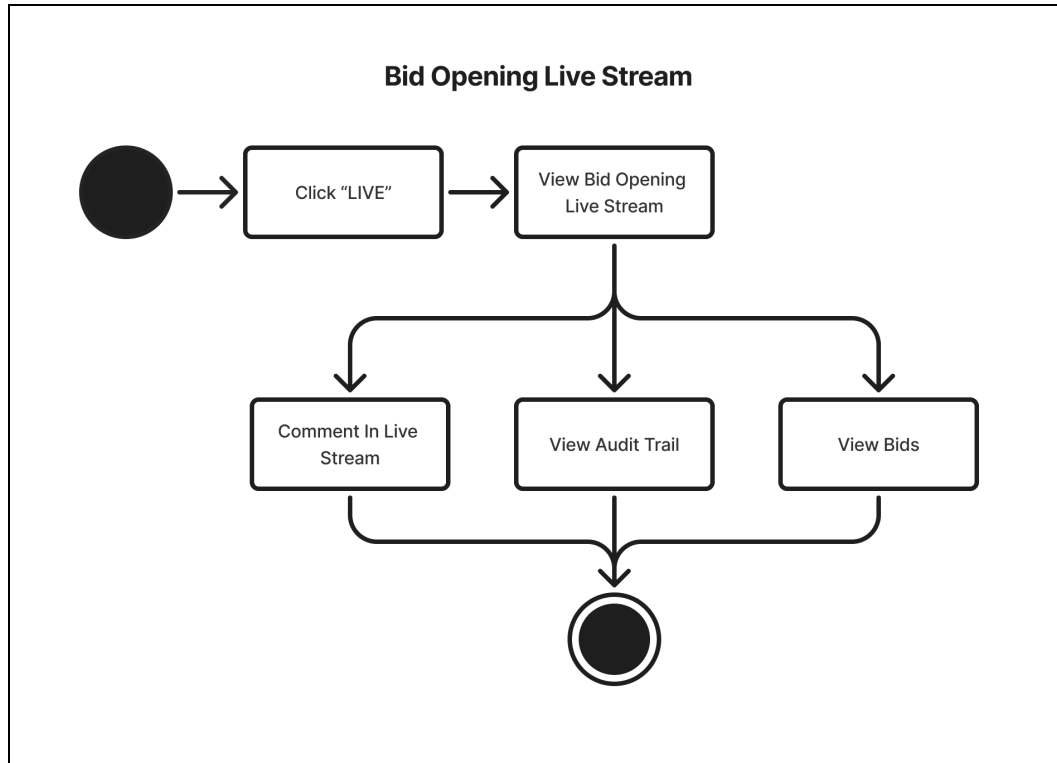


Figure 5. Bid Opening Live Stream Activity Diagram

Figure 5 shows how a user can access and interact with the Bid Opening Live Stream within the system. The process begins when the user clicks the "LIVE" button, which directs them to the Bid Opening Live Stream page. Once on the page, the system simultaneously presents three available actions the user can perform: Comment in the Live Stream, View Audit Trail, and View Bids. The user may engage in any of these activities during the live stream session. After completing any of the available actions, the process reaches its end.

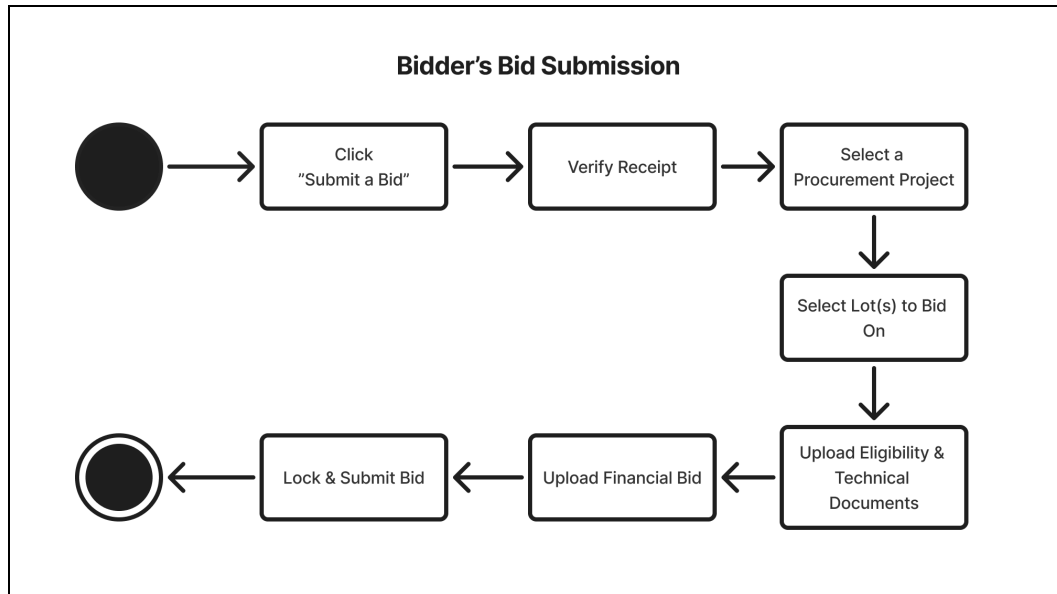


Figure 6. Bidder's Bid Submission Activity Diagram

Figure 6 illustrates the process of how a bidder can submit a bid within the system. The process begins when the bidder clicks the "Submit a Bid" button. The system then verifies the receipt of the bidder before proceeding. Once verified, the bidder selects a Procurement Project they wish to bid on, followed by selecting the specific Lot(s) under that project. The bidder then uploads the required Eligibility and Technical Documents, followed by uploading the Financial Bid. After all necessary documents have been submitted, the bidder clicks "Lock & Submit Bid" to finalize the submission, completing the process.

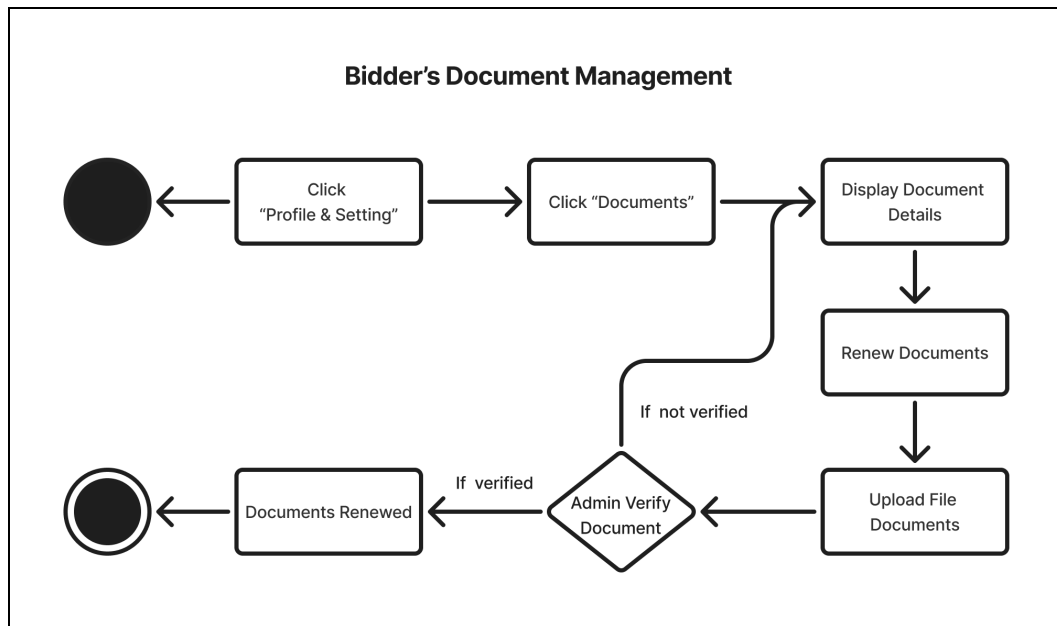


Figure 7. Bidder's Document Management Activity Diagram

Figure 7 illustrates how a bidder can manage and renew their documents within the system. The process begins when the bidder clicks "Profile & Settings", which then navigates them to the profile page. The bidder then clicks "Documents", where the system displays the current document details. From there, the bidder proceeds to renew their documents by uploading the updated file documents. Once uploaded, the Admin verifies the submitted document. If the document is verified, the system confirms that the documents have been successfully renewed, completing the process. If the document is not verified, the bidder is redirected back to the renewal step to re-upload the required documents.

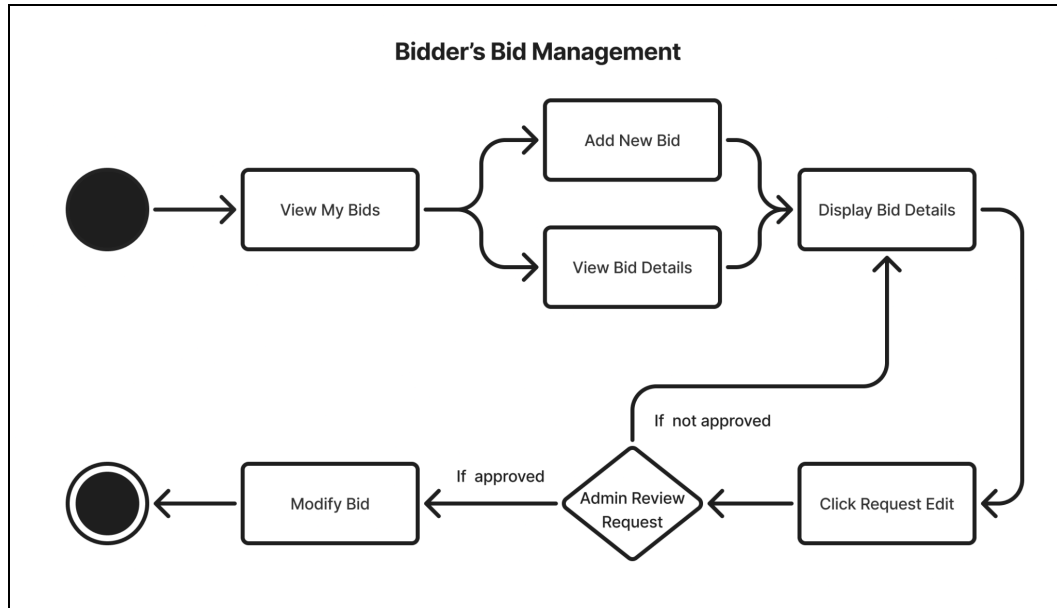


Figure 8. Bidder's Bid Management Activity Diagram

Figure 8 illustrates how a bidder can manage their bids within the system. The process begins when the bidder navigates to "View My Bids", where they are presented with two options: Add New Bid or View Bid Details. Both actions lead to the system displaying the full Bid Details. From the bid details page, the bidder may click "Request Edit" if they wish to make modifications to their bid. The Admin then reviews the edit request. If the request is approved, the bidder is granted access to Modify the Bid, completing the process. If the request is not approved, the system redirects the bidder back to the bid details page, where they may submit another request if needed.

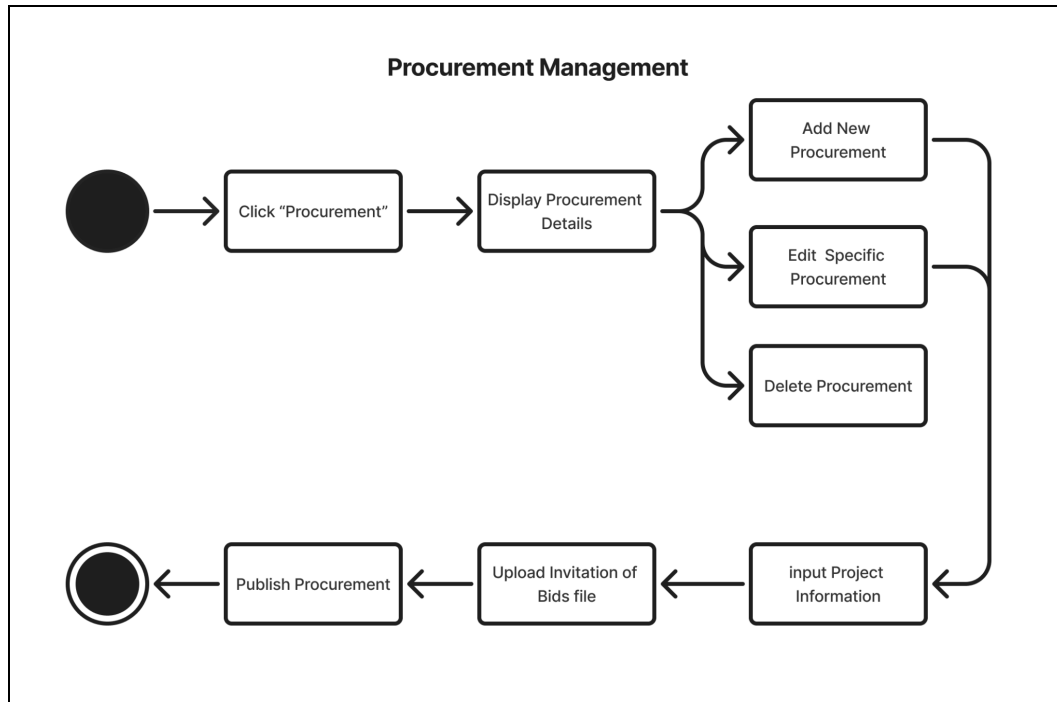


Figure 9. Procurement Management Activity Diagram

Figure 9 illustrates how the admin can manage procurement projects within the system. The process begins when the admin clicks the "Procurement" navigation, which directs them to the procurement page where all procurement details are displayed. From there, the admin is presented with three available actions: Add New Procurement, Edit Specific Procurement, or Delete Procurement. Regardless of which action is selected, the admin proceeds to input the Project Information, followed by uploading the Invitation of Bids file. Once all required information and files have been provided, the admin clicks "Publish Procurement" to make the procurement project visible within the system, completing the process.

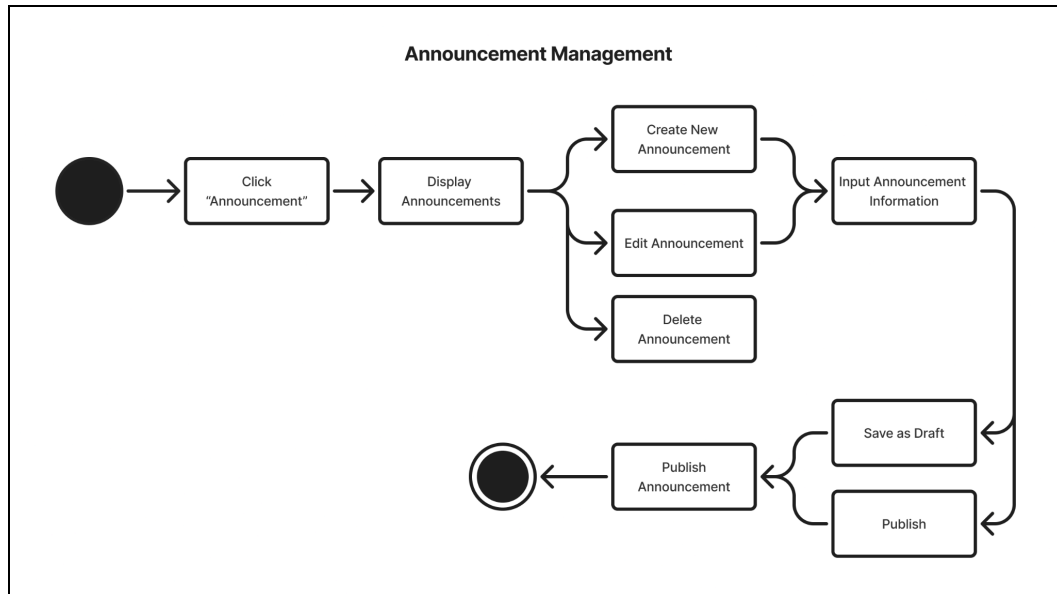


Figure 10. Announcement Management Activity Diagram

Figure 10 illustrates how the admin can manage announcements within the system. The process begins when the admin clicks the "Announcement" navigation, which directs them to the announcements page where all existing announcements are displayed. From there, the admin is presented with three available actions: Create New Announcement, Edit Announcement, or Delete Announcement. For both creating and editing, the admin proceeds to input the Announcement Information. After filling in the necessary details, the admin is given two options on how to proceed: Save as Draft to temporarily store the announcement for later, or Publish to make it immediately visible. Either option leads to the Publish Announcement step, completing the process.

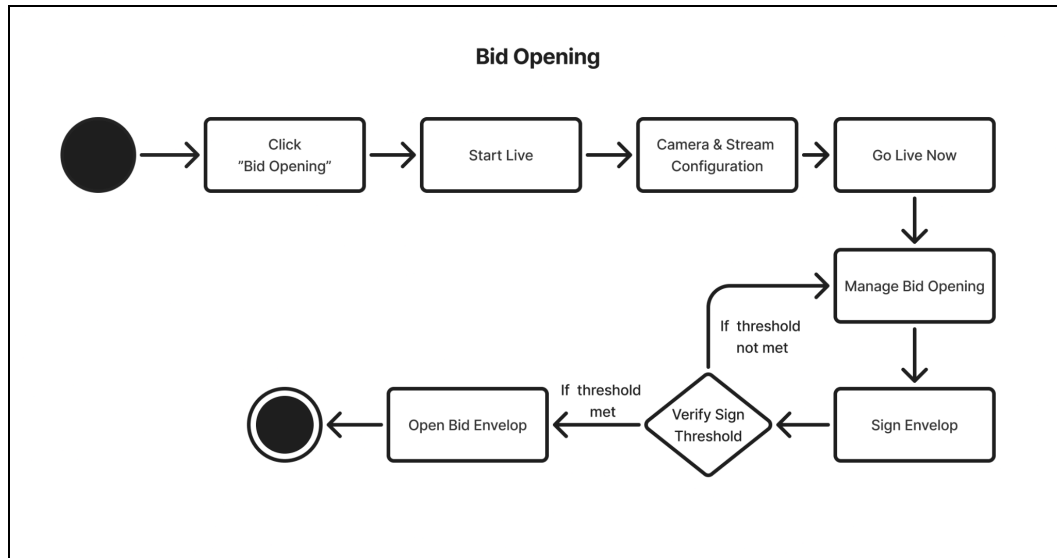


Figure 11. Bid Opening Activity Diagram

Figure 11 illustrates how the admin conducts the Bid Opening process within the system. The process begins when the admin clicks "Bid Opening", which directs them to the bid opening page. The admin then clicks "Start Live" and proceeds to configure the Camera & Stream Settings. Once configured, the admin clicks "Go Live Now" to begin the live session. During the live stream, the admin manages the Bid Opening proceedings and proceeds to Sign the Envelope. The system then verifies the Sign Threshold. If the threshold is met, meaning the required number of signatories have signed, the system proceeds to Open the Bid Envelope, completing the process. If the threshold is not met, the admin is redirected back to the Manage Bid Opening step to continue until the required threshold is satisfied.

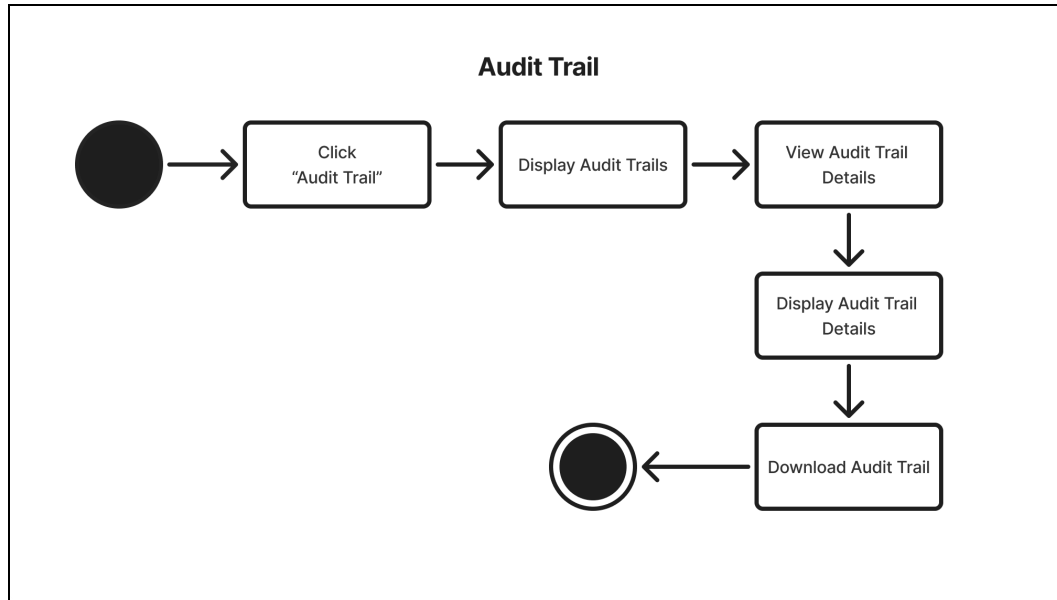


Figure 12. Audit Trail Activity Diagram

Figure 12 illustrates how the admin can access and manage the audit trail records within the system. The process begins when the admin clicks "Audit Trail" from the navigation, which directs them to the audit trail page where all recorded logs are displayed. The admin can then select a specific entry to View the Audit Trail Details, upon which the system displays the complete details of the selected record. After reviewing the details, the admin may then Download the Audit Trail to save a copy of the records, completing the process.

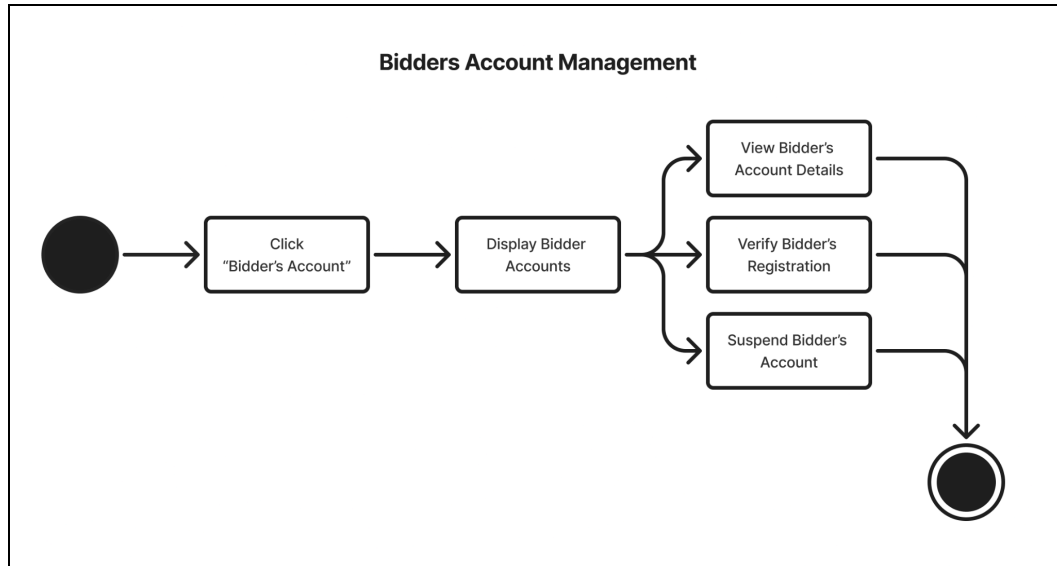


Figure 13. Bidders Account Management Activity Diagram

Figure 13 illustrates how the admin can manage bidder accounts within the system. The process begins when the admin clicks "Bidder's Account" from the navigation, which directs them to the page where all registered bidder accounts are displayed. From there, the admin is presented with three available actions: View Bidder's Account Details to review the full profile and information of a specific bidder, Verify Bidder's Registration to approve or validate a bidder's submitted registration, or Suspend Bidder's Account to restrict a bidder's access to the system. After performing any of the three actions, the process reaches its end.

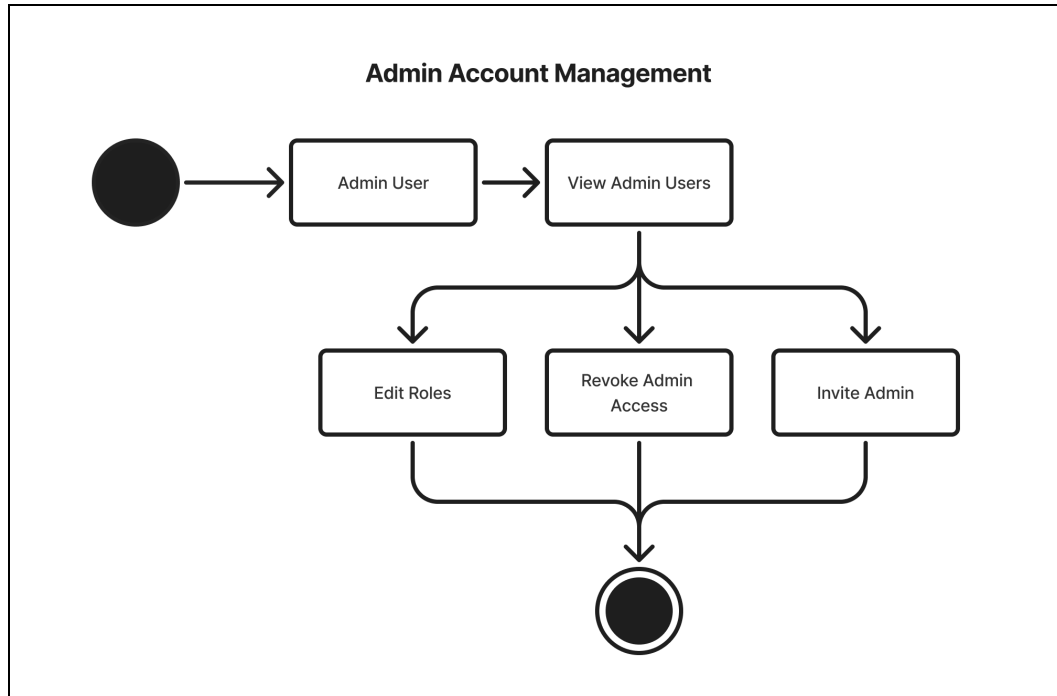


Figure 14. Admin Account Management Activity Diagram

Figure 14 illustrates how the super admin can manage administrator accounts within the system. The process begins when the super admin navigates to "Admin User", which directs them to the page where all existing admin users are displayed. From there, the super admin is presented with three available actions: Edit Roles to modify the permissions and responsibilities assigned to a specific admin, Revoke Admin Access to remove an admin's privileges and restrict their access to the system, or Invite Admin to add a new administrator by sending an invitation. After performing any of the three actions, the process reaches its end.

Sequence Diagram

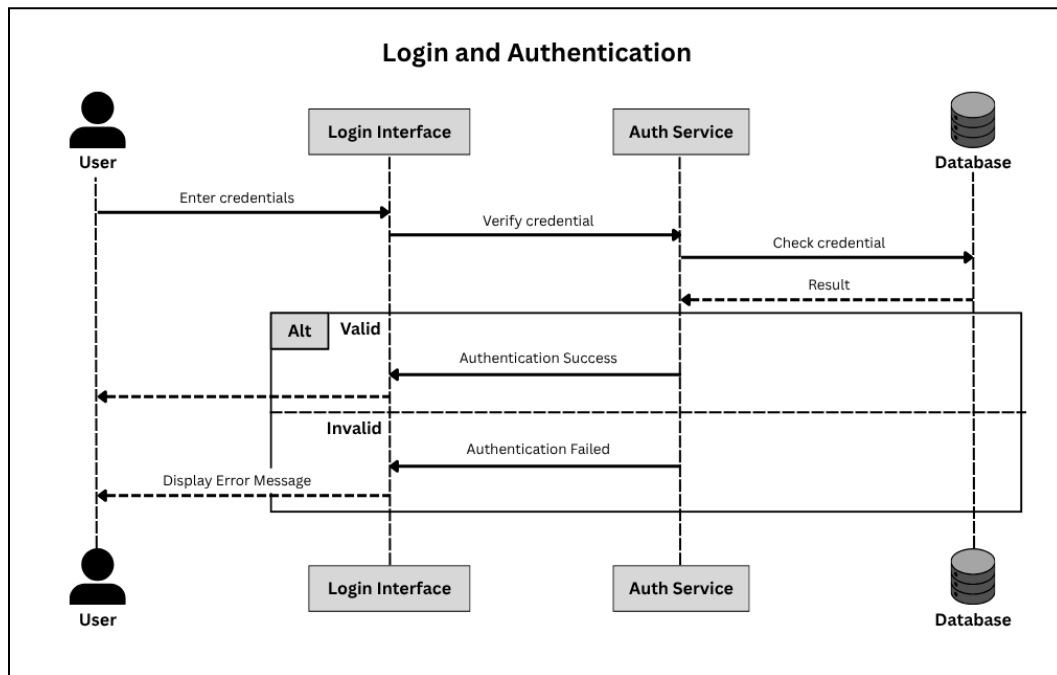


Figure 15. Login and Authentication Sequence Diagram

Figure 15 shows the login and authentication sequence of YesParency, where the user inputs their username and password into the login interface, which forwards the credentials to the authentication service to check against existing records in the database. The database returns the results to the authentication service, which then evaluates two possible outcomes: if the credentials are valid, an authentication success response is sent back and the login interface displays a success message granting the user access to their role-based dashboard; if the credentials are invalid, an authentication failed response is returned and the interface displays an error message, prompting the user to try again.

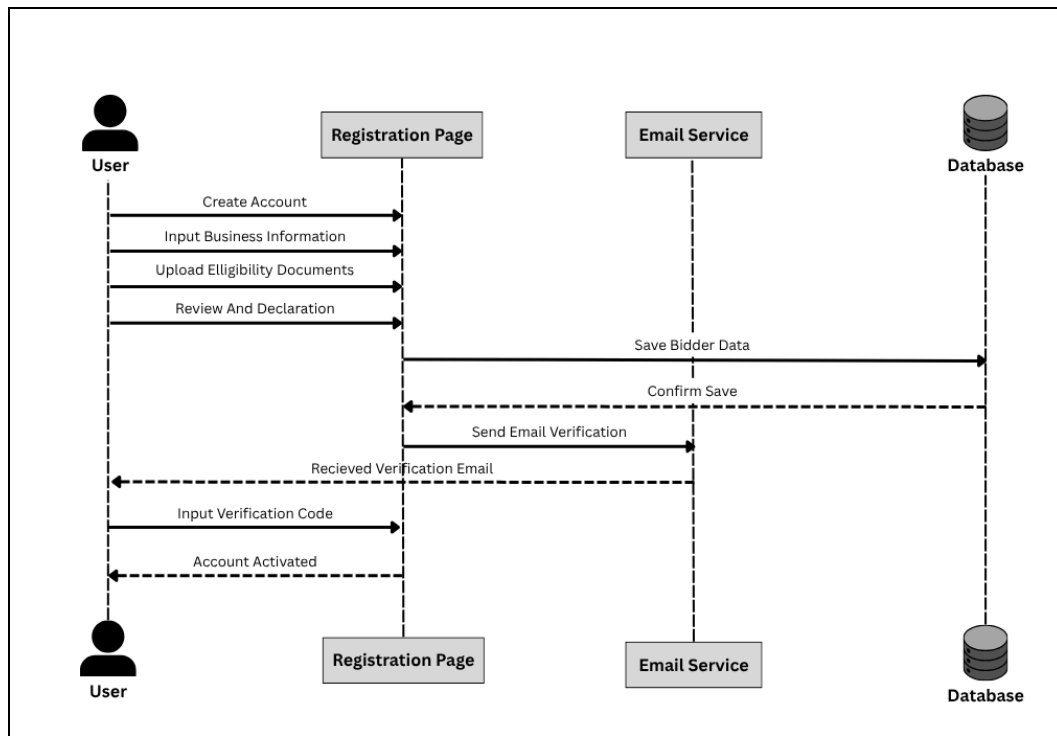


Figure 16. Bidder Registration Sequence Diagram

Figure 16 shows the bidder registration sequence of YesParenity, where a prospective bidder creates an account by inputting their business information, uploading eligibility documents, and completing the review and declaration through the registration page, which then saves the submitted data to the database and triggers the email service to send a verification email to the bidder. Once the bidder clicks the verification link, the registration page confirms the account activation and redirects the bidder to the login page, completing the registration process.

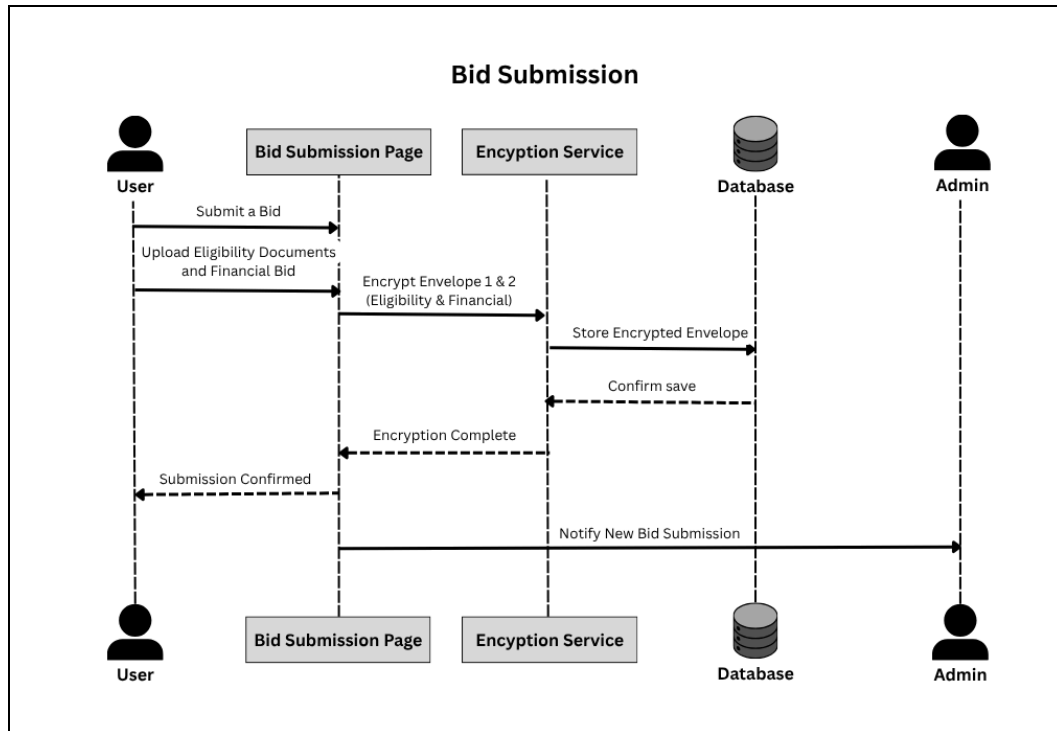


Figure 17. Bid Submission Sequence Diagram

Figure 17 shows the bid submission sequence of YesParenity, where a registered bidder selects a procurement project and the specific lot(s) to bid on through the bid portal, uploads the eligibility and technical documents along with the financial bid, and locks the submission, after which the portal forwards both sets of documents to the encryption service to be separately encrypted into two digital envelopes and stored in the database. The portal then confirms the submission to the bidder with a receipt, logs the timestamp and submission order in the database, and notifies the procurement admin of the new bid submission, completing the two-envelope bidding process in a secure and traceable manner.

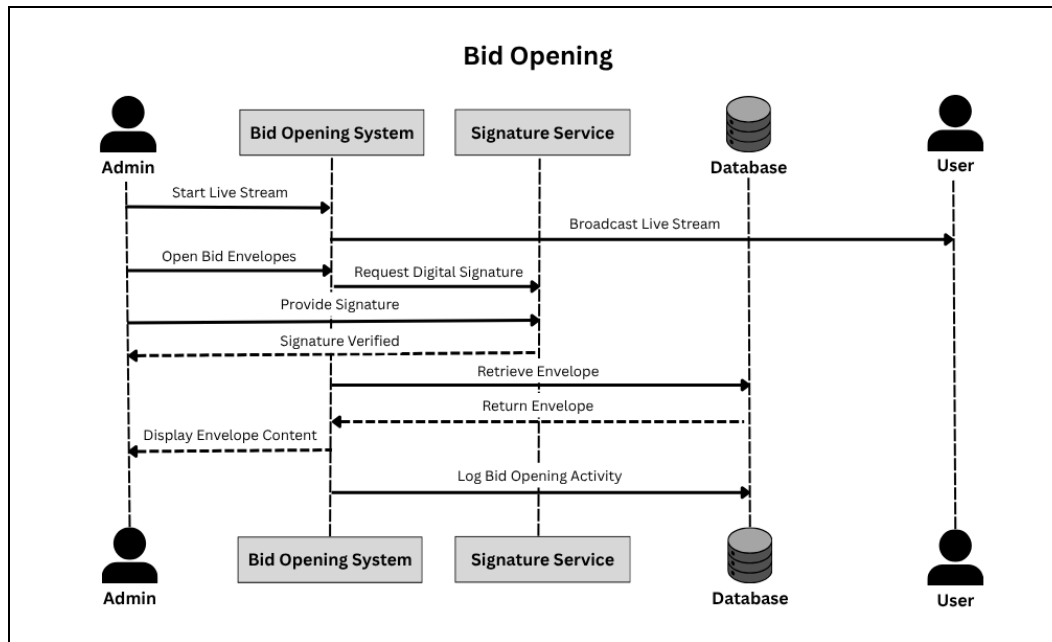


Figure 18. Bid Opening Sequence Diagram

Figure 18 shows the bid opening sequence of YesParency, where the BAC officer initiates the session by starting the live stream through the bid opening system, which immediately broadcasts it to public observers for transparency. The BAC officer then opens the bid envelopes, prompting the system to request a digital signature, which the officer provides and the signature service verifies before granting access to the submitted documents. The system retrieves and displays the eligibility documents from the first envelope, followed by the financial bid amounts from the second envelope, and finally logs the entire bid opening activity in the database to maintain a complete and tamper-evident audit record.

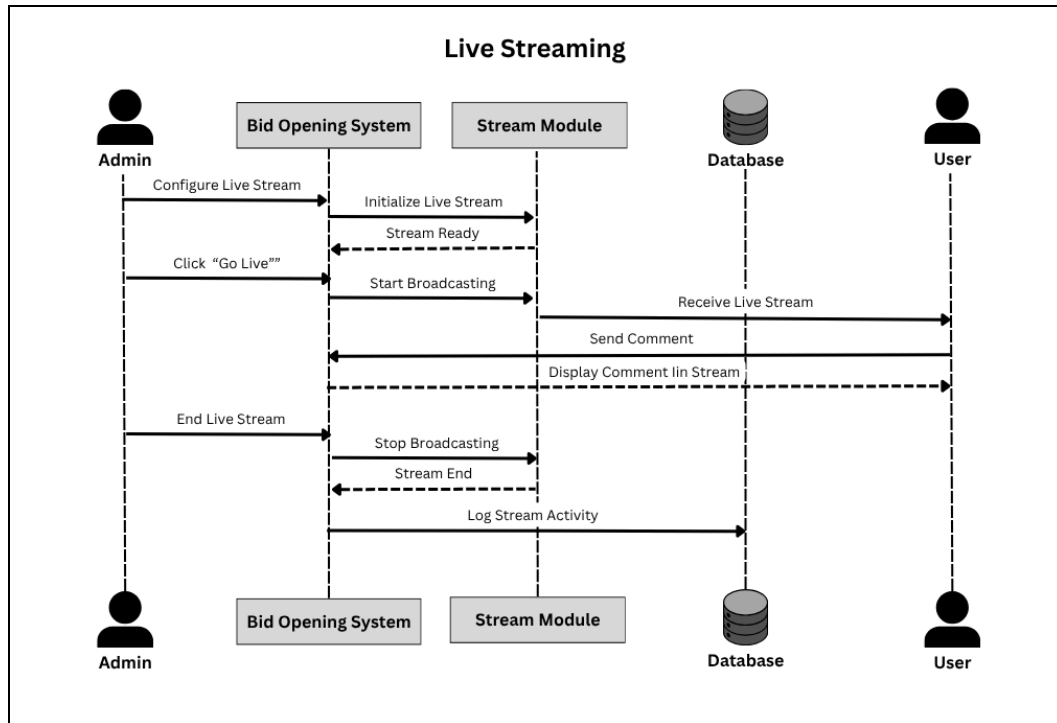


Figure 19. Live Streaming Sequence Diagram

Figure 19 shows the live streaming sequence of YesParenity, where the admin configures the camera and stream settings through the bid opening system, invites all admins, and initiates the live stream by instructing the streaming module to begin broadcasting, which public observers can then access in real time and interact with by sending comments that are displayed within the stream. Once the bid opening proceedings are concluded, the admin ends the live stream, the streaming module stops broadcasting and confirms the stream has ended, and the system logs the entire stream activity in the database for audit and accountability purposes.

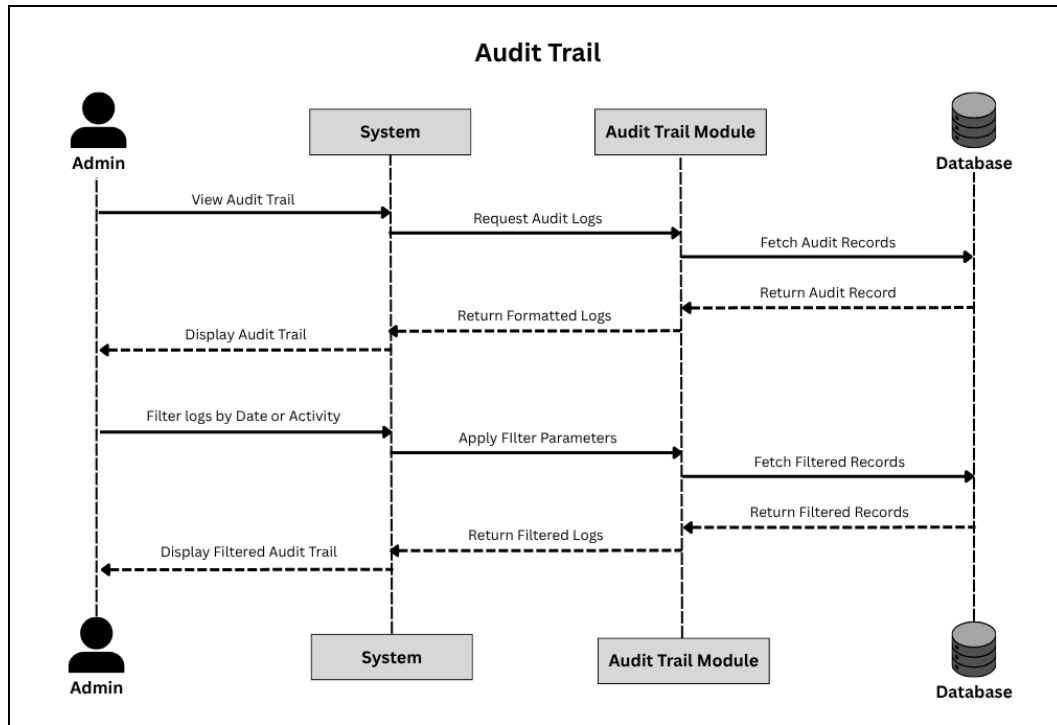


Figure 20. Audit Trail Sequence Diagram

Figure 20 shows the audit trail sequence of YesParenity, where the admin requests to view the audit trail through the system, which forwards the request to the audit trail module to fetch all audit records from the database and return them as formatted logs displayed to the admin. The admin can then filter the logs by date or activity type, prompting the system to apply the filter parameters through the audit trail module, which retrieves the filtered records from the database and returns them for display, allowing the admin to review specific system activities in a precise and accountable manner.

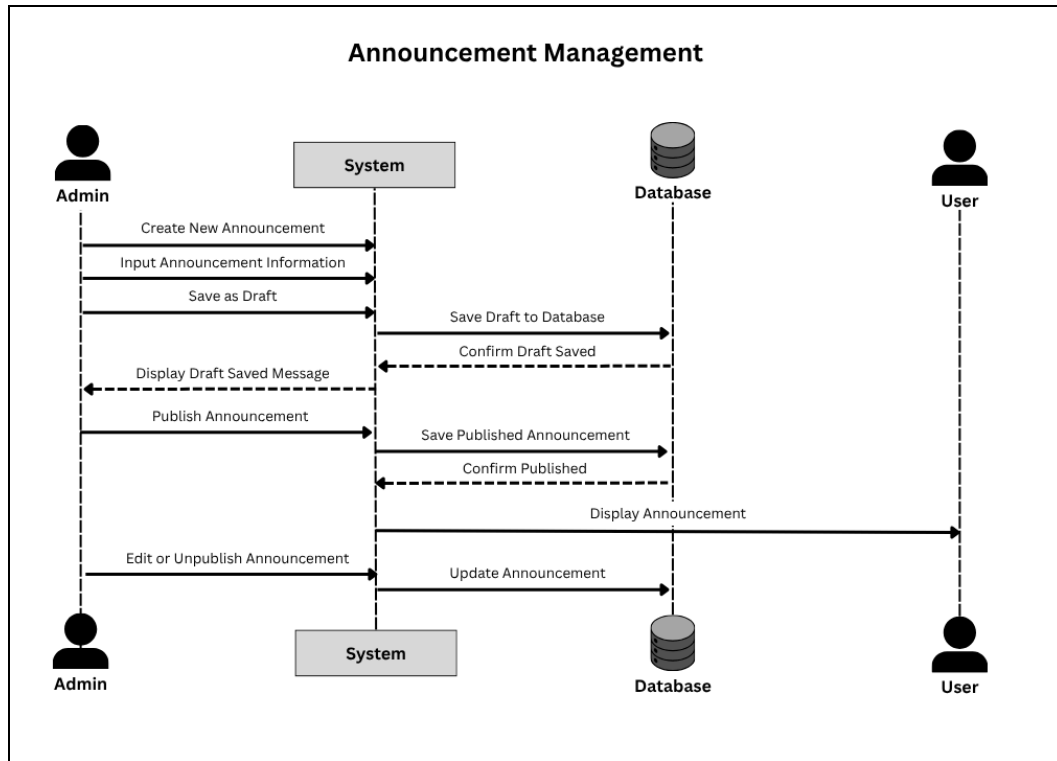


Figure 21. Announcement Management Sequence Diagram

Figure 21 shows the announcement management sequence of YesParenity, where the admin creates a new announcement by inputting the announcement information and publish date through the announcement system, then saves it as a draft which is stored in the database and confirmed back to the admin. Once the admin is ready, the announcement is published, saved to the database, and displayed to the public, after which the system confirms the successful publication to the admin. The admin can also edit or unpublish the announcement at any time, giving the procurement office full control over the information made available to the public.

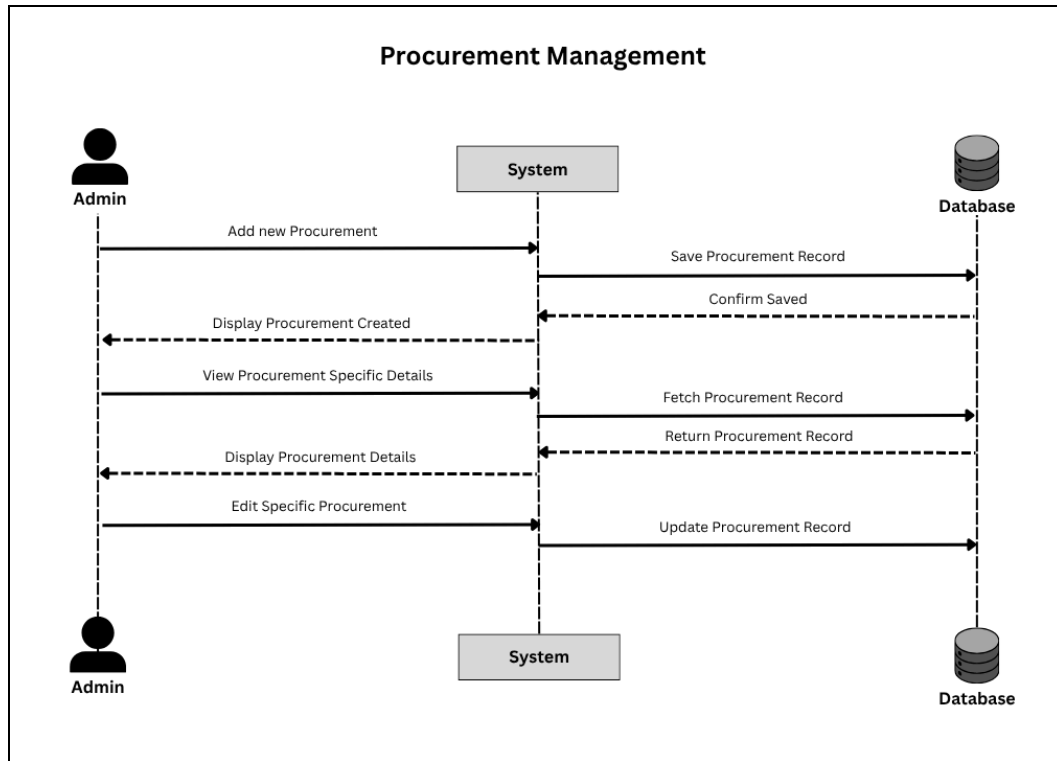


Figure 22. Manage Procurement Sequence Diagram

Figure 22 shows the procurement management sequence of YesParency, where the admin adds a new procurement by inputting the project information, procurement schedule, budget, and uploading the invitation to bid file through the procurement system, which then saves the record to the database and confirms the successful creation back to the admin. The admin can also view specific procurement details by requesting the system to fetch the corresponding record from the database and display it, as well as edit a specific procurement, with the system updating the record in the database to reflect the changes made.

Class Diagram

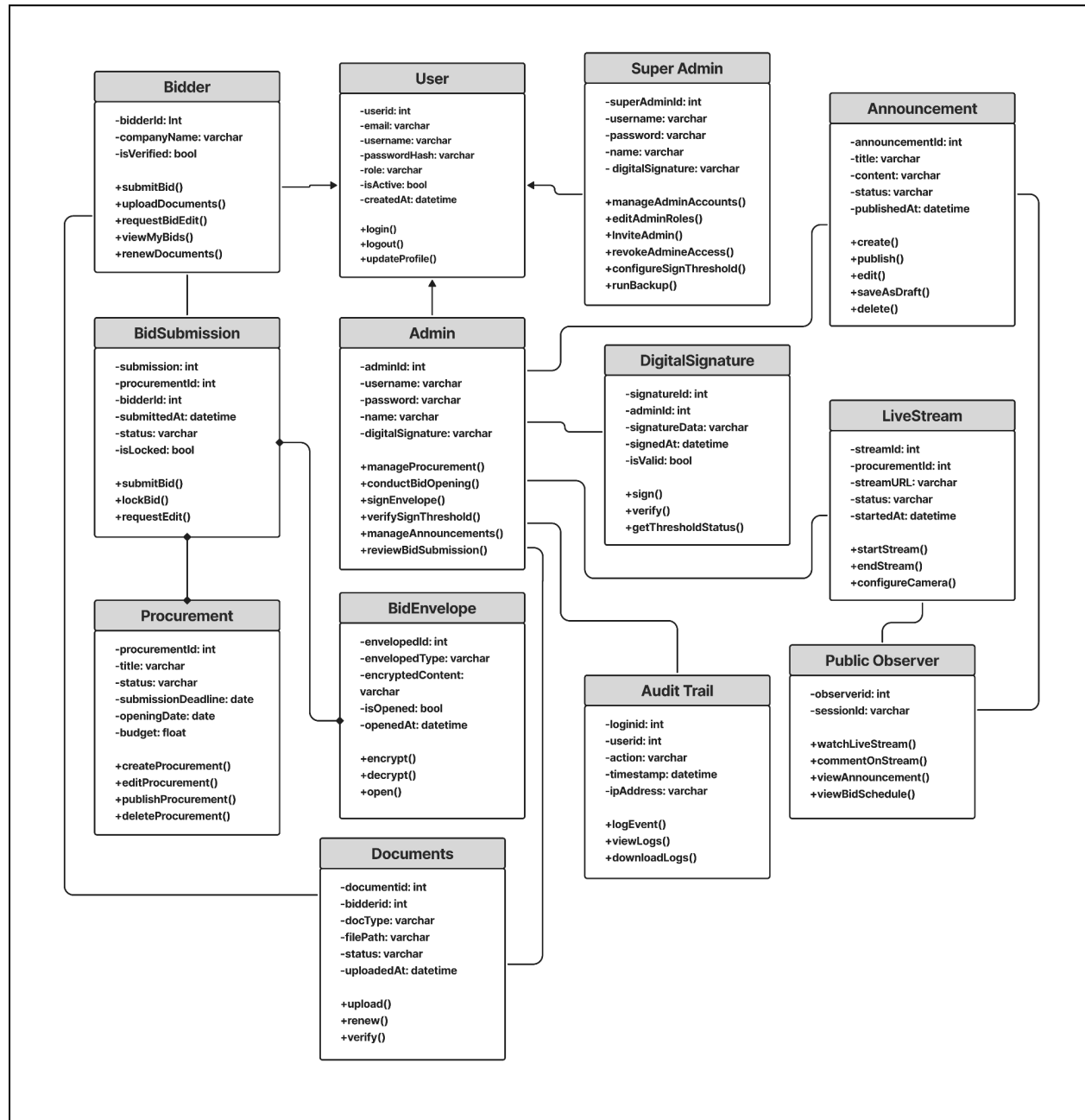


Figure 23. Class Diagram of YesParency: A Web-Based Bidding System.

Figure 23 illustrates the class diagram of the web-based system entitled YesParency: A Web-Based Bidding System. This diagram shows the classes and its activity and processes per

module or per functionalities or features. This illustration also enumerates the data types and variables necessary to perform each given function or system activity, where the '+' and '-' symbols indicate public and private accessibility.

Database Diagram

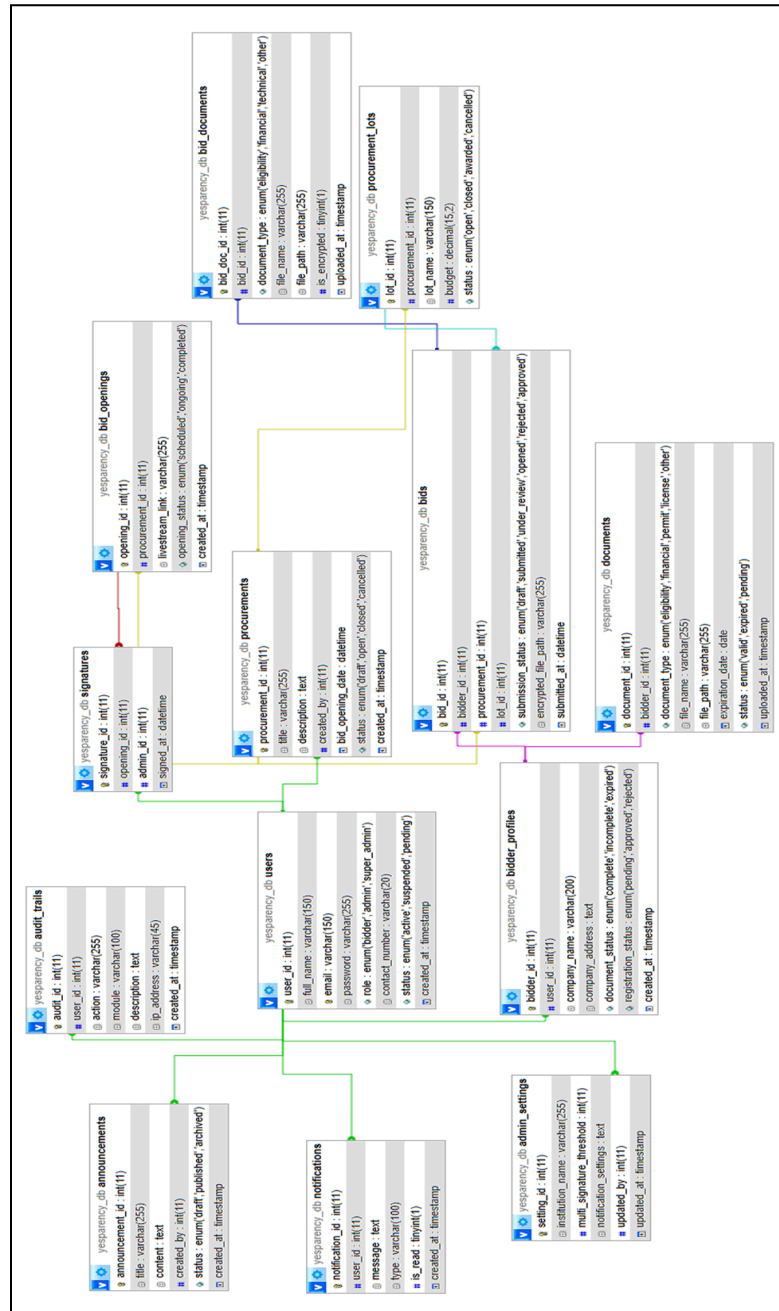


Figure 24. Database Diagram of YesParency: A Web-Based Bidding System.

Figure 24 presents the relationships existing among the different entities of the YesParency database, whereby each table is linked via its respective primary and foreign keys. A

rectangle is used for each database table that has attributes along with their specific data types and SQL constraints. Lines connecting various tables signify the relationships between them be it one-to-one relationship or a one-to-many relationship. Primary keys are depicted using the yellow keys, whereas the foreign keys are indicated using the red keys. The database diagram provides a clear understanding of the database design in regard to its various functionalities, such as user management, bidding, procurement process, document management, and auditing capabilities.

Deployment Diagram

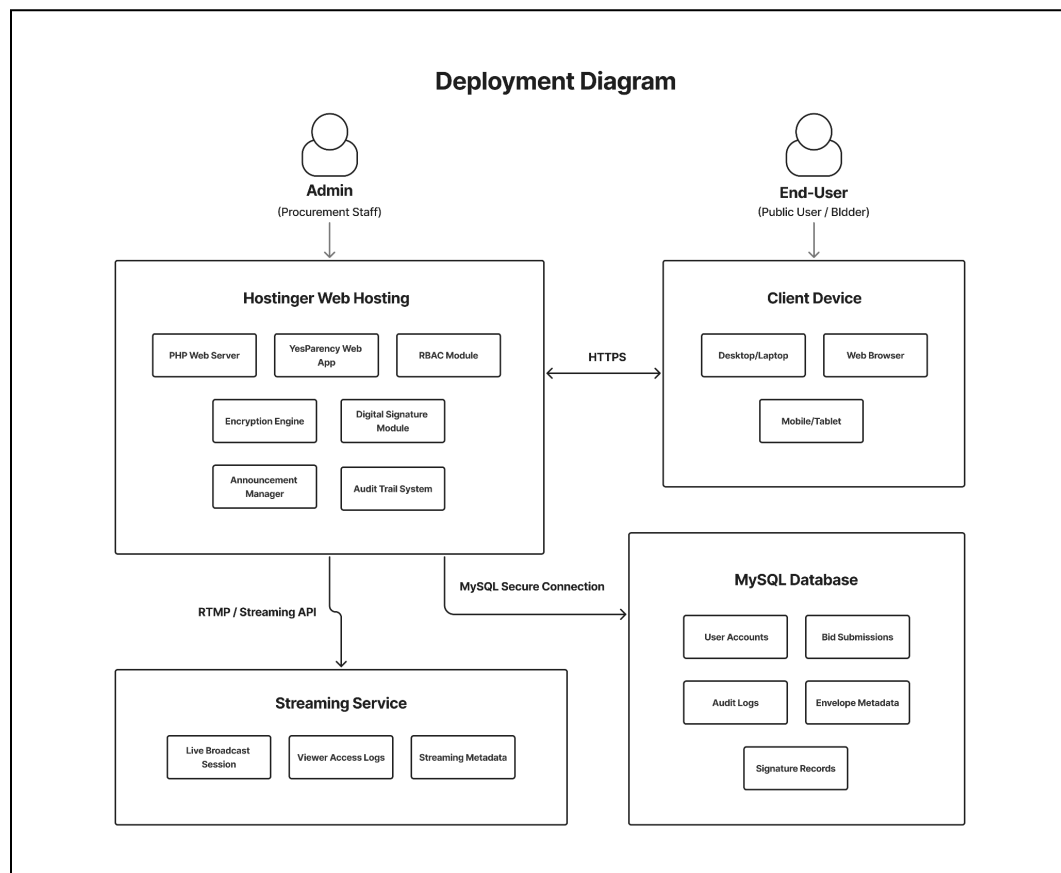


Figure 25. Deployment Diagram of YesParenCy: A Web-Based Bidding System.

Figure 25 illustrates the physical architecture of the YesParenCy Web-Based Bidding System, showing how procurement administrators, bidders, and public observers access the system through client devices such as desktops, laptops, and mobile devices using web browsers connected via secure HTTPS communication. The system is hosted on a Hostinger web server where the YesParenCy web application and its core modules including the RBAC module, encryption engine, digital signature module, audit trail system, and announcement manager are deployed to manage secure bid submission and role based access control. The application communicates with a MySQL database server through a secure connection to store user

accounts, bid submissions, envelope metadata, audit logs, and signature records to support the digitized two envelope bidding process. A streaming service connected through an RTMP or streaming API is also included to enable real time broadcasting of bid opening sessions and support transparency and public monitoring of procurement activities.

Development and Testing Procedure

Development

During the development phase of the YesParenity system, the researchers followed a structured and collaborative process to ensure that the system's features and functionalities were properly implemented. The team maintained continuous communication with their adviser and co-adviser, conducting regular consultations and weekly interviews to gather insights and validate the features to be included in the system. These interactions helped ensure that the system design and functionalities aligned with the intended objectives of the study. The initial design process involved planning the system structure, organizing features, and preparing the necessary components for development. Tasks were then divided among the researchers, with each member assigned specific responsibilities. A shared documentation table was used to manage and monitor all tasks, where each task was labeled as Not Started, In Progress, or Completed. This allowed the team to track progress and maintain organization throughout the development process.

The system was developed using Visual Studio Code as the primary development environment, utilizing web technologies such as HTML, CSS, and JavaScript. Each member worked on their assigned components, while one member was responsible for developing the

system prototype, which served as the initial working version of the system. Once completed, the prototype was shared with the team for further evaluation.

Throughout development, outputs were regularly reviewed by the team leader, and feedback was provided to ensure quality and consistency. Necessary revisions and improvements were applied based on the feedback received. This process reflects the implementation phase of the Software Development Lifecycle (SDLC), where continuous development, monitoring, and refinement are carried out to achieve the desired system functionality.

Testing

During the testing phase of the YesParenity system, the researchers followed a systematic process to evaluate the system's functionality and overall usability. Testing was conducted continuously during and after development to ensure that the implemented features were working as intended. The researchers focused on identifying errors, inconsistencies, and areas that required improvement within the system. The system prototype was reviewed by the team, where each component was checked based on its expected output and behavior. The researchers evaluated the system's interface, navigation, and responsiveness to ensure that it was user-friendly and aligned with the intended design. Feedback was also gathered through consultations with the adviser and co-adviser to further validate the system's performance and functionality.

Any issues identified during testing were documented and addressed accordingly. Necessary corrections and improvements were applied to enhance the system's reliability and usability. This process was repeated as needed, allowing the system to be gradually refined. This

reflects the testing and evaluation phase of the Software Development Lifecycle (SDLC), where the system is assessed and improved to ensure that it meets the required standards and performs effectively.

Description of Prototype

Landing Page

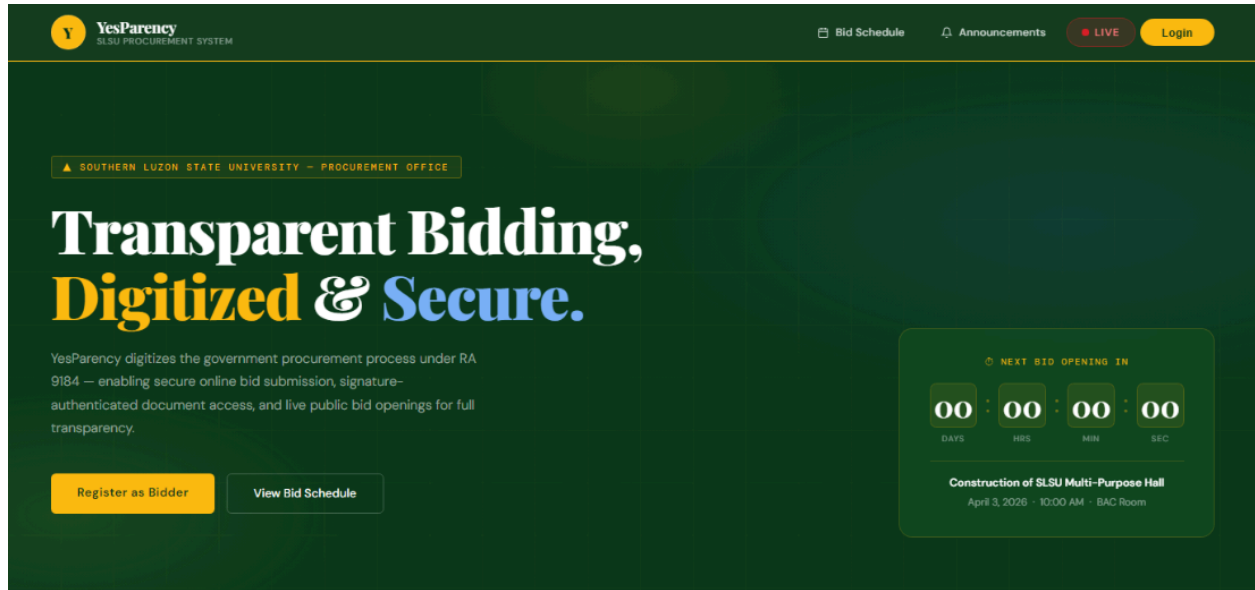


Figure 26. Landing/Information Page

Figure 26 shows the landing page which serves as the public-facing entry point of YesParenCy. The navigation bar contains the system logo and name, with links redirecting users to the Bid Schedule and Announcements pages, a Live indicator button that activates during ongoing bid opening broadcasts, and a Login button for registered users and administrators. The hero section displays the headline "Transparent Bidding, Digitized & Secure" with a brief system description and two call-to-action buttons, namely "Register as Bidder" and "View Bid Schedule," that redirect users to their respective pages. On the right side is a countdown timer widget showing the upcoming bid opening's project title, date, time, and venue, allowing public users to monitor procurement schedules without logging in.

PROCUREMENT DETAILS

Active & Upcoming Bid Openings

All procurement activities are conducted in accordance with Republic Act No. 9184. Click on any item to view full procurement details.

LIVE NOW

Supply and Delivery of Science Laboratory Equipment for College of Arts and Sciences

Opening Today — March 26, 2026 at 10:00 AM

BAC Conference Room, Admin Building

ABC: ₱2,450,000.00

Ref No: SLSU-BAC-2026-003 · PhilGEPS: 8821047

OPEN FOR SUBMISSION

Construction of New Multi-Purpose Hall at SLSU Main Campus

Bid Opening: April 3, 2026 at 10:00 AM

Submission Deadline: April 2, 2026, 5:00 PM

ABC: ₱18,750,000.00

Ref No: SLSU-BAC-2026-007 · PhilGEPS: 8834502

UPCOMING

Supply of Information Technology Equipment and Peripherals for CIT

Bid Opening: April 10, 2026 at 2:00 PM

Submission Deadline: April 9, 2026, 5:00 PM

ABC: ₱3,200,000.00

Ref No: SLSU-BAC-2026-011 · PhilGEPS: 8841289

CLOSED

Procurement of Janitorial and Maintenance Services FY 2026

Opened: March 15, 2026

Notice of Award Released

ABC: ₱1,800,000.00

Ref No: SLSU-BAC-2026-001

View All Bid Openings

Figure 27. YesParenity's Landing Page – Active and Upcoming Bid Openings Section

Figure 27 presents the Active and Upcoming Bid Openings section displayed when scrolling down the landing page. This section features procurement cards containing each project's title, bid opening date, submission deadline, Approved Budget for the Contract (ABC), and reference numbers. Each card is labeled with a status indicator such as “Live Now,” “Open for Submission,” “Upcoming,” or “Closed,” allowing public users to quickly identify the current status of each procurement activity. A “View All Bid Openings” button at the bottom redirects users to the complete bid schedule page for further details.

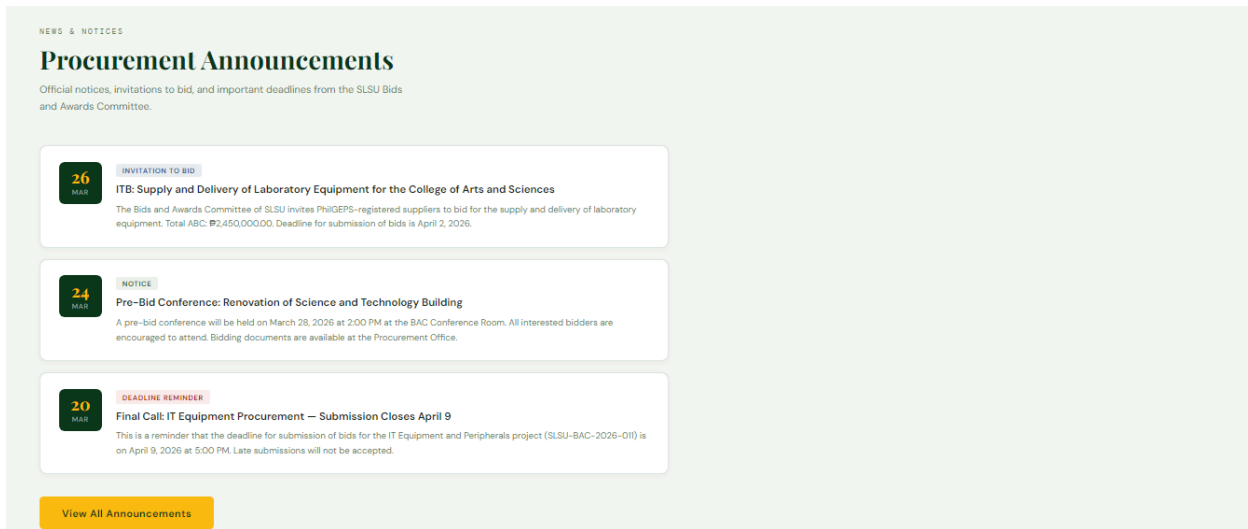


Figure 28. YesParenCy's Landing Page – Procurement Announcements Section

Figure 28 presents the Procurement Announcements section located further down the landing page, which displays official notices, invitations to bid, and important deadlines from the SLSU Bids and Awards Committee. Each announcement card includes the publication date, a category tag such as “Invitation to Bid,” “Notice,” or “Deadline Reminder,” the announcement title, and a brief description. A “View All Announcements” button at the bottom redirects users to the complete announcements page for additional information.

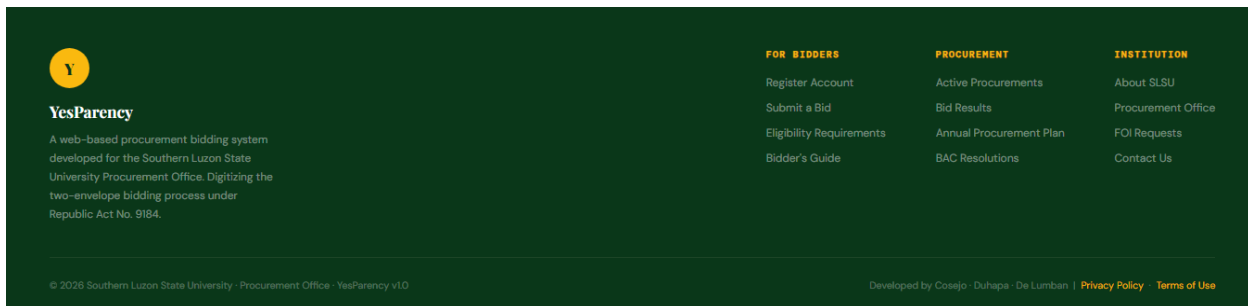


Figure 29. YesParenCy's Landing Page – Footer

Figure 29 presents the footer section located at the bottom of the landing page, which displays the YesParenthood logo and a brief system description on the left side. On the right side are three grouped link columns, namely “For Bidders,” “Procurement,” and “Institution,” which provide quick access to pages such as Register Account, Submit a Bid, Active Procurements, Bid Results, About SLSU, and Contact Us, among others. At the very bottom of the footer is a copyright notice and developer credits, along with links to the Privacy Policy and Terms of Use.

Bid Schedule Page

The screenshot displays the 'YesParenity' SLSU Procurement System interface. At the top, there's a navigation bar with a 'Bid Schedule' button, 'Announcements', a 'LIVE' indicator, and a 'Login' button. Below this, a header section titled 'Active & Upcoming Bid Openings' states that all procurement activities are conducted in accordance with Republic Act No. 9184 and provides a link to view full details. A filter bar allows users to select between 'All', 'Live Now', 'Open for Submission', 'Upcoming', and 'Closed' categories. The main content area features six procurement cards arranged in a grid. Each card includes a status tag (e.g., 'LIVE NOW', 'OPEN FOR SUBMISSION', 'UPCOMING', 'CLOSED'), a project title, a calendar icon for the bid opening date and time, a clock icon for the submission deadline, the estimated amount in ABC (Philippine Pesos), and the specific reference numbers (SLSU-BAC and PhilGEPS).

Status	Project Title	Bid Opening Date & Time	Submission Deadline	Estimated Amount (ABC)	Ref No.	PhilGEPS
LIVE NOW	Supply and Delivery of Science Laboratory Equipment for College of Arts and Sciences	March 26, 2026 at 10:00 AM	BAC Conference Room, Admin Building	₱2,450,000.00	SLSU-BAC-2026-003	8821047
OPEN FOR SUBMISSION	Construction of New Multi-Purpose Hall at SLSU Main Campus	April 3, 2026 at 10:00 AM	April 2, 2026, 5:00 PM	₱18,750,000.00	SLSU-BAC-2026-007	8834502
UPCOMING	Supply of Information Technology Equipment and Peripherals for CIT	April 10, 2026 at 2:00 PM	April 9, 2026, 5:00 PM	₱3,200,000.00	SLSU-BAC-2026-011	8841289
UPCOMING	Procurement of Office Supplies and Materials for All Departments FY 2026	April 17, 2026 at 10:00 AM	April 16, 2026, 5:00 PM	₱980,000.00	SLSU-BAC-2026-014	8849301
UPCOMING	Repair and Rehabilitation of SLSU Gymnasium Roofing	April 24, 2026 at 2:00 PM	April 23, 2026, 5:00 PM	₱5,600,000.00	SLSU-BAC-2026-016	8852774
CLOSED	Procurement of Janitorial and Maintenance Services FY 2026	Opened: March 15, 2026	Notice of Award Released	₱1,800,000.00	SLSU-BAC-2026-001	

At the bottom of the page, a footer contains the copyright notice '© 2026 Southern Luzon State University - Procurement Office - YesParenity v1.0' and a 'Back to Home' link.

Figure 30. YesParenity's Bid Schedule Page

Figure 30 presents the Bid Schedule page, which displays all active and upcoming procurement activities in a card-based layout. At the top of the page are filter buttons that allow users to sort procurements by status, namely “All,” “Live Now,” “Open for Submission,” “Upcoming,” and “Closed.” Each procurement card includes the project title, bid opening date

and time, submission deadline, Approved Budget for the Contract (ABC), and SLSU-BAC and PhilGEPS reference numbers, along with a color-coded status tag for quick identification. Users can click any card to view the full procurement details of the selected bidding activity.

Procurement Announcement Page

YesParency
SLSU PROCUREMENT SYSTEM

Bid Schedule Announcements LIVE Login

NEWS & NOTICES

Procurement Announcements

Official notices, invitations to bid, and important deadlines from the SLSU Bids and Awards Committee.

All Invitation to Bid Notices Deadlines Awards

26
MAR

INVITATION TO BID

ITB: Supply and Delivery of Laboratory Equipment for the College of Arts and Sciences

The Bids and Awards Committee of SLSU invites PhilGEPS-registered suppliers to bid for the supply and delivery of laboratory equipment. Total ABC: ₱2,450,000.00. Deadline for submission of bids is April 2, 2026.

12
MAR

INVITATION TO BID

ITB: Construction of New Multi-Purpose Hall at SLSU Main Campus

SLSU invites eligible contractors to bid for the construction of a new multi-purpose hall. Total ABC: ₱18,750,000.00. Pre-bid conference on March 20, 2026. Bid opening on April 3, 2026 at 10:00 AM.

© 2026 Southern Luzon State University - Procurement Office - YesParency v1.0

← Back to Home

Figure 31. YesParency's Procurement Announcements Page

Figure 31 presents the Procurement Announcements page, which displays all official notices, invitations to bid, and important deadlines from the SLSU Bids and Awards Committee. Filter buttons at the top of the page allow users to sort announcements by category, namely “All,” “Invitation to Bid,” “Notices,” “Deadlines,” and “Awards.” Each announcement card

includes the publication date, category tag, title, and a brief description of the procurement notice. Users can click any card to view the full details of the selected announcement.

Live Bid Opening Page

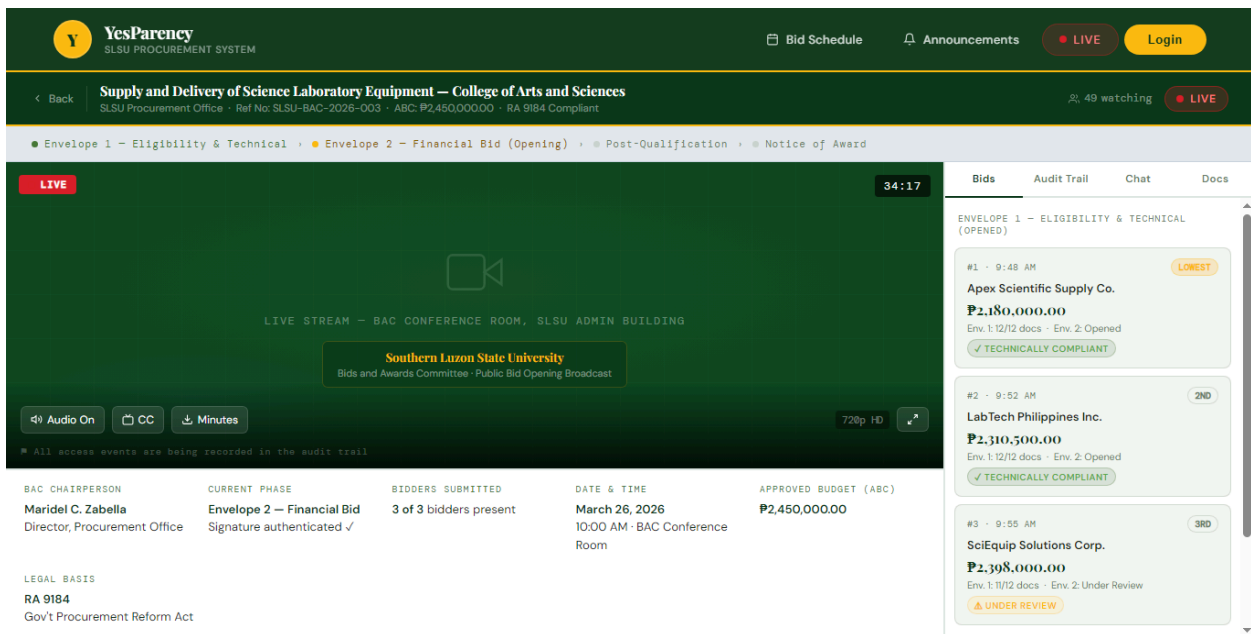


Figure 32. YesParenCy's Live Bid Opening Page

Figure 32 presents the Live Bid Opening page, which provides a real-time public broadcast of the bid opening proceedings conducted by the Bids and Awards Committee. The page displays the procurement title, reference number, Approved Budget for the Contract (ABC), RA 9184 compliance status, viewer count, and a Live indicator, along with a progress bar showing the current phase of the process. The main section contains the live stream feed from the BAC Conference Room with controls for audio, closed captions, and minutes download, while the right panel includes tabs such as Bids, Audit Trail, Chat, and Docs, where the Bids tab

shows each bidder's submitted bid amount, document status, and compliance results in ranked order.

Login Modal

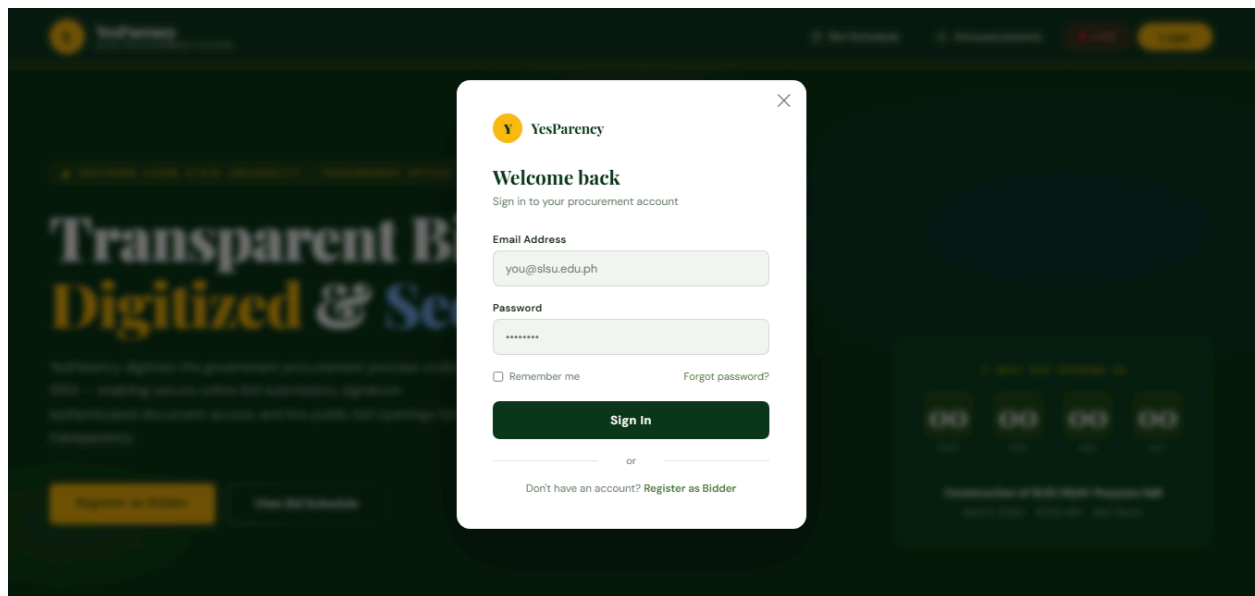


Figure 33. YesParency's Login Modal

Figure 33 presents the login modal, which appears as an overlay on the landing page when the Login button is clicked. It contains fields for email address and password, a “Remember me” checkbox, and a “Forgot password?” link. A “Sign In” button submits the credentials, while a “Register as Bidder” link at the bottom redirects users without an account to the registration page.

Bidder Registration Page

YesParenity
SLSU PROCUREMENT SYSTEM

Bid Schedule Announcements **LIVE** Login

1 2 3 4 5
ACCOUNT BUSINESS DOCUMENTS VERIFY REVIEW

STEP 1 OF 5
Create Your Account
Set up your login credentials for the YesParenity bidder portal.

First Name * Last Name *
Juan Dela Cruz

Email Address *
juan@company.com
A verification code will be sent to this email.

Password * Confirm Password *
Min. 8 characters Re-enter password

Mobile Number * Role / Position
+63 9XX XXX XXXX e.g. Procurement Officer

ⓘ Your account will be linked to your business entity in the next step. Only PhilGEPS-registered suppliers are eligible to submit bids under RA 9184.

← Back Step 1 of 5 Continue →

Figure 34. YesParenity's Bidder Registration Page – Step 1: Account

Figure 34 presents the Bidder Registration page, which follows a five-step process indicated by a progress bar at the top: Account, Business, Documents, Verify, and Review. Step 1 requires bidders to enter their first name, last name, email address, password, mobile number, and role or position. A notice at the bottom reminds users that only PhilGEPS-registered suppliers are eligible to submit bids under RA 9184, and a “Continue” button proceeds to the next step of the registration process.

YesParency
SLSU PROCUREMENT SYSTEM

Bid Schedule Announcements **LIVE** Login

1 ACCOUNT 2 **BUSINESS** 3 DOCUMENTS 4 VERIFY 5 REVIEW

STEP 2 OF 5
Business Information
Provide your company or organization details as registered with PhilGEPS.

Registered Business Name *
e.g. Apex Scientific Supply Co.

PhilGEPS Registration No. * **TIN (Tax Identification No.) ***
e.g. 1234567 XXX-XXX-XXX-XXX
Required for bid submission under RA 9184.

Business Type * **Year Established**
Select type... e.g. 2010

Registered Business Address *
Unit/Floor, Building, Street, City, Province, ZIP Code

Business Email **Business Phone**
info@company.com +63 2 XXXX XXXX

Nature of Business / Products & Services
Briefly describe what your company supplies or provides...

← Back Step 2 of 5 Continue →

Figure 35. YesParency's Bidder Registration Page – Step 2: Business Information

Figure 35 presents Step 2 of the Bidder Registration page, which collects the bidder's company or organization details as registered with PhilGEPS. The form requires the registered business name, PhilGEPS registration number, Tax Identification Number, business type, year established, registered business address, business email, business phone, and a brief description

of the nature of the business. A “Continue” button proceeds to the next step, while a “Back” button returns to the previous step.

Y

YesParenCy
SLSU PROCUREMENT SYSTEM

Bid Schedule

Announcements

LIVE

Login

✓

✓

3

4

5

ACCOUNT

BUSINESS

DOCUMENTS

VERIFY

REVIEW

STEP 3 OF 5

Eligibility Documents

Upload the required legal and eligibility documents. Accepted formats: PDF, JPG, PNG (max 5MB each).

⚠

All documents must be valid and current. Expired or falsified documents will result in disqualification and may be subject to legal action under RA 9184.

DTI / SEC / CDA Registration Certificate *

↑

Click to upload or drag & drop

PDF, JPG, PNG · Max 5MB

Mayor's / Business Permit (current year) *

↑

Click to upload or drag & drop

PDF, JPG, PNG · Max 5MB

BIR Certificate of Registration (Form 2303) *

↑

Click to upload or drag & drop

PDF, JPG, PNG · Max 5MB

PhilGEPS Certificate of Registration *

↑

Click to upload or drag & drop

PDF, JPG, PNG · Max 5MB

Valid Government-Issued ID of Authorized Representative *

↑

Click to upload or drag & drop

PDF, JPG, PNG · Max 5MB

← Back

Step 3 of 5

Continue >

Figure 36. YesParenity's Bidder Registration Page – Step 3: Eligibility Documents

Figure 36 presents Step 3 of the Bidder Registration page, which requires bidders to upload the necessary legal and eligibility documents. These include the DTI/SEC/CDA Registration Certificate, Mayor's or Business Permit for the current year, BIR Certificate of Registration (Form 2303), PhilGEPS Certificate of Registration, and a valid government-issued ID of the Authorized Representative. Each document has a drag-and-drop upload area supporting PDF, JPG, and PNG formats, with a maximum file size of 5MB. A warning at the top reminds bidders that all documents must be valid and current, as expired or falsified documents may lead to disqualification under RA 9184.

The screenshot displays the 'Email Verification' step of the bidder registration process. At the top, a dark green header contains the 'YesParency SLSU PROCUREMENT SYSTEM' logo, navigation links for 'Bid Schedule' and 'Announcements', and a 'LIVE' status indicator with a 'Login' button. Below the header, a progress bar shows five steps: ACCOUNT, BUSINESS, DOCUMENTS, VERIFY (current step), and REVIEW. The main content area features a dark green box with the title 'Email Verification' and the instruction 'Enter the 6-digit code sent to your registered email address.' Below this, an email icon is shown with the text 'Code sent to your email' and a note to 'Check your inbox and spam folder. Code expires in 09:54'. A row of six input boxes is provided for the code. A 'Resend Code' link is available below the input boxes. A light blue information box at the bottom states: 'This verification confirms your email address. Your account will also undergo manual review by the SLSU Procurement Office before full bidder access is granted.' At the very bottom, there are 'Back' and 'Verify & Continue' buttons, with 'Step 4 of 5' centered between them.

Figure 37. YesParency's Bidder Registration Page – Step 4: Email Verification

Figure 37 presents Step 4 of the Bidder Registration page, which requires bidders to verify their email address by entering the 6-digit code sent to their registered email. The code expires within a set time displayed by a countdown timer, and a “Resend Code” option is available if the code is not received. A notice at the bottom informs bidders that, in addition to email verification, the account will undergo manual review by the SLSU Procurement Office before full bidder access is granted.

YesParency
SLSU PROCUREMENT SYSTEM

Bid Schedule Announcements **LIVE** Login

ACCOUNT BUSINESS DOCUMENTS VERIFY REVIEW

STEP 5 OF 5
Review & Declaration
Read and confirm the following declarations before submitting your registration.

☐ **Accuracy of Information**
I certify that all information and documents submitted are true, accurate, and complete. I understand that any false declaration may result in disqualification and legal action.

☐ **PhilGEPS Registration Compliance**
I confirm that my business is registered with PhilGEPS and that the registration is current and valid, as required under Republic Act No. 9184.

☐ **No Conflict of Interest**
I declare that I have no direct or indirect conflict of interest with any SLSU official or procurement officer involved in the bidding process.

☐ **Terms of Use & Privacy Policy**
I have read and agree to the YesParency Terms of Use and Privacy Policy, including the collection and processing of my personal and business data for procurement purposes.

☐ **Audit Trail Consent**
I acknowledge that all my actions within the YesParency system — including document submissions, access events, and signature authentications — will be recorded in an audit trail for accountability purposes.

← Back Step 5 of 5 Submit Registration ✓

Figure 38. YesParency's Bidder Registration Page – Step 5: Review and Declaration

Figure 38 presents Step 5 of the Bidder Registration page, the final step of the process, where bidders must read and confirm five declarations before submitting their registration. The declarations cover Accuracy of Information, PhilGEPS Registration Compliance, No Conflict of Interest, Terms of Use and Privacy Policy, and Audit Trail Consent, each acknowledged by checking the corresponding checkbox. Once all declarations are confirmed, the bidder can click the “Submit Registration” button to complete the registration process.

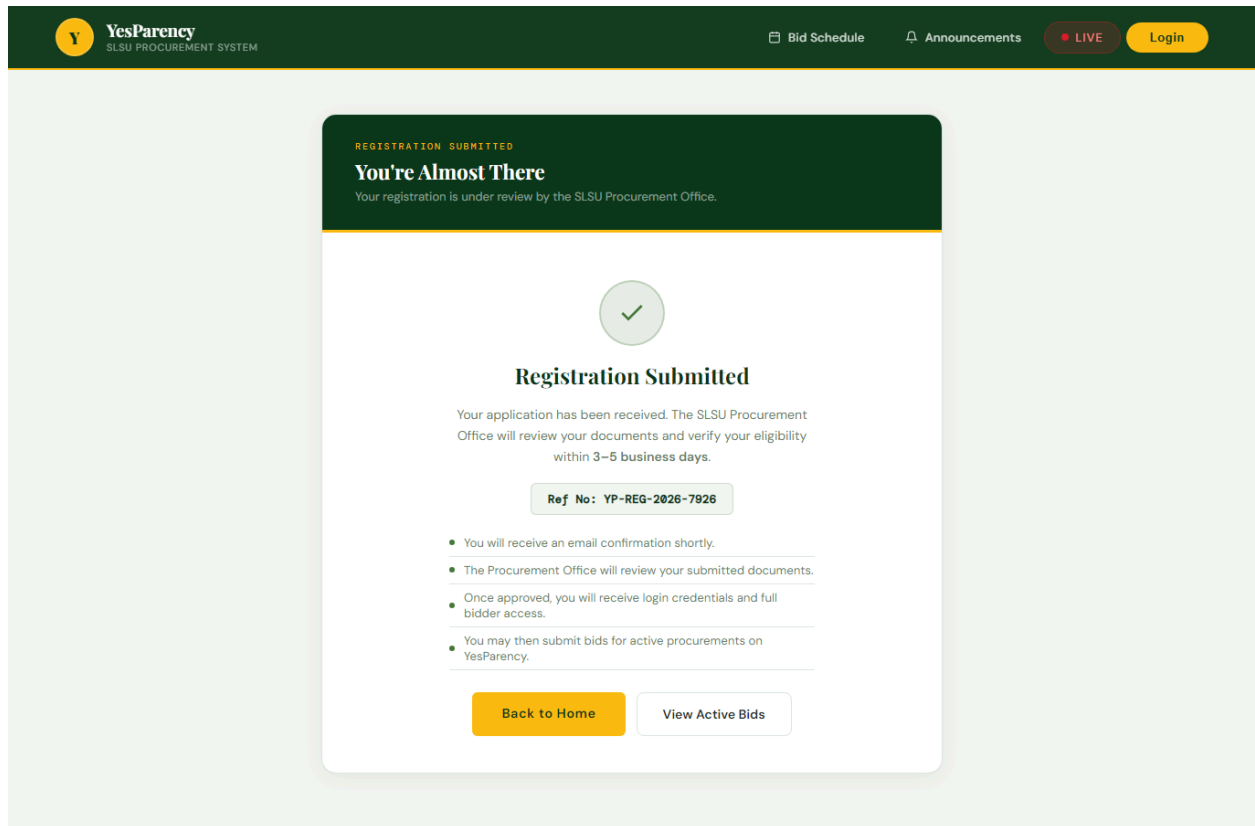


Figure 39. YesParency's Bidder Registration – Registration Submitted Page

Figure 39 presents the registration confirmation screen, displayed after completing all five steps, which informs the bidder that their registration has been successfully submitted and is under review by the SLSU Procurement Office. A reference number is provided for tracking purposes, along with a summary of the next steps, including receiving an email confirmation, document review by the Procurement Office, and obtaining login credentials and full bidder access once approved. Buttons for “Back to Home” and “View Active Bids” are provided at the bottom.

Bidder Dashboard Page

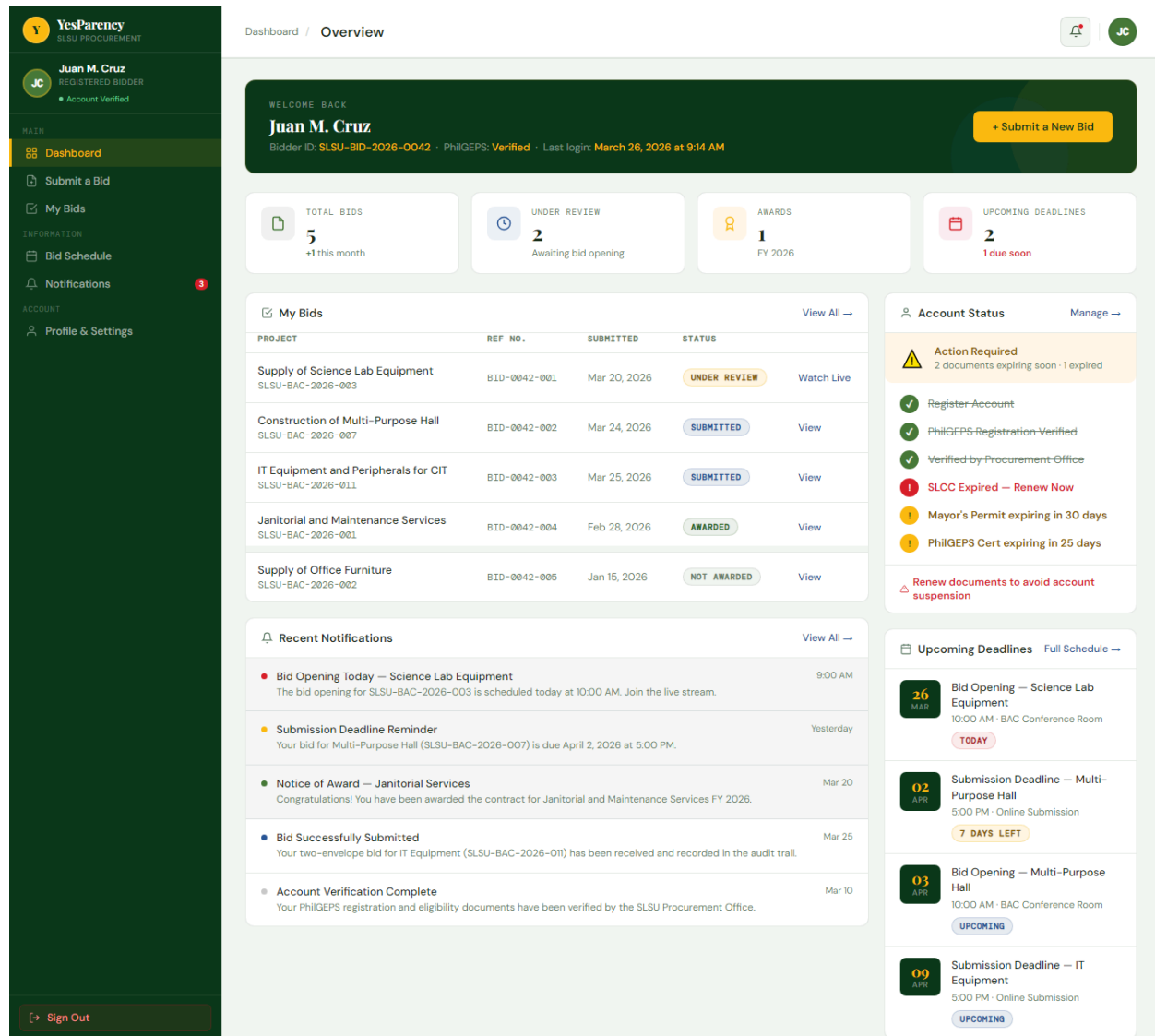


Figure 40. YesParency's Bidder Dashboard

Figure 40 presents the Bidder Dashboard, which serves as the main hub for registered bidders after logging in. The top of the dashboard displays the bidder's name, Bidder ID, PhilGEPS verification status, and last login details, along with a "Submit a New Bid" button. Summary cards show the total bids submitted, bids under review, awards received, and upcoming

deadlines. The main area contains the My Bids table, listing each submitted bid with its project name, reference number, submission date, and status. A Recent Notifications panel displays system alerts, while the Account Status panel shows document compliance with alerts for expiring or expired documents. The Upcoming Deadlines panel lists scheduled bid openings and submission deadlines with their respective dates and venues.

Submit a Bid Page

The screenshot shows the 'Submit a Bid' page on the YesParency SLSU Procurement portal. The user is Juan M. Cruz, a registered bidder with a verified account. The page is divided into a left sidebar with navigation links (Dashboard, Submit a Bid, My Bids, Bid Schedule, Notifications, Profile & Settings) and a main content area. The main content area has a progress bar with five steps: 1. Verify Receipt (active), 2. Select Project, 3. Select Lots, 4. Bid Details, and 5. Review & Submit. The 'Verify Receipt' step is highlighted, showing a green box with the title 'Verify Your Bid Documents Receipt'. Below this, a notice states that bid documents are available at the SLSU Procurement Office, Admin Building during office hours (8:00 AM – 5:00 PM, Mon–Fri). The receipt reference number is printed on the official receipt (OR). A form field labeled 'Official Receipt Reference Number' is shown with the example 'E.G. OR-2026-004821' and a 'Verify' button. A note below the form states that each receipt number is valid for one bid submission only and is tied to a specific procurement project. At the bottom, there is a link to 'Continue' and a note for users who don't have a receipt to visit the Procurement Office.

YesParency
SLSU PROCUREMENT

Dashboard / Submit a Bid

Juan M. Cruz
REGISTERED BIDDER
Account Verified

MAIN

- Dashboard
- Submit a Bid
- My Bids

INFORMATION

- Bid Schedule
- Notifications

ACCOUNT

- Profile & Settings

Sign Out

1 Verify Receipt

2 Select Project

3 Select Lots

4 Bid Details

5 Review & Submit

STEP 1 OF 5 – BID DOCUMENT RECEIPT

Verify Your Bid Documents Receipt

Before submitting a bid, you must purchase the official Bid Documents from the SLSU Procurement Office and enter the reference number from your official receipt.

ⓘ Bid Documents are available at the SLSU Procurement Office, Admin Building during office hours (8:00 AM – 5:00 PM, Mon–Fri). The receipt reference number is printed on your Official Receipt (OR).

Official Receipt Reference Number *

E.G. OR-2026-004821

Verify

ⓘ Each receipt number is valid for one bid submission only and is tied to a specific procurement project. Ensure your receipt corresponds to the project you intend to bid on.

Don't have a receipt? Visit the Procurement Office.

Continue >

Figure 41. YesParency's Submit a Bid Page – Step 1: Verify Receipt

Figure 41 presents Step 1 of the Bid Submission process, which requires the bidder to verify their Official Receipt reference number before proceeding. This confirms that the official bid documents have been purchased from the SLSU Procurement Office. A notice informs the bidder that each receipt number is valid for one bid submission only and is tied to a specific procurement project. Once the reference number is entered and verified, the bidder can proceed to the next step.

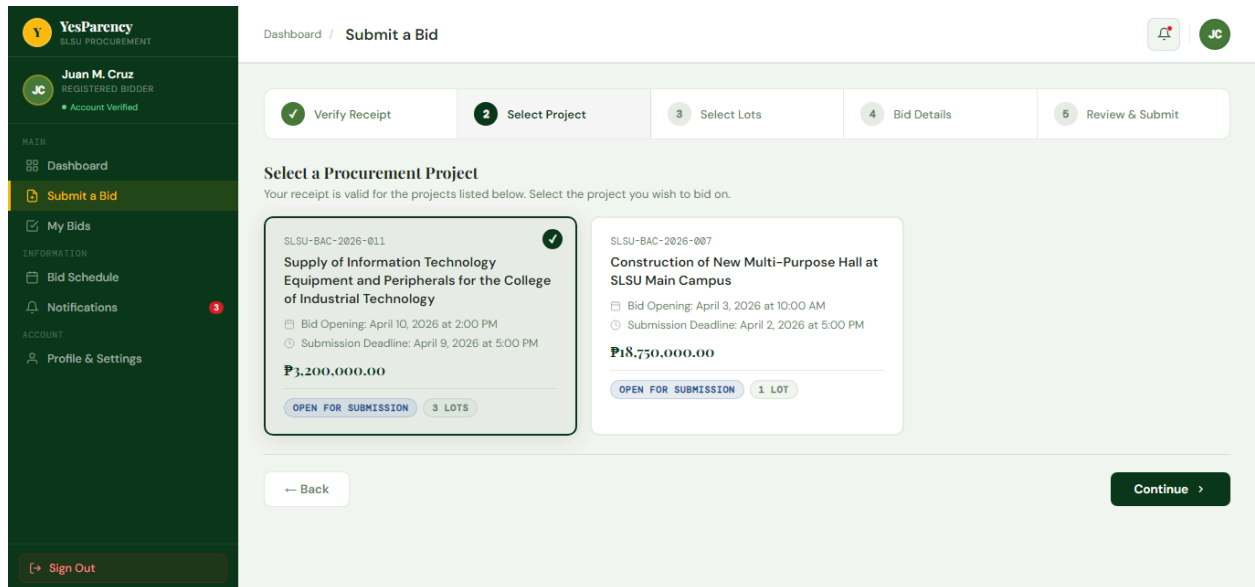


Figure 42. YesParency's Submit a Bid Page – Step 2: Select Project

Figure 42 presents Step 2 of the Bid Submission process, which displays the procurement projects associated with the verified receipt, allowing the bidder to select the project they wish to bid on. Each project card shows the reference number, project title, bid opening date, submission deadline, ABC, status, and number of lots available. The bidder selects the applicable project before proceeding to the next step.

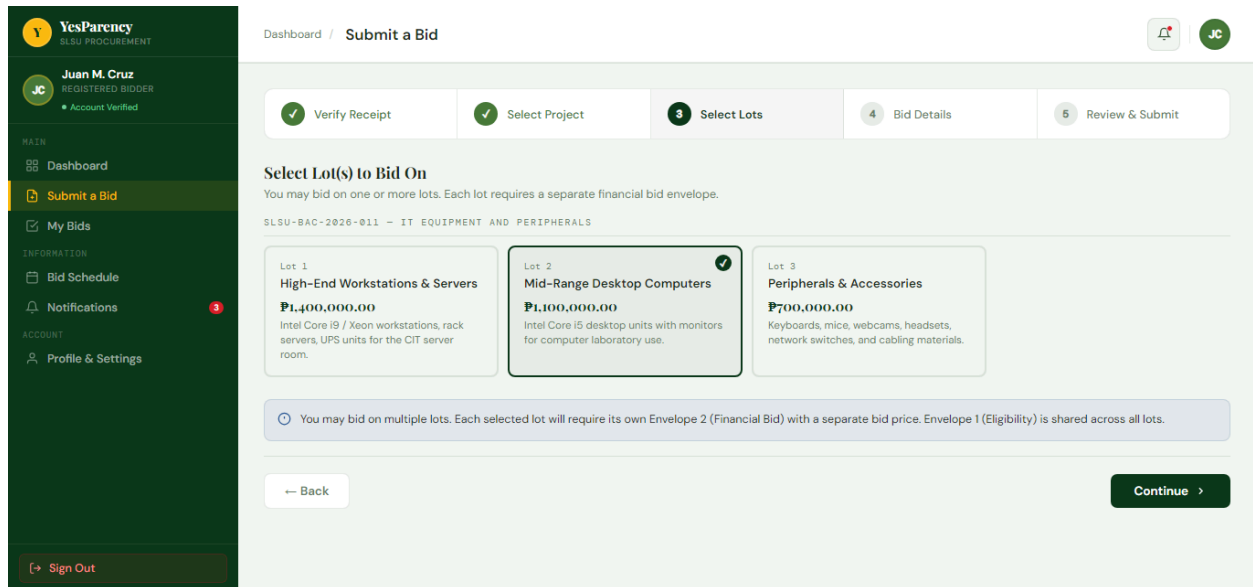


Figure 43. YesParency's Submit a Bid Page – Step 3: Select Lots

Figure 43 presents Step 3 of the Bid Submission process, which allows the bidder to select the lot or lots they wish to bid on within the chosen procurement project. Each lot card displays the lot number, title, ABC, and a brief description of the items included. A notice informs the bidder that each selected lot requires its own Envelope 2 (Financial Bid) with a separate bid price, while Envelope 1 (Eligibility) is shared across all lots. The bidder may select one or more lots before proceeding to the next step.

YesParenCy
SLSU PROCUREMENT

Juan M. Cruz
REGISTERED BIDDER
Account Verified

Dashboard / Submit a Bid

Verify Receipt Select Project Select Lots **4 Bid Details** 5 Review & Submit

Envelope 1 — Eligibility & Technical Documents

ENVELOPE 1 • ELIGIBILITY & TECHNICAL

Legal Eligibility Documents *

Upload PDF files
DTI/SEC, Mayor's Permit, BIR Form 2303, PhilGEPS Cert

Technical Specifications Compliance *

Upload PDF files
Technical specs, brochures, compliance statements

Statement of Ongoing & Completed Contracts

Upload PDF
SLCC / Audited Financial Statements

\$ Envelope 2 — Financial Bid

\$ ENVELOPE 2 • LOT 2 — MID-RANGE DESKTOPS

Total Bid Price (PHP) *

0.00

ABC: ₱1,100,000.00 — Bids exceeding the ABC will be disqualified.

Bid Security Form *

Select type...

Bid Security Amount (PHP) *

Min. 2% of bid price

Minimum 2% of your total bid price as required under RA 9184 IRR.

Financial Bid Document (Signed) *

Upload signed Financial Bid Form (PDF)
Must be signed by authorized representative

BID SUMMARY

Project Supply of IT Equipment and Peripherals f...

Ref No. SLSU-BAC-2026-011

Receipt No. 1

Selected Lots Lot 2

Total ABC for Selected Lots
₱1,100,000.00

Once submitted, your bid is locked. To request an amendment, you must submit a formal Edit Request to the BAC before the submission deadline.

Back Review Bid Sign Out

Figure 44. YesParenCy's Submit a Bid Page – Step 4: Bid Details

Figure 44 presents Step 4 of the Bid Submission process, where the bidder submits the two-envelope bid documents. Envelope 1 requires uploading Legal Eligibility Documents, Technical Specifications Compliance files, and a Statement of Ongoing and Completed Contracts. Envelope 2 requires entering the Total Bid Price, selecting the Bid Security Form type, entering the Bid Security Amount, and uploading the signed Financial Bid Document. A Bid Summary panel on the right displays the selected project, reference number, receipt number,

selected lot, and total ABC. A notice informs the bidder that once submitted, the bid is locked, and any amendments require a formal Edit Request to the BAC before the submission deadline.

Y

YesParenCy

SLSU PROCUREMENT

JC

Juan M. Cruz

REGISTERED BIDDER

Account Verified

MAIN

Dashboard

Submit a Bid

My Bids

INFORMATION

Bid Schedule

Notifications

ACCOUNT

Profile & Settings

Sign Out

Dashboard / Submit a Bid

Verify Receipt

Select Project

Select Lots

Bid Details

5 Review & Submit

Review Your Bid Before Submitting

PROCUREMENT DETAILS

Project

Supply of IT Equipment and Peripherals for CIT

Ref No.

SLSU-BAC-2026-011

Bid Opening

April 10, 2026

Deadline

April 9, 2026 at 5:00 PM

BIDDER INFORMATION

Bidder

Juan M. Cruz

Bidder ID

SLSU-BID-2026-0042

Receipt No.

1

PhilGEPS

Verified ✓

SELECTED LOTS & BID PRICES

Lot 2 — Mid-Range Desktops

(not entered)

Declaration & Certification

☐ I certify that all information and documents submitted in this bid are true, accurate, and complete. I understand that any false declaration may result in disqualification and legal action under RA 9184.

☐ I confirm that the bid prices submitted in Envelope 2 are firm and valid for the period stated in the Bidding Documents, and that I have not colluded with any other bidder.

☐ I acknowledge that once submitted, my bid is **locked and cannot be modified** without a formal Edit Request approved by the BAC before the submission deadline.

☐ I understand that this submission will be recorded in the YesParenCy audit trail with a timestamp, and that the BAC will open my bid during the scheduled public bid opening.

Back

Lock & Submit Bid

Figure 45. YesParenCy's Submit a Bid Page – Step 5: Review and Submit

Figure 45 presents Step 5 of the Bid Submission process, which provides a final summary of the bid before submission. It displays the procurement details, bidder information, and selected lots with bid prices. The bidder must confirm four Declaration and Certification checkboxes before finalizing the submission. A “Lock and Submit Bid” button at the bottom encrypts and submits the bid to the system.

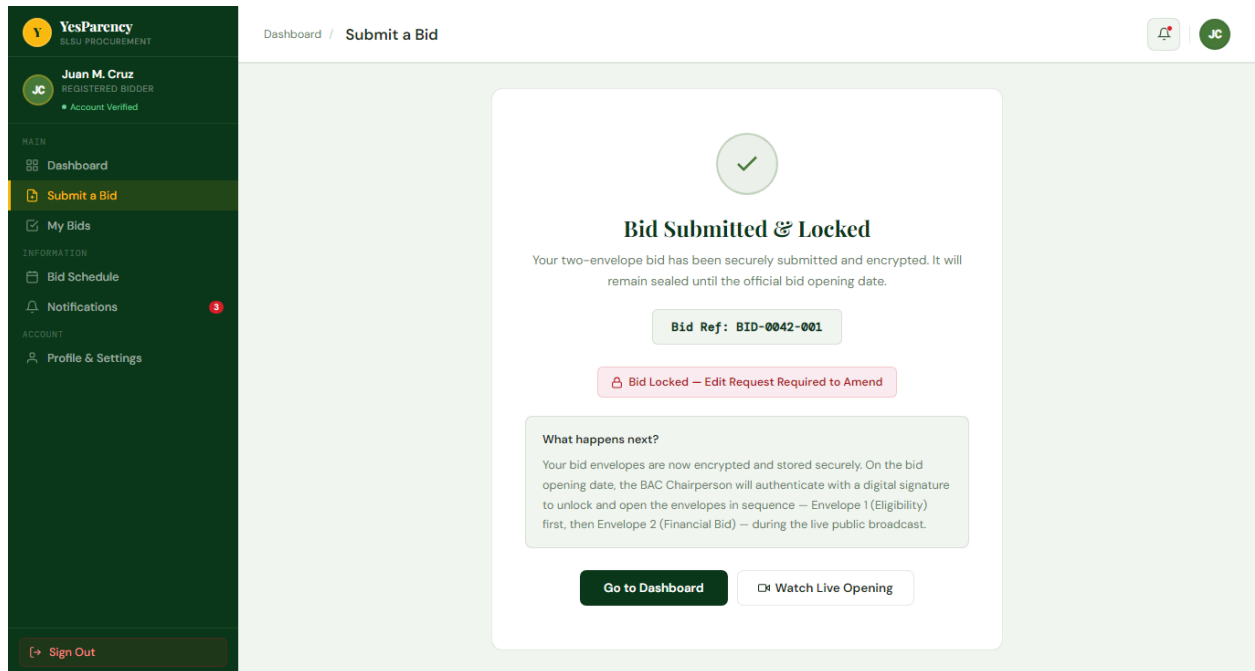


Figure 46. YesParency's Bid Submitted and Locked Confirmation Page

Figure 46 presents the bid submission confirmation screen, which displays the bid reference number and informs the bidder that their two-envelope bid has been securely encrypted and will remain sealed until the official bid opening date. A brief explanation of the next steps is provided, along with buttons for “Go to Dashboard” and “Watch Live Opening.”

My Bids Page

The screenshot displays the 'My Bids' page on the YesParency SLSU Procurement platform. The user, Juan M. Cruz, is logged in as a Registered Bidder. The page features a sidebar with navigation options: Dashboard, Submit a Bid, My Bids (selected), Bid Schedule, Notifications, and Profile & Settings. The main content area shows a list of four bids, each with its reference number, status, and a 'Locked' indicator. The bids are:

- BID-0042-001**: UNDER REVIEW, LOCKED. Project: Supply of Information Technology Equipment and Peripherals for CIT. Submitted: March 25, 2026 at 4:47 PM. 2 Lots. Lot 1: High-End Workstations (P1,350,000.00), Lot 2: Mid-Range Desktops (P1,050,000.00). Actions: View Details, Watch Live, Request Edit.
- BID-0042-002**: SUBMITTED, LOCKED, EDIT REQUEST PENDING. Project: Construction of New Multi-Purpose Hall at SLSU Main Campus. Submitted: March 24, 2026 at 3:22 PM. 1 Lot. Lot 1: Full Construction (P18,200,000.00). Actions: View Details, Request Edit.
- BID-0042-003**: AWARDED, LOCKED. Project: Procurement of Janitorial and Maintenance Services FY 2026. Submitted: February 28, 2026 at 2:10 PM. 1 Lot. Lot 1: Janitorial Services (P1,720,000.00). Action: View Details.
- BID-0042-004**: NOT AWARDED, LOCKED. Project: Supply of Office Furniture and Fixtures. Submitted: January 15, 2026 at 1:45 PM. 1 Lot. Lot 1: Office Furniture (P980,000.00). Action: View Details.

Figure 47. YesParency's My Bids Page

Figure 47 presents the My Bids page, which displays all bids submitted by the bidder in a list format. Each entry shows the bid reference number, project title, submission date and time, number of lots, selected lots with corresponding bid prices, and status tags such as “Submitted,” “Under Review,” “Awarded,” and “Not Awarded,” along with a “Locked” indicator. Filter buttons at the top allow sorting by status, and a search bar helps locate specific bids. Action buttons for viewing details, watching the live opening, and requesting an edit are provided for each entry.

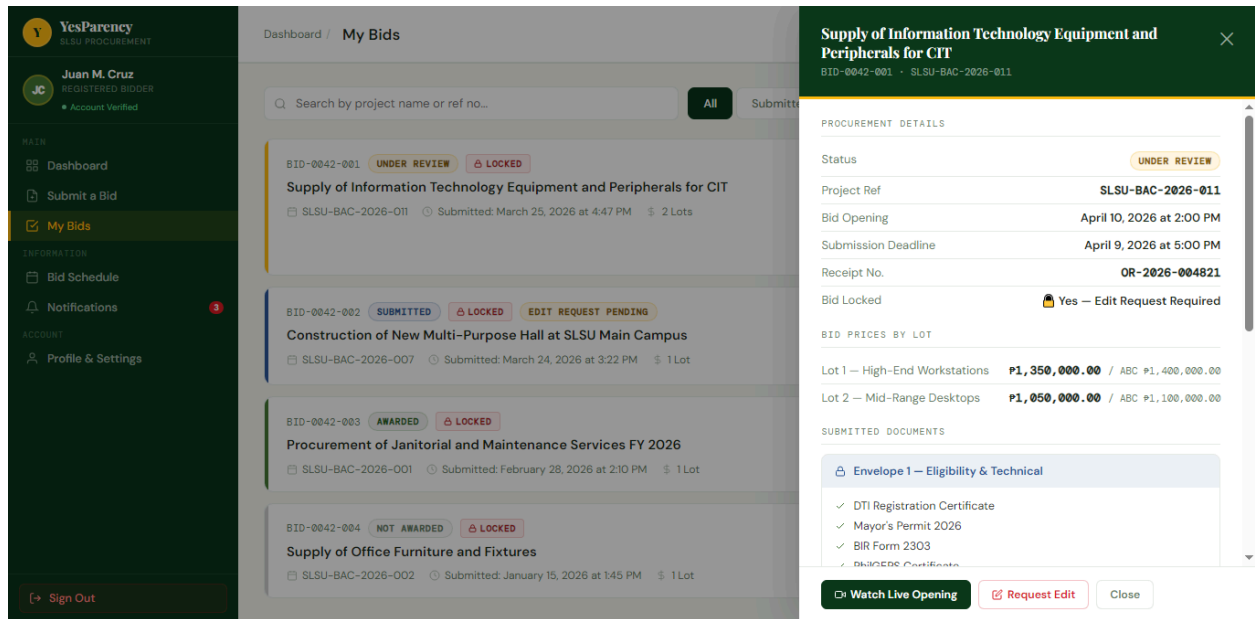


Figure 48. YesParenCy's My Bids Page – View Details Panel

Figure 48 presents the bid details side panel, which opens when the “View Details” button is clicked on a bid entry. It displays the procurement status, project reference number, bid opening date, submission deadline, receipt number, and bid lock status. The panel also shows the submitted bid prices per lot alongside their respective ABCs and lists the submitted documents under Envelope 1 for verification. Buttons for “Watch Live Opening” and “Request Edit” are available at the bottom of the panel.

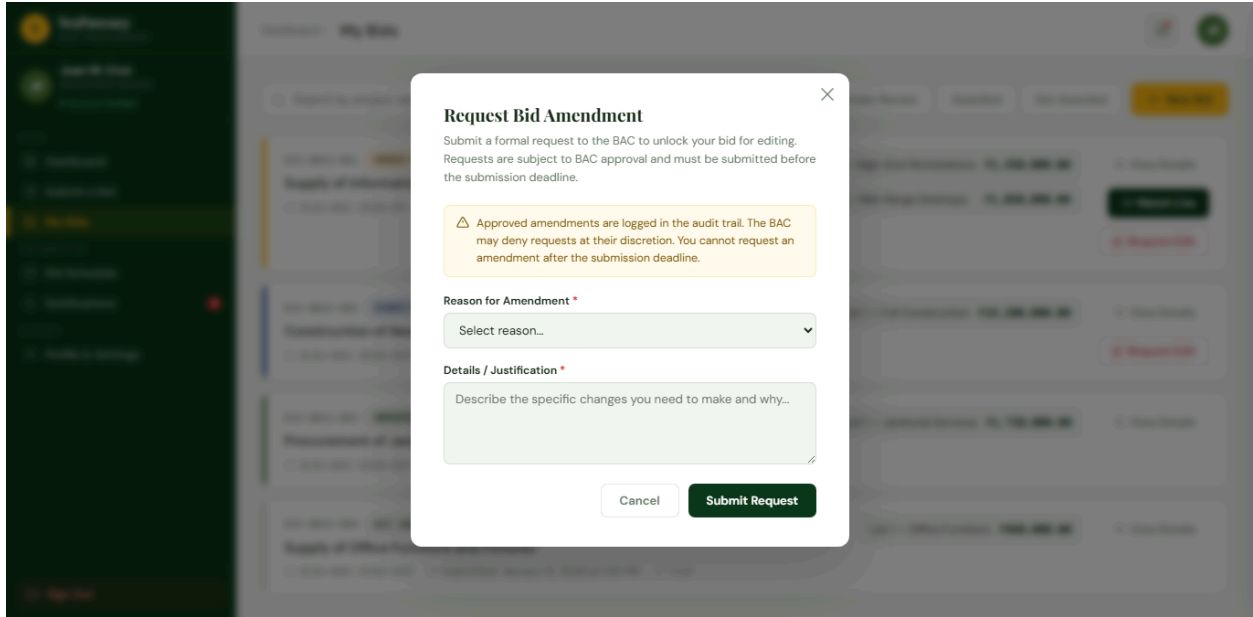


Figure 49. YesParency's Request Bid Amendment Modal

Figure 49 presents the Request Bid Amendment modal, which opens when the “Request Edit” button is clicked. It allows the bidder to submit a formal amendment request to the BAC by selecting a reason for the amendment and providing details or justification. A notice informs the bidder that approved amendments are logged in the audit trail, that the BAC may deny requests at their discretion, and that amendments cannot be requested after the submission deadline.

Bid Schedule Page

YesParenCy
SLSU PROCUREMENT

Juan M. Cruz
REGISTERED BIDDER
Account Verified

MAIN

- Dashboard
- Submit a Bid
- My Bids

INFORMATION

- Bid Schedule**
- Notifications 3

ACCOUNT

- Profile & Settings

[Sign Out](#)

Dashboard / Bid Schedule

NEXT BID OPENING

Supply of IT Equipment and Peripherals for CIT
April 10, 2026 - 2:00 PM - BAC Conference Room

03 DAYS 15 HRS 22 MIN 09 SEC

Search by project name or ref no...

All Live Now Open Upcoming Closed

26 MAR 10:00 AM	SLSU-BAC-2026-003 · Ph110EPS: 8821047 Supply and Delivery of Science Laboratory Equipment for College of Arts and Sciences Deadline: March 25, 2026 at 5:00 PM · BAC Conference Room, Admin Building LIVE NOW 1 LOT MY BID: BID-0042-001	ABC P2,450,000.00 1 lot	Details Watch Live
03 APR 10:00 AM	SLSU-BAC-2026-007 · Ph110EPS: 8834502 Construction of New Multi-Purpose Hall at SLSU Main Campus Deadline: April 2, 2026 at 5:00 PM · BAC Conference Room, Admin Building OPEN FOR SUBMISSION 1 LOT MY BID: BID-0042-002	ABC P18,750,000.00 1 lot	Details Submit Bid
10 APR 2:00 PM	SLSU-BAC-2026-011 · Ph110EPS: 8841289 Supply of Information Technology Equipment and Peripherals for the College of Industrial Technology Deadline: April 9, 2026 at 5:00 PM · BAC Conference Room, Admin Building UPCOMING 3 LOTS MY BID: BID-0042-001	ABC P3,200,000.00 3 lots	Details Submit Bid
17 APR 10:00 AM	SLSU-BAC-2026-014 · Ph110EPS: 8849301 Procurement of Office Supplies and Materials for All Departments FY 2026 Deadline: April 16, 2026 at 5:00 PM · BAC Conference Room, Admin Building UPCOMING 1 LOT	ABC P980,000.00 1 lot	Details Submit Bid
15 MAR 10:00 AM	SLSU-BAC-2026-001 · Ph110EPS: 8810022 Procurement of Janitorial and Maintenance Services FY 2026 Deadline: March 14, 2026 at 5:00 PM · BAC Conference Room, Admin Building CLOSED 1 LOT MY BID: BID-0042-004	ABC P1,800,000.00 1 lot	Details

Figure 50. YesParenCy's Bid Schedule Page (Bidder View)

Figure 50 presents the Bid Schedule page in the bidder view, which displays all procurement activities along with a countdown timer for the next upcoming bid opening. Each entry shows the project details, submission deadline, ABC, status tag, and the bidder's corresponding bid reference number when applicable. Action buttons for viewing details, submitting a bid, or watching the live opening are provided depending on each procurement's status.

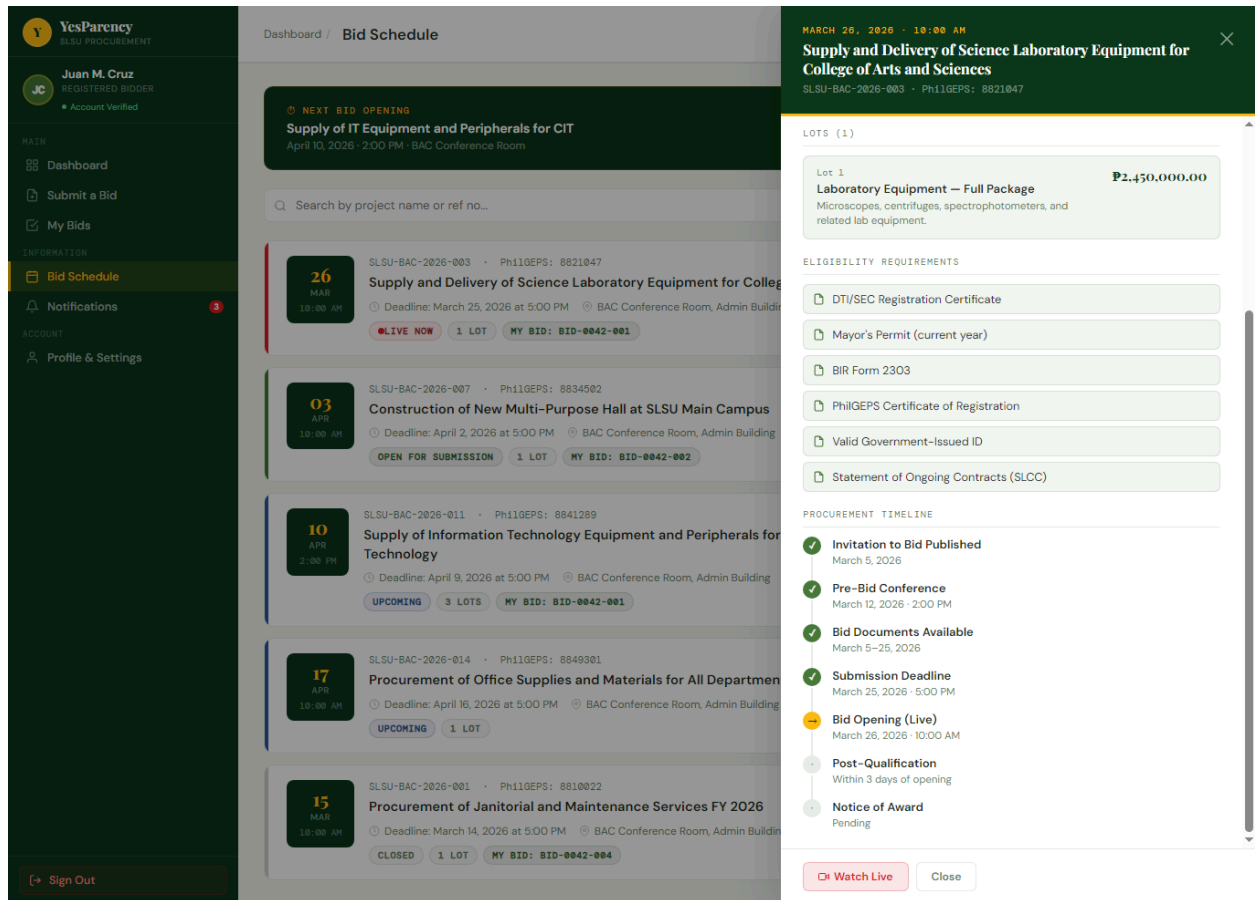


Figure 51. YesParenity's Bid Schedule Page – View Details Panel (Bidder View)

Figure 51 presents the procurement details side panel, which opens when a procurement entry is clicked. The panel displays the lot information with corresponding ABC, the eligibility requirements listing all required documents for submission, and a procurement timeline showing key milestones from the invitation to bid publication through to the notice of award. A “Watch Live” button is available at the bottom for procurements with an active live session.

Notification Page

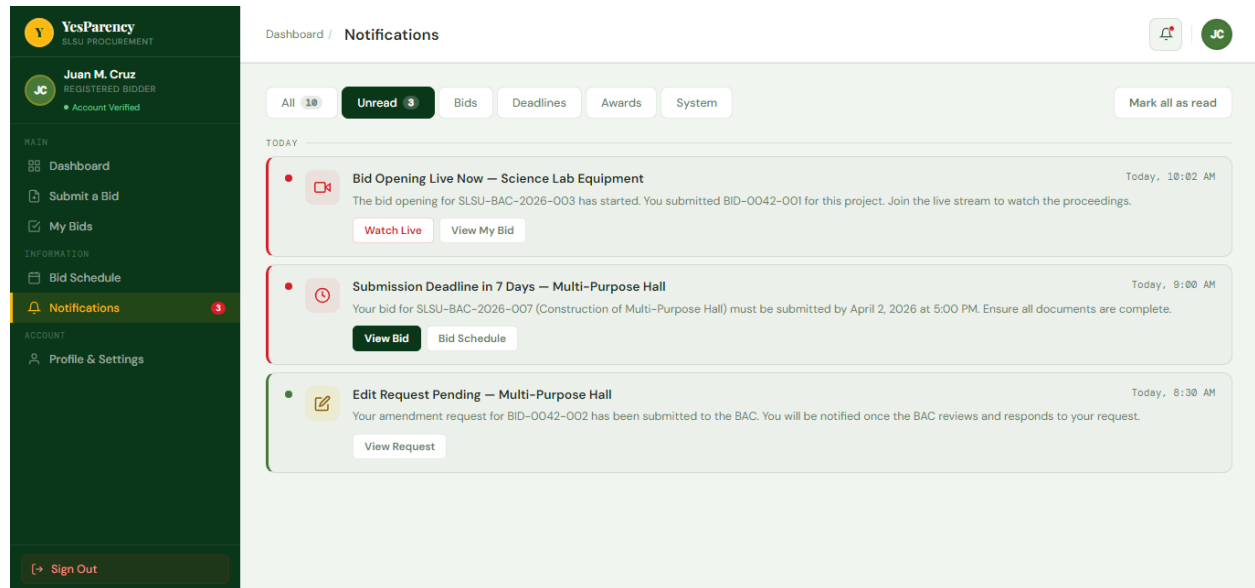
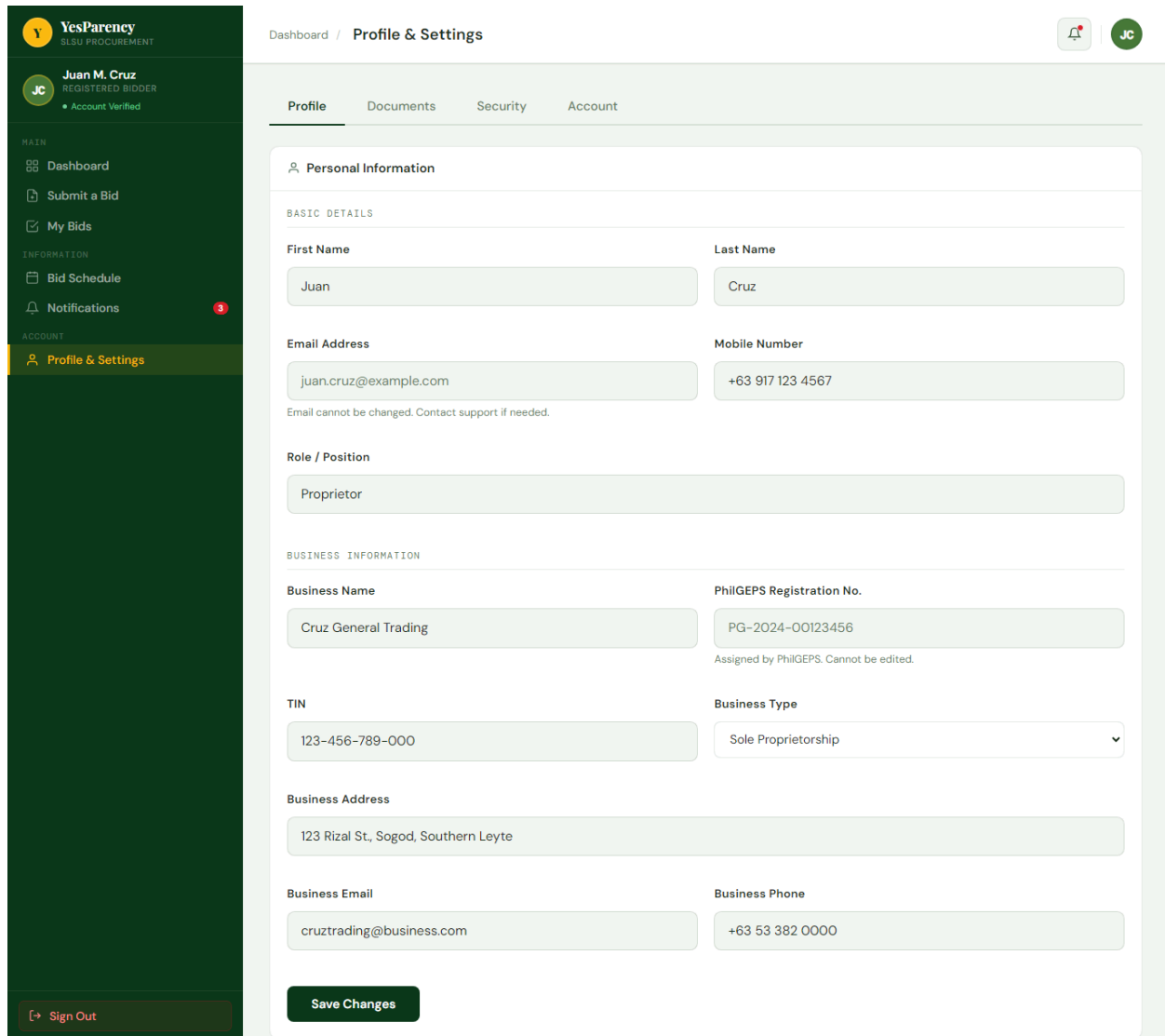


Figure 52. YesParency's Notifications Page

Figure 52 presents the Notifications page, which displays system alerts sent to the bidder, organized under tabs: “All,” “Unread,” “Bids,” “Deadlines,” “Awards,” and “System.” Each notification shows the alert type, message, timestamp, and relevant action buttons such as “Watch Live,” “View My Bid,” or “View Request.” A “Mark all as read” button is available at the top right for convenience.

Bidder's Profile & Settings Page



The screenshot displays the 'Profile & Settings' page for a bidder named Juan M. Cruz. The interface includes a dark green sidebar on the left with navigation links: Dashboard, Submit a Bid, My Bids, Bid Schedule, Notifications (with a red badge), and Profile & Settings (highlighted). The main content area has a breadcrumb 'Dashboard / Profile & Settings' and a notification bell icon. Below the breadcrumb are four tabs: Profile, Documents, Security, and Account. The 'Profile' tab is active, showing a 'Personal Information' section with 'BASIC DETAILS'. Fields include First Name (Juan), Last Name (Cruz), Email Address (juan.cruz@example.com), Mobile Number (+63 917 123 4567), and Role / Position (Proprietor). A note states 'Email cannot be changed. Contact support if needed.' The 'BUSINESS INFORMATION' section includes Business Name (Cruz General Trading), PhilGEPS Registration No. (PG-2024-00123456, with a note 'Assigned by PhilGEPS. Cannot be edited.'), TIN (123-456-789-000), Business Type (Sole Proprietorship), Business Address (123 Rizal St., Sogod, Southern Leyte), Business Email (cruztrading@business.com), and Business Phone (+63 53 382 0000). A 'Save Changes' button is at the bottom, and a 'Sign Out' button is in the sidebar.

Personal Information	
BASIC DETAILS	
First Name	Last Name
Juan	Cruz
Email Address	Mobile Number
juan.cruz@example.com	+63 917 123 4567
Email cannot be changed. Contact support if needed.	
Role / Position	
Proprietor	

BUSINESS INFORMATION	
Business Name	PhilGEPS Registration No.
Cruz General Trading	PG-2024-00123456
Assigned by PhilGEPS. Cannot be edited.	
TIN	Business Type
123-456-789-000	Sole Proprietorship
Business Address	
123 Rizal St., Sogod, Southern Leyte	
Business Email	Business Phone
cruztrading@business.com	+63 53 382 0000

Figure 53. YesParenity's Profile and Settings Page – Profile Tab

Figure 53 presents the Profile and Settings page, which allows the bidder to view and update their account information across four tabs: Profile, Documents, Security, and Account. The Profile tab displays the bidder's personal information, including name, email address, mobile number, and role, as well as business information such as business name, PhilGEPS

registration number, TIN, business type, address, and contact details. A “Save Changes” button is provided at the bottom to apply any updates.

Dashboard / Profile & Settings

Profile Documents Security Account

Document Renewal Required
2 documents are expiring within 30 days. Please upload renewed copies to avoid account suspension.

Required Documents 1 EXPIRED

DOCUMENT	STATUS	EXPIRY DATE	DAYS LEFT	ACTION
DTI Registration Certificate Last updated: Jan 10, 2025	VALID	Dec 15, 2026	264 days	Renew Document
Mayor's Permit (2026) Last updated: Jan 5, 2026	EXPIRING SOON	Apr 25, 2026	30 days	Renew Document
BIR Form 2303 Last updated: Feb 2, 2025	VALID	Dec 31, 2026	280 days	Renew Document
PhilGEPS Certificate Last updated: Apr 20, 2025	EXPIRING SOON	Apr 20, 2026	25 days	Renew Document
Valid Government-Issued ID Last updated: Mar 10, 2024	VALID	Mar 10, 2028	714 days	Renew Document
Statement of Ongoing Contracts (SLCC) Last updated: Mar 1, 2025	EXPIRED	Mar 1, 2026	EXPIRED	Renew Document

Your account will be revoked if expired documents are not renewed within 15 days of expiration. The SLSU Procurement Office will notify you before expiration.

Figure 54. YesParenency's Profile and Settings Page – Documents Tab

Figure 54 presents the Documents tab of the Profile and Settings page, which displays the status of all required eligibility documents submitted by the bidder. Each document entry shows the document name, last updated date, validity status, expiry date, days remaining, and a “Renew Document” button. Status tags indicate whether a document is “Valid,” “Expiring Soon,” or “Expired.” A warning notice at the top alerts the bidder when documents are nearing expiration, and a reminder at the bottom informs the bidder that expired documents not renewed within 15 days may result in account revocation.

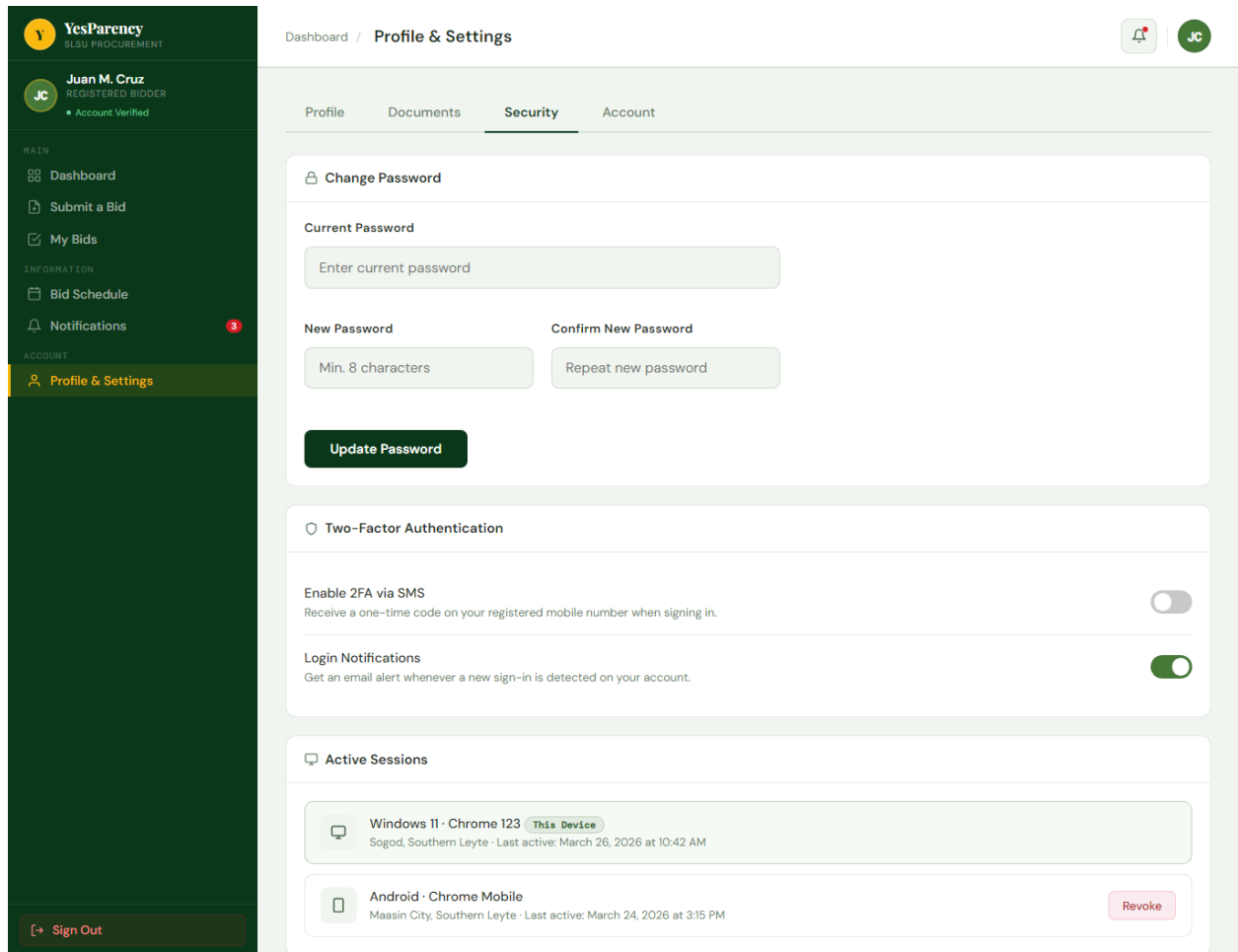


Figure 55. YesParency's Profile and Settings Page – Security Tab

Figure 55 presents the Security tab of the Profile and Settings page, which allows the bidder to manage their account security settings. It includes a Change Password section requiring the current and new password, a Two-Factor Authentication section with toggles for enabling SMS-based 2FA and login notifications, and an Active Sessions section displaying all devices currently logged into the account, with an option to revoke access for unrecognized sessions.

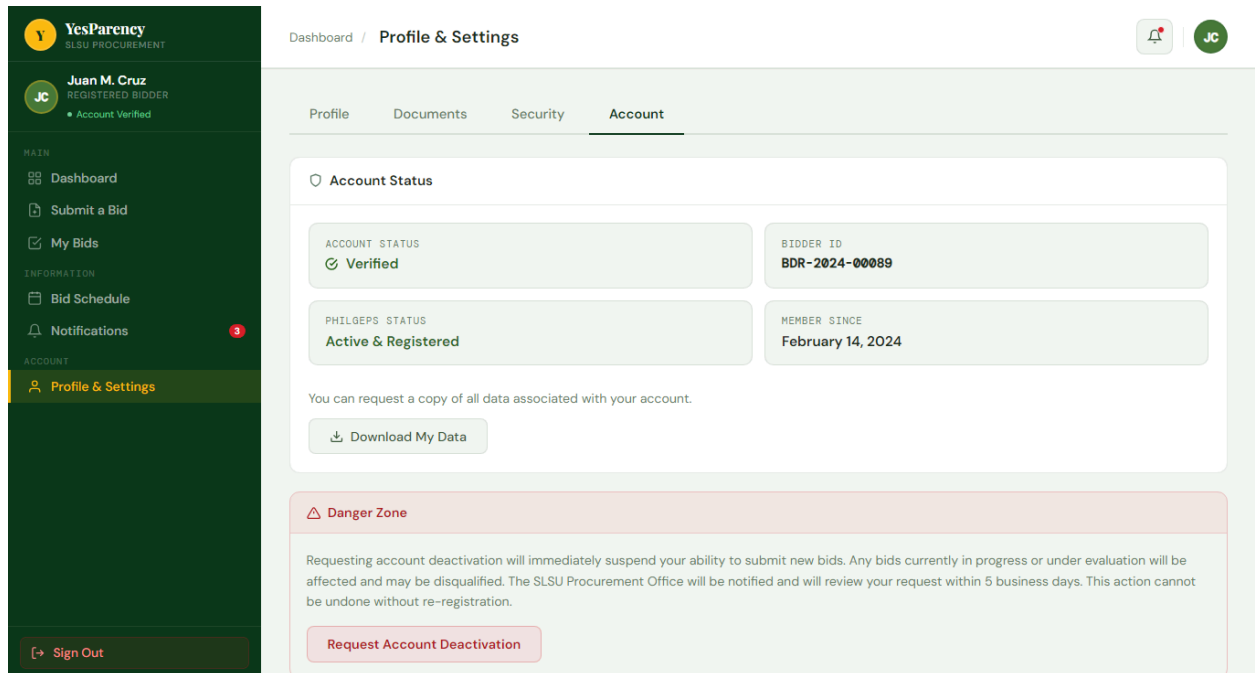


Figure 56. YesParency's Profile and Settings Page – Account Tab

Figure 56 presents the Account tab of the Profile and Settings page, which displays the bidder’s account status, Bidder ID, PhilGEPS status, and membership date. A “Download My Data” button allows the bidder to request a copy of all account-associated data. At the bottom, a Danger Zone section provides a “Request Account Deactivation” option, with a warning that deactivation will immediately suspend bid submission privileges and may affect bids currently under evaluation.

Admin Dashboard Page

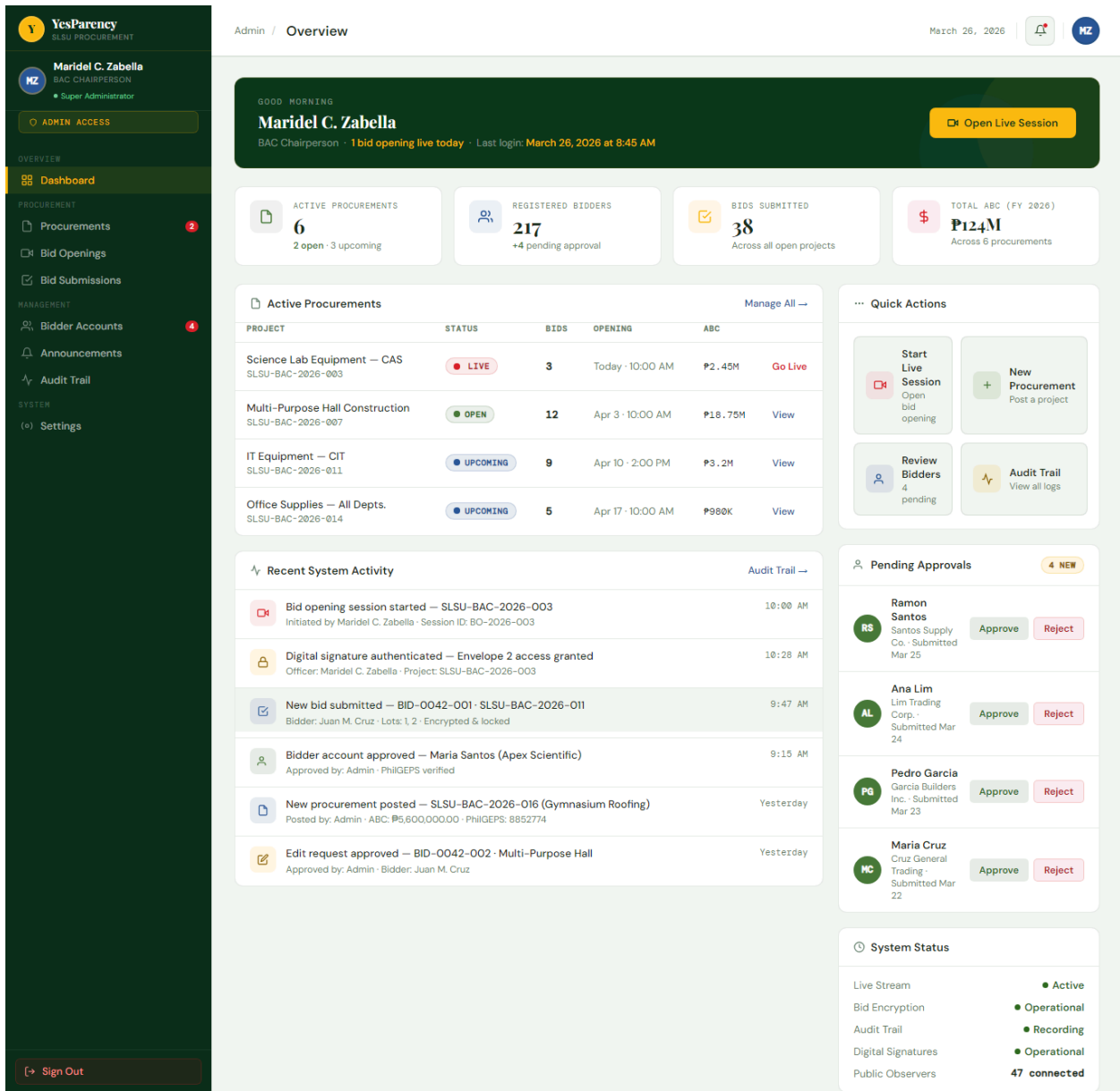


Figure 57. YesParenCy's Admin Dashboard

Figure 57 presents the Admin Dashboard, which serves as the main hub for the BAC Chairperson and procurement administrators. Summary cards at the top display the number of active procurements, registered bidders, bids submitted, and total ABC for the fiscal year. The

Active Procurements table lists ongoing and upcoming projects with their status, number of bids, opening date, and ABC. A Recent System Activity panel logs the latest system events, such as bid openings, digital signature authentications, and bidder approvals. On the right side, Quick Actions provide shortcuts for starting a live session, posting a new procurement, reviewing bidders, and viewing the audit trail. A Pending Approvals panel allows administrators to approve or reject bidder registration requests, while a System Status panel displays the operational status of key components including live stream, bid encryption, audit trail, and digital signatures.

Procurement Management Page

The screenshot displays the 'Procurements' page in the YesParency system, viewed from an administrator's perspective. The interface includes a dark green sidebar on the left with navigation options: ADMIN ACCESS, OVERVIEW (Dashboard), PROCUREMENT (Procurements), MANAGEMENT (Bid Openings, Bid Submissions, Bidder Accounts, Announcements, Audit Trail), and SYSTEM (Settings). The main content area is titled 'Admin / Procurements' and features a search bar and filter buttons (All, Live, Open, Upcoming, Closed, + New Procurement). A list of procurement projects is shown, each with a reference number, PhilGEPS number, title, bid opening date, submission deadline, total bids, number of lots, ABC, and status. The projects are: 1. SLSU-BAC-2026-003: Supply and Delivery of Science Laboratory Equipment for College of Arts and Sciences (LIVE NOW, 3 bidders, 1 lot, P2,450,000.00). 2. SLSU-BAC-2026-007: Construction of New Multi-Purpose Hall at SLSU Main Campus (OPEN FOR SUBMISSION, 12 bidders, 1 lot, P18,750,000.00). 3. SLSU-BAC-2026-011: Supply of Information Technology Equipment and Peripherals for CIT (UPCOMING, 9 bidders, 3 lots, P3,200,000.00). 4. SLSU-BAC-2026-014: Procurement of Office Supplies and Materials for All Departments FY 2026 (UPCOMING, 5 bidders, 1 lot, P980,000.00). 5. SLSU-BAC-2026-016: Repair and Rehabilitation of SLSU Gymnasium Roofing (UPCOMING, 0 bidders, 1 lot, P5,600,000.00). 6. SLSU-BAC-2026-001: Procurement of Janitorial and Maintenance Services FY 2026 (CLOSED, 7 bidders, 1 lot, P1,800,000.00). Each entry has 'Details', 'Manage', and 'Edit' buttons. A 'Go Live' button is present for the first project.

Reference Number	PhilGEPS Number	Project Title	Bid Opening Date	Submission Deadline	Total Bids	Number of Lots	ABC	Status
SLSU-BAC-2026-003	8821047	Supply and Delivery of Science Laboratory Equipment for College of Arts and Sciences	March 26, 2026 - 10:00 AM	March 25, 2026 - 5:00 PM	3	1	P2,450,000.00	LIVE NOW
SLSU-BAC-2026-007	8834502	Construction of New Multi-Purpose Hall at SLSU Main Campus	April 3, 2026 - 10:00 AM	April 2, 2026 - 5:00 PM	12	1	P18,750,000.00	OPEN FOR SUBMISSION
SLSU-BAC-2026-011	8841289	Supply of Information Technology Equipment and Peripherals for CIT	April 10, 2026 - 2:00 PM	April 9, 2026 - 5:00 PM	9	3	P3,200,000.00	UPCOMING
SLSU-BAC-2026-014	8849301	Procurement of Office Supplies and Materials for All Departments FY 2026	April 17, 2026 - 10:00 AM	April 16, 2026 - 5:00 PM	5	1	P980,000.00	UPCOMING
SLSU-BAC-2026-016	8852774	Repair and Rehabilitation of SLSU Gymnasium Roofing	April 24, 2026 - 2:00 PM	April 23, 2026 - 5:00 PM	0	1	P5,600,000.00	UPCOMING
SLSU-BAC-2026-001	8810022	Procurement of Janitorial and Maintenance Services FY 2026	March 15, 2026 - 10:00 AM	March 14, 2026 - 5:00 PM	7	1	P1,800,000.00	CLOSED

Figure 58. YesParency's Procurements Page (Admin View)

Figure 58 presents the Procurements page, which displays all procurement projects managed by the administrator. Each entry shows the reference number, PhilGEPS number, project title, bid opening date, submission deadline, total bids, number of lots, ABC, and status tag. Filter buttons and a search bar allow administrators to sort procurements by status. Action buttons for viewing details, managing, editing, and going live are provided for each entry depending on its current status, while a “New Procurement” button at the top allows administrators to post a new project.

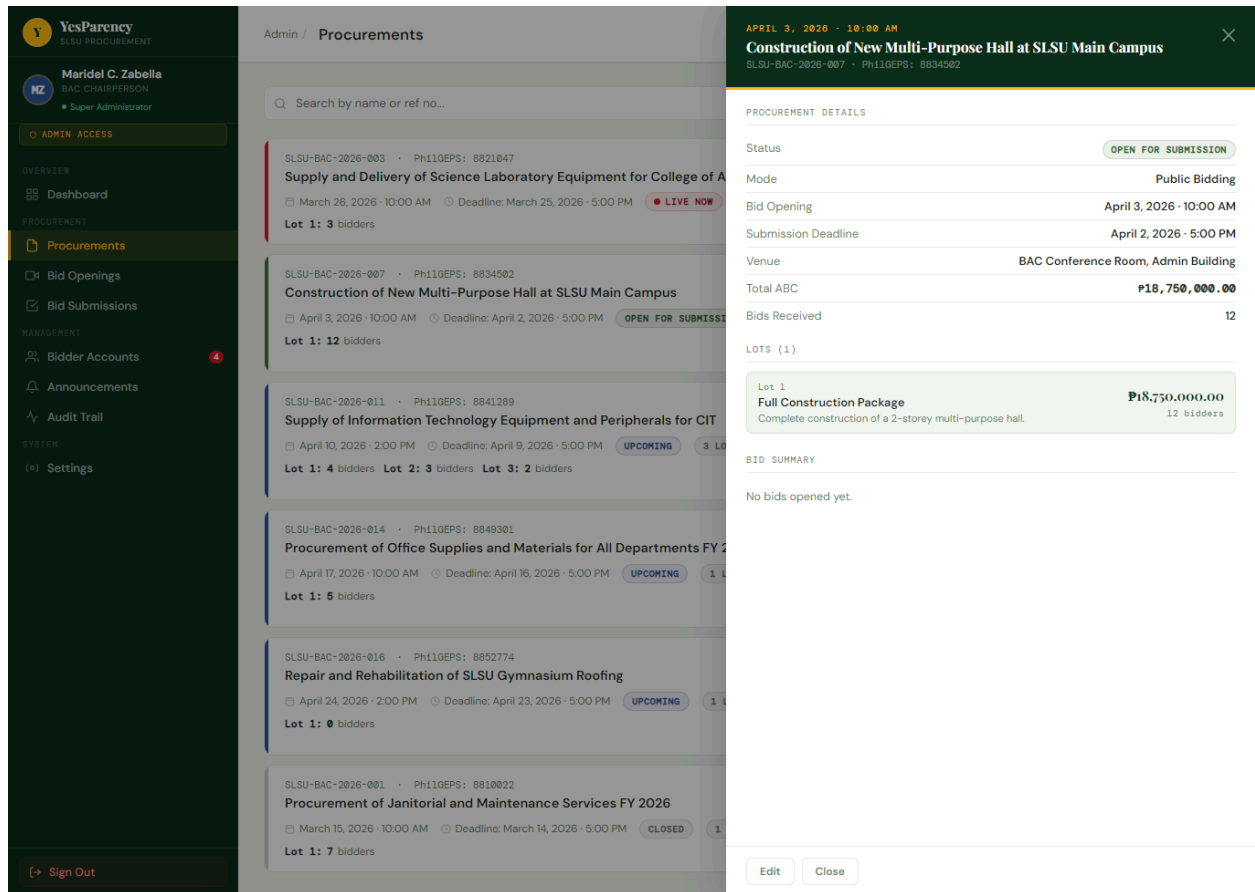


Figure 59. YesParenCy's Procurements Page – View Details Panel (Admin View)

Figure 59 presents the procurement details side panel, which opens when the “Details” button is clicked on a procurement entry. The panel displays the status, bidding mode, bid opening date, submission deadline, venue, total ABC, and number of bids received. It also shows the lot details and bid summary, with Edit and Close buttons available at the bottom of the panel.

New Procurement

PROJECT INFORMATION

Project Title *

e.g. Supply of Laboratory Equipment for CAS

Reference No. * **PhilGEPS Ref No.**

SLSU-BAC-2026-XXX e.g. 8821047

Procurement Mode * **Category**

Public Bidding Goods

Description

Brief description of the procurement...

SCHEDULE

Pre-Bid Conference **Submission Deadline ***

dd/mm/yyyy --:-- -- dd/mm/yyyy --:-- --

Bid Opening Date * **Venue**

dd/mm/yyyy --:-- -- BAC Conference Room, Admin

BUDGET

Approved Budget (ABC) * **Number of Lots**

0.00 1

Cancel Create Procurement

Figure 60. YesParenCy's New Procurement Modal

Figure 60 presents the New Procurement modal, which requires the administrator to enter the project title, reference number, PhilGEPS reference number, procurement mode, category, and description. The Schedule section requires the pre-bid conference date, submission deadline, bid opening date, and venue. The Budget section requires the Approved Budget for the Contract and the number of lots. A “Create Procurement” button at the bottom finalizes and posts the new procurement project to the system.

Bid Opening Management Page

The screenshot displays the 'Bid Openings' page in the YesParenity system. The left sidebar contains navigation links for Dashboard, Procurements, Bid Openings (selected), Bid Submissions, Bidder Accounts, Announcements, Audit Trail, and Settings. The main content area shows a banner for 'Science Lab Equipment — Bid Opening In Progress' with a 'Join Live Session' button. Below this, a list of bid opening sessions is shown, each with a date, time, session number, title, venue, BAC Chairperson, number of bidders and observers, status, and a progress tracker.

Date	Time	Session Number	Title	Venue	BAC Chairperson	Bidders	Observers	Status	Progress Tracker	Action
26 MAR	10:00 AM	BO-2026-003	Supply and Delivery of Science Laboratory Equipment — CAS	BAC Conference Room, Admin Building	Maridel C. Zaballa	3	47	LIVE NOW	Session Started — Envelope 1 Opened — Envelope 2 Opening — Post-Qualification Notice	Join Live, Manage
03 APR	10:00 AM	BO-2026-007	Construction of New Multi-Purpose Hall at SLSU Main Campus	BAC Conference Room, Admin Building	Maridel C. Zaballa	12	0	SCHEDULED	Session Start — Envelope 1 Opening — Envelope 2 Opening — Post-Qualification Notice	Start Live
10 APR	2:30 PM	BO-2026-011	Supply of IT Equipment and Peripherals for CIT	BAC Conference Room, Admin Building	Maridel C. Zaballa	9	0	SCHEDULED	Session Start — Envelope 1 Opening — Envelope 2 Opening — Post-Qualification Notice	Start Live
15 MAR	10:00 AM	BO-2026-001	Procurement of Janitorial and Maintenance Services FY 2026	BAC Conference Room, Admin Building	Maridel C. Zaballa	7	31	COMPLETED	Session Started — Envelope 1 Opened — Envelope 2 Opened — Post-Qualification Issued	View Results

Figure 61. YesParenity's Bid Openings Page (Admin View)

Figure 61 presents the Bid Openings page, which displays all bid opening sessions managed by the administrator. A banner at the top alerts the admin of any currently active bid opening session, with quick access buttons to join or manage it. Each entry shows the session reference number, project title, scheduled date and time, venue, BAC Chairperson, number of bidders and observers, status tag, and a progress tracker indicating the current phase of the opening process. Filter buttons allow sorting by status, and action buttons for joining live, managing, starting, or viewing results are provided depending on each session's status.

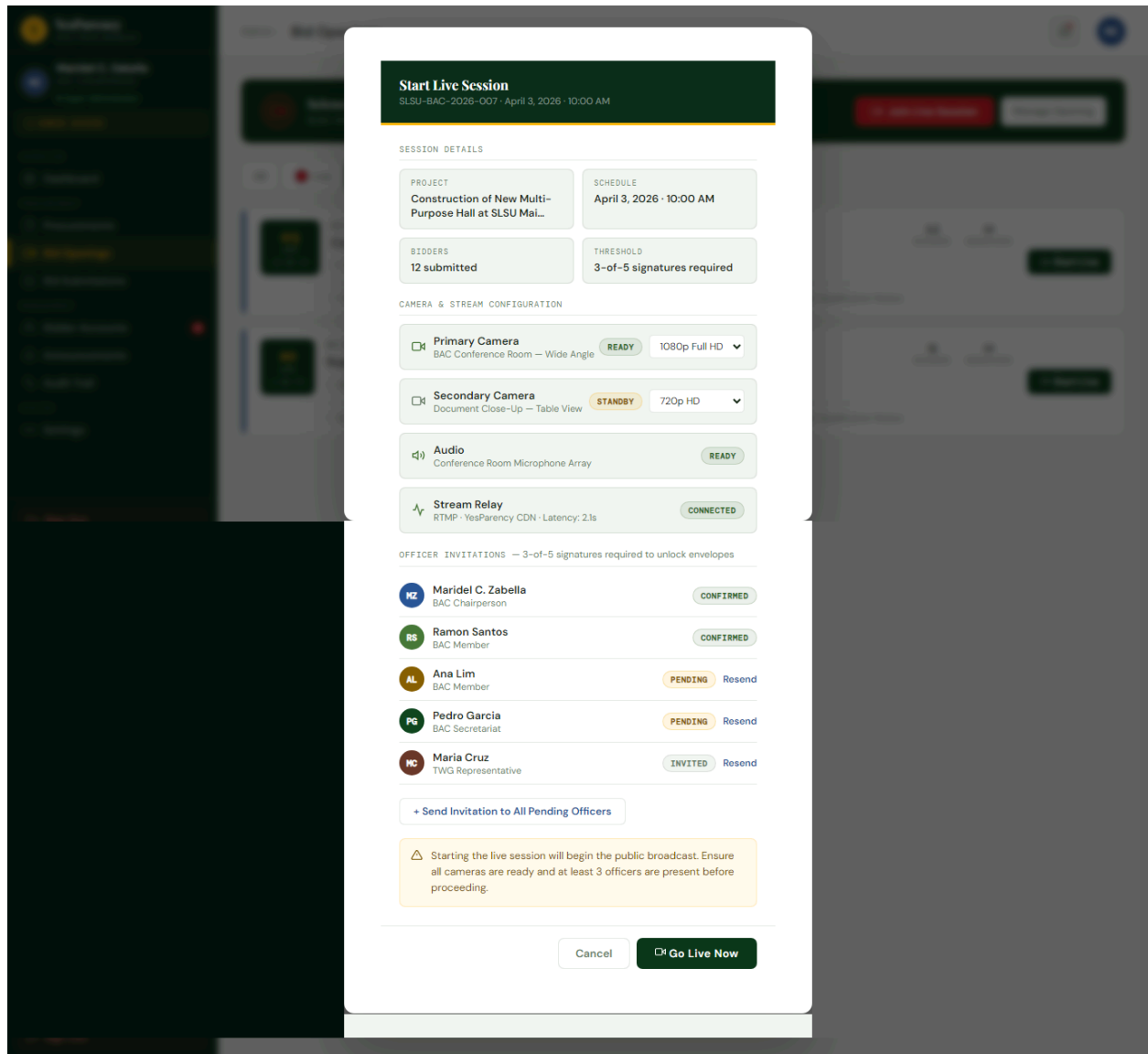


Figure 62. YesParenCy's Start Live Session Modal

Figure 62 presents the Start Live Session modal, which displays the session details, including the project name, schedule, number of submitted bids, and the required digital signature threshold. The Camera and Stream Configuration section shows the status of the primary camera, secondary camera, audio, and stream relay. The Officer Invitations section lists

all BAC officers with their confirmation status, and a “Send Invitation to All Pending Officers” button is available for officers who have yet to confirm. A warning notice at the bottom reminds the administrator to ensure all cameras are ready and that at least the required number of officers are present before clicking “Go Live Now.”

Admin’s Live Bid Opening Page

The screenshot displays the YesParency Live Session Page in Admin View. The interface is divided into several sections:

- Header:** Shows the project title "Supply and Delivery of Science Laboratory Equipment — CAS" and session details "SLSU-BAC-2026-003 · March 26, 2026 · 10:00 AM · BAC Conference Room". It also indicates "46 watching" and a "LIVE" status.
- Left Sidebar:** Contains the user profile of Maridel C. Zaballa (BAC Chairperson, Super Administrator) and navigation links for "ADMIN ACCESS", "SESSION INFO", "PROJECT" (Science Lab Equipment — CAS), "SESSION ID" (BO-2026-003), "SIGNATURES" (2 / 3 required), and "CONTROLS" (Public View, Audit Trail, Bid Submissions). It also includes buttons for "End Live Session", "Exit to Bid Openings", and "Sign Out".
- Main Content Area:** Features a live stream feed from the BAC Conference Room. Below the stream, there is a digital signature authentication panel for "Envelope 1 - ELIGIBILITY & TECHNICAL DOCUMENTS", which requires "3-OF-5 SIGNATURES". The panel shows the status of five officers: Maridel C. Zaballa (You) (Sign), Ramon Santos (BAC Member) (checked), Ana Lim (BAC Member) (checked), Pedro Garcia (BAC Secretariat) (checked), and Maria Cruz (TWG Representative) (unchecked). A message at the bottom of the panel states "Waiting for 1 more signature...".
- Right Sidebar:** Displays a list of bids under the "Bids" tab, including "Apex Scientific Supply Co.", "LabTech Philippines Inc.", and "SciEquip Solutions Corp.". It also has tabs for "Audit Trail" and "Chat".

Figure 63. YesParency's Live Session Page (Admin View)

Figure 63 presents the Live Session page, which serves as the administrator’s interface during an active bid opening. The top of the page displays the project title, session details, and a live viewer count. The main area shows the live stream feed from the BAC Conference Room alongside the digital signature authentication panel, which displays each BAC officer’s signing status and tracks the required signature threshold before the envelope can be unlocked. The current phase tabs for Envelope 1 and Envelope 2 are shown at the top of the panel. On the right

side, tabs for Bids, Audit Trail, and Chat are available, with the Bids tab listing submitted bidders in timestamped order. Controls for ending the live session and accessing the audit trail and bid submissions are provided on the left sidebar.

Bid Submission Page

YesParency
SLSU PROCUREMENT

Maridel C. Zaballa
BAC CHAIRPERSON
Super Administrator

ADMIN ACCESS

OVERVIEW

- Dashboard

PROCUREMENT

- Procurements
- Bid Openings
- Bid Submissions**

MANAGEMENT

- Bidder Accounts
- Announcements
- Audit Trail

SYSTEM

- Settings

Admin / Bid Submissions

Search by bidder name, ref no., or project...

All Projects **Live** **Open** **Edit Requests** **Closed**

SLSU-BAC-2026-003 - Opening: March 25, 2026 - 10:00 AM
Supply and Delivery of Science Laboratory Equipment – CAS **LIVE** **P2,450,000.00** **3 BIDS**

BIDDER	BID REF	LOTS	BID PRICE	RANK	STATUS	SUBMITTED
Apex Scientific Supply Co. BDR-2024-00012	BID-0001-003	Lot 1	Lot 1: P2,180,000.00	1	COMPLIANT	Mar 25, 2026 - 9:48 AM
LabTech Philippines Inc. BDR-2024-00034	BID-0002-003	Lot 1	Lot 1: P2,310,500.00	2	COMPLIANT	Mar 25, 2026 - 9:52 AM
SciEquip Solutions Corp. BDR-2024-00078	BID-0003-003	Lot 1	Lot 1: P2,398,000.00	3	UNDER REVIEW	Mar 25, 2026 - 9:55 AM

SLSU-BAC-2026-007 - Opening: April 3, 2026 - 10:00 AM
Construction of New Multi-Purpose Hall at SLSU Main Campus **OPEN** **P10,750,000.00** **2 BIDS**

BIDDER	BID REF	LOTS	BID PRICE	RANK	STATUS	SUBMITTED
Juan M. Cruz (Cruz General Trading) BDR-2024-00089	BID-0042-002	Lot 1	Lot 1: [REDACTED]	-	SUBMITTED	Mar 24, 2026 - 3:22 PM
Garcia Builders Inc. BDR-2024-00101	BID-0011-007	Lot 1	Lot 1: [REDACTED]	-	SUBMITTED	Mar 25, 2026 - 11:10 AM

SLSU-BAC-2026-011 - Opening: April 10, 2026 - 2:00 PM
Supply of IT Equipment and Peripherals for CIT **OPEN** **P3,200,000.00** **2 BIDS**

BIDDER	BID REF	LOTS	BID PRICE	RANK	STATUS	SUBMITTED
Juan M. Cruz (Cruz General Trading) BDR-2024-00089	BID-0042-001	Lot 1 Lot 2	Lot 1: [REDACTED] Lot 2: [REDACTED]	-	SUBMITTED	Mar 25, 2026 - 4:47 PM
TechSource Philippines BDR-2024-00055	BID-0015-011	Lot 1 Lot 2 Lot 3	Lot 1: [REDACTED] Lot 2: [REDACTED] Lot 3: [REDACTED]	-	SUBMITTED	Mar 26, 2026 - 8:30 AM

SLSU-BAC-2026-001 - Opening: March 15, 2026 - 10:00 AM
Procurement of Janitorial and Maintenance Services FY 2026 **CLOSED** **P1,000,000.00** **2 BIDS**

BIDDER	BID REF	LOTS	BID PRICE	RANK	STATUS	SUBMITTED
Juan M. Cruz (Cruz General Trading) BDR-2024-00089	BID-0042-004	Lot 1	Lot 1: P1,720,000.00	1	AWARDED	Feb 28, 2026 - 2:10 PM
CleanPro Services Corp. BDR-2024-00062	BID-0008-001	Lot 1	Lot 1: P1,755,000.00	2	NOT AWARDED	Mar 1, 2026 - 10:22 AM

[Sign Out](#)

Figure 64. YesParency's Bid Submissions Page (Admin View)

Figure 64 presents the Bid Submissions page, which displays all submitted bids grouped by procurement project. Each project group shows the reference number, opening date, ABC, status, and total number of bids. Within each group, individual bid entries display the bidder

name, Bidder ID, bid reference number, selected lots, bid price, rank, submission status, and submission timestamp, with bid prices of unopened submissions hidden until the official bid opening. Administrators can approve or deny edit requests directly from this page, and a “View” button is available for each entry to access the full bid details.

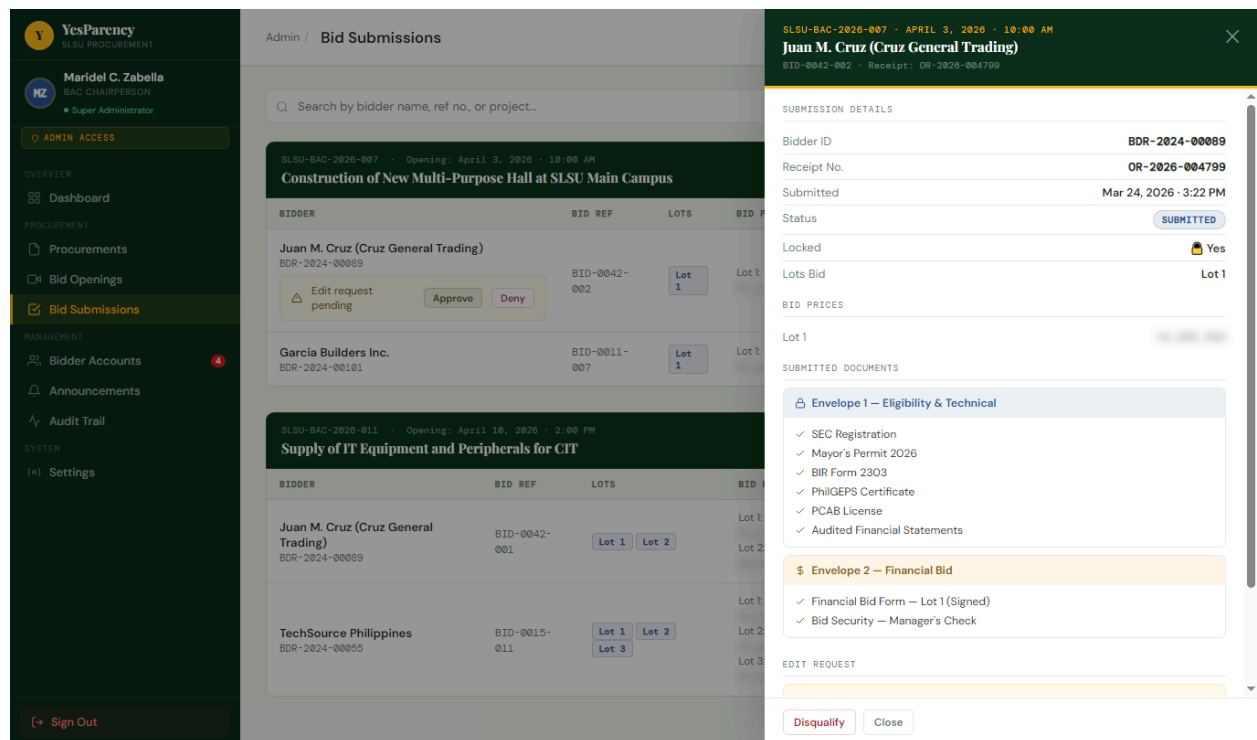
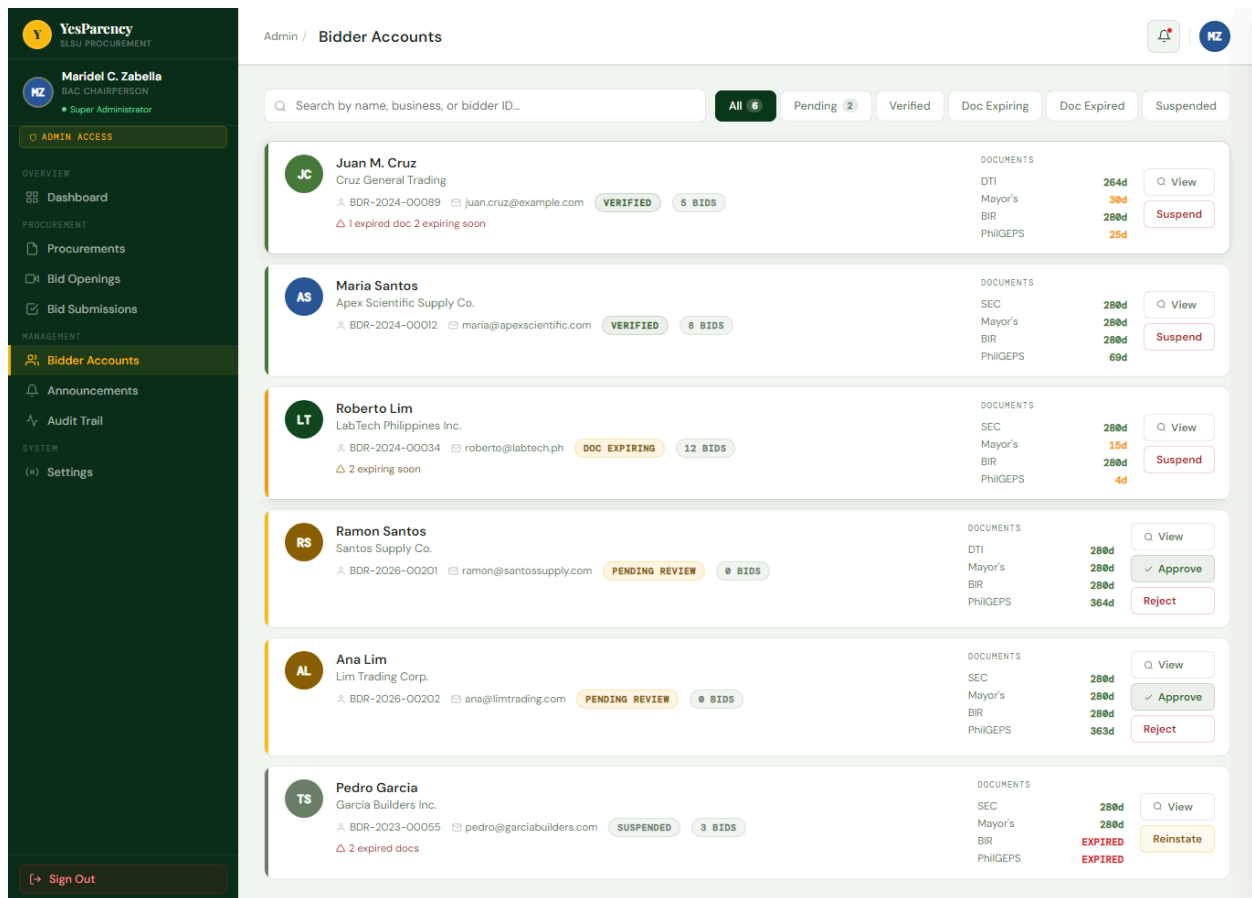


Figure 65. YesParenCy's Bid Submissions Page – View Details Panel (Admin View)

Figure 65 presents the bid submission details side panel, which opens when the “View” button is clicked on a bid entry. It displays the Bidder ID, receipt number, submission date and time, status, lock status, and lots bid. Submitted documents are listed under Envelope 1 (Eligibility and Technical) and Envelope 2 (Financial Bid), while bid prices remain hidden until

the official bid opening. A Disqualify button is available at the bottom for administrators to act on a submission when necessary.

Bidder's Account Management Page



The screenshot displays the 'Bidder Accounts' page in the YesParency system. The left sidebar shows the user 'Maridel C. Zaballa' as a 'Super Administrator'. The main content area lists six bidder accounts, each with a profile card showing their details and a table of document expiration dates.

Bidder ID	Name	Business	Email	Status	Bids	DTI	SEC	Mayor's	BIR	PHIGEPS
BDR-2024-00089	Juan M. Cruz	Cruz General Trading	juan.cruz@example.com	VERIFIED	5	264d	280d	30d	280d	25d
BDR-2024-00012	Maria Santos	Apex Scientific Supply Co.	maria@apexscientific.com	VERIFIED	8	280d	280d	280d	280d	69d
BDR-2024-00034	Roberto Lim	LabTech Philippines Inc.	roberto@labtech.ph	DOC EXPIRING	12	280d	280d	15d	280d	4d
BDR-2026-00201	Ramon Santos	Santos Supply Co.	ramon@santossupply.com	PENDING REVIEW	0	280d	280d	280d	280d	364d
BDR-2026-00202	Ana Lim	Lim Trading Corp.	ana@limtrading.com	PENDING REVIEW	0	280d	280d	280d	280d	363d
BDR-2023-00055	Pedro Garcia	Garcia Builders Inc.	pedro@garciabuilders.com	SUSPENDED	3	280d	280d	EXPIRED	EXPIRED	EXPIRED

Figure 66. YesParency's Bidder Accounts Page (Admin View)

Figure 66 presents the Bidder Accounts page, which displays all registered bidders and allows administrators to manage their account statuses. Each entry shows the bidder's name, business name, Bidder ID, email, account status tag, number of bids submitted, and the validity status of required documents with days remaining. Filter buttons at the top allow sorting by

status, such as Pending, Verified, Doc Expiring, Doc Expired, and Suspended. Action buttons for viewing, approving, rejecting, suspending, or reinstating accounts are provided for each entry depending on the bidder’s current status.

The screenshot displays the 'Bidder Accounts' page in the YesParenCy Admin interface. The left sidebar contains navigation links for Overview, Procurement, and Management. The main area shows a list of bidder accounts with their status and bid counts. A side panel is open for the account of Juan M. Cruz, showing detailed information and a table of eligibility documents.

DOCUMENT	EXPIRY	STATUS
DTI Registration	Dec 15, 2026	2640
Mayor's Permit (2026)	Apr 25, 2026	300 LEFT
BIR Form 2303	Dec 31, 2026	2880
PhilGEPS Certificate	Apr 20, 2026	250 LEFT
Valid Gov. ID	Mar 10, 2028	7140
SLCC Statement	Mar 1, 2026	EXPIRED

Figure 67. YesParenCy's Bidder Accounts Page – View Details Panel (Admin View)

Figure 67 presents the bidder account details side panel, which opens when the “View” button is clicked on a bidder entry. It displays the account status, membership date, contact details, TIN, business type, address, total bids, and awards. The panel also shows the eligibility documents table with expiry dates and validity status, as well as the bidder’s bid history with

submission dates and statuses. A “Suspend Account” button is available at the bottom for administrators to act on the account when necessary.

Announcement Management Page

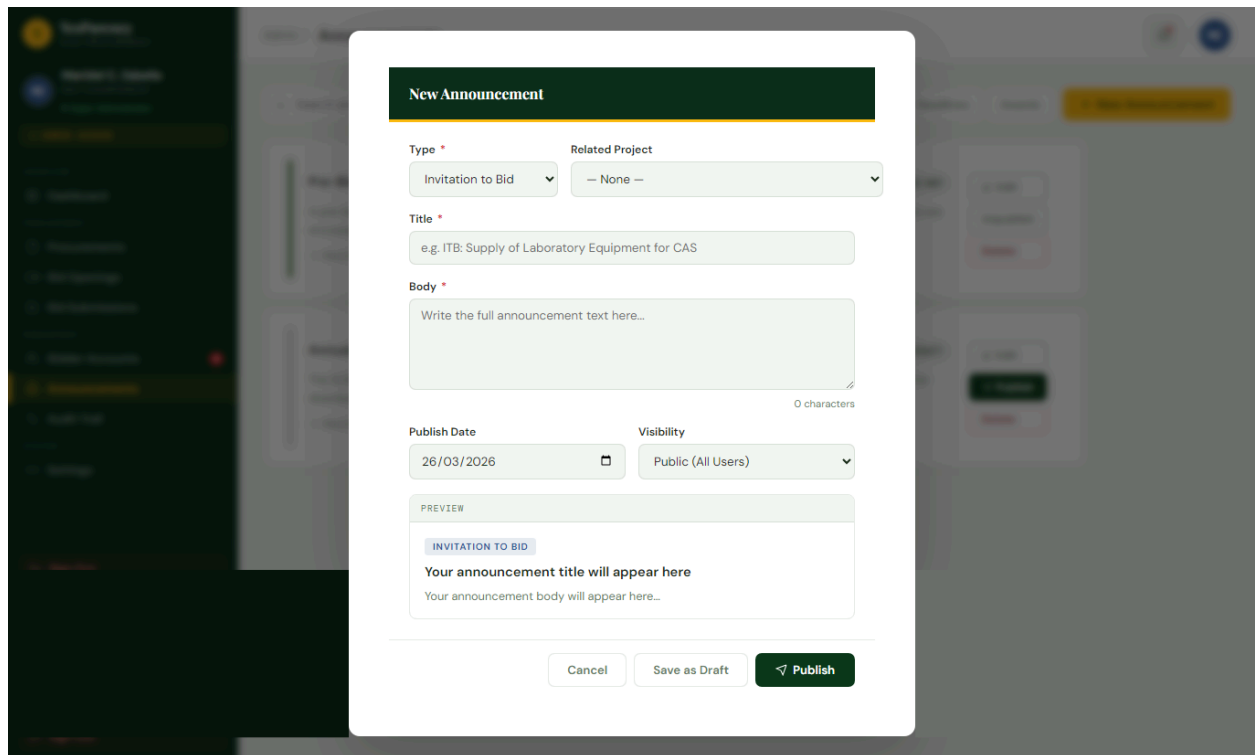
The screenshot displays the 'Announcements' management interface in the YesParency SLSU Procurement system. The left sidebar shows the user 'Maridel C. Zaballa' as a 'Super Administrator' with access to various system functions. The main content area lists six announcements, each with a title, category tag, description, date, visibility, and view count. Actions like 'Edit', 'Unpublish', and 'Delete' are available for each entry. The announcements include an invitation to bid for laboratory equipment, a pre-bid conference notice, a deadline reminder for IT equipment, a notice of award for janitorial services, an invitation to bid for a new hall, and a published annual procurement plan.

Title	Category	Description	Date	Visibility	Views	Actions
ITB: Supply and Delivery of Laboratory Equipment for the College of Arts and Sciences	INVITATION TO BID	The Bids and Awards Committee of SLSU invites PhilGEPS-registered suppliers to bid for the supply and delivery of laboratory equipment. Total ABC: ₱2,450,000.00.	March 26, 2026	Public	142	Edit, Unpublish, Delete
Pre-Bid Conference: Renovation of Science and Technology Building	NOTICE	A pre-bid conference will be held on March 28, 2026 at 2:00 PM at the BAC Conference Room. All interested bidders are encouraged to attend. Bidding documents ar...	March 24, 2026	Public	89	Edit, Unpublish, Delete
Final Call: IT Equipment Procurement — Submission Closes April 9	DEADLINE REMINDER	This is a reminder that the deadline for submission of bids for the IT Equipment and Peripherals project (SLSU-BAC-2026-011) is on April 9, 2026 at 5:00 PM. Lat...	March 20, 2026	Bidders Only	67	Edit, Unpublish, Delete
Notice of Award: Janitorial and Maintenance Services Contract FY 2026	NOTICE OF AWARD	The BAC has issued a Notice of Award for the Janitorial and Maintenance Services Contract (SLSU-BAC-2026-001). The awarded bidder has been notified and is requi...	March 20, 2026	Public	203	Edit, Unpublish, Delete
ITB: Construction of New Multi-Purpose Hall at SLSU Main Campus	INVITATION TO BID	SLSU invites eligible contractors to bid for the construction of a new multi-purpose hall. Total ABC: ₱18,750,000.00. Pre-bid conference on March 20, 2026. Bid ...	March 12, 2026	Public	318	Edit, Unpublish, Delete
Annual Procurement Plan (APP) FY 2026 — Published	DRAFT	The SLSU Procurement Office has published the Annual Procurement Plan for Fiscal Year 2026. The APP is available for download at the Procurement Office and on L...	March 8, 2026	Public		Edit, Publish, Delete

Figure 68. YesParency's Announcements Page (Admin View)

Figure 68 presents the Announcements page, which allows administrators to manage all procurement-related announcements. Each entry displays the announcement title, category tag, linked procurement reference, brief description, publication date, visibility setting, and view

count. Filter buttons allow sorting by category, such as Published, Drafts, ITB, Notices, Deadlines, and Awards. Action buttons for editing, unpublishing, publishing, and deleting are provided for each entry, while a “New Announcement” button at the top allows administrators to create and post new announcements.



The image shows a 'New Announcement' modal form. At the top is a dark green header with the text 'New Announcement'. Below this, there are two dropdown menus: 'Type' with 'Invitation to Bid' selected, and 'Related Project' with '— None —' selected. The 'Title' field contains the text 'e.g. ITB: Supply of Laboratory Equipment for CAS'. The 'Body' field is a large text area with the placeholder 'Write the full announcement text here...' and a character count of '0 characters'. Below the body field are two more dropdowns: 'Publish Date' with '26/03/2026' and a calendar icon, and 'Visibility' with 'Public (All Users)' selected. A 'PREVIEW' section shows a blue button labeled 'INVITATION TO BID', followed by the text 'Your announcement title will appear here' and 'Your announcement body will appear here...'. At the bottom are three buttons: 'Cancel', 'Save as Draft', and a green 'Publish' button with a checkmark icon.

Figure 69. YesParency's New Announcement Modal

Figure 69 presents the New Announcement modal, which opens when the “New Announcement” button is clicked. The modal allows administrators to create and publish procurement announcements by entering the announcement type, related project, title, and body text, along with a publish date and visibility setting. A live preview at the bottom shows how the

announcement will appear once published, and buttons for canceling, saving as a draft, or publishing the announcement are provided at the bottom.

The screenshot shows a modal window titled "Edit Announcement" with a dark green header. The modal is set against a blurred background of a web application. The form inside the modal includes the following elements:

- Type ***: A dropdown menu with "Notice" selected.
- Related Project**: A dropdown menu with "SLSU-BAC-2026-007 · Multi-Purpose Hall" selected.
- Title ***: A text input field containing "Pre-Bid Conference: Renovation of Science and Technology Building".
- Body ***: A text area containing the text: "A pre-bid conference will be held on March 28, 2026 at 2:00 PM at the BAC Conference Room. All interested bidders are encouraged to attend. Bidding documents are available at the Procurement Office." A character count "198 characters" is visible at the bottom right of the text area.
- Publish Date**: A date input field showing "26/03/2026" with a calendar icon.
- Visibility**: A dropdown menu with "Public (All Users)" selected.
- PREVIEW**: A section showing a live preview of the announcement, including a "NOTICE" tag, the title, and the body text.
- Buttons**: At the bottom of the modal are three buttons: "Cancel", "Save as Draft", and "Publish" (which is highlighted in dark green).

Figure 70. YesParency's Edit Announcement Modal

Figure 70 presents the Edit Announcement modal, which opens when the “Edit” button is clicked on an existing announcement. The modal follows the same layout as the New Announcement form, with fields pre-populated with the existing announcement details. Administrators can modify the type, related project, title, body, publish date, and visibility. A live preview updates in real time as changes are made, and buttons for canceling, saving as a draft, or republishing are available at the bottom.

Audit Trail Page

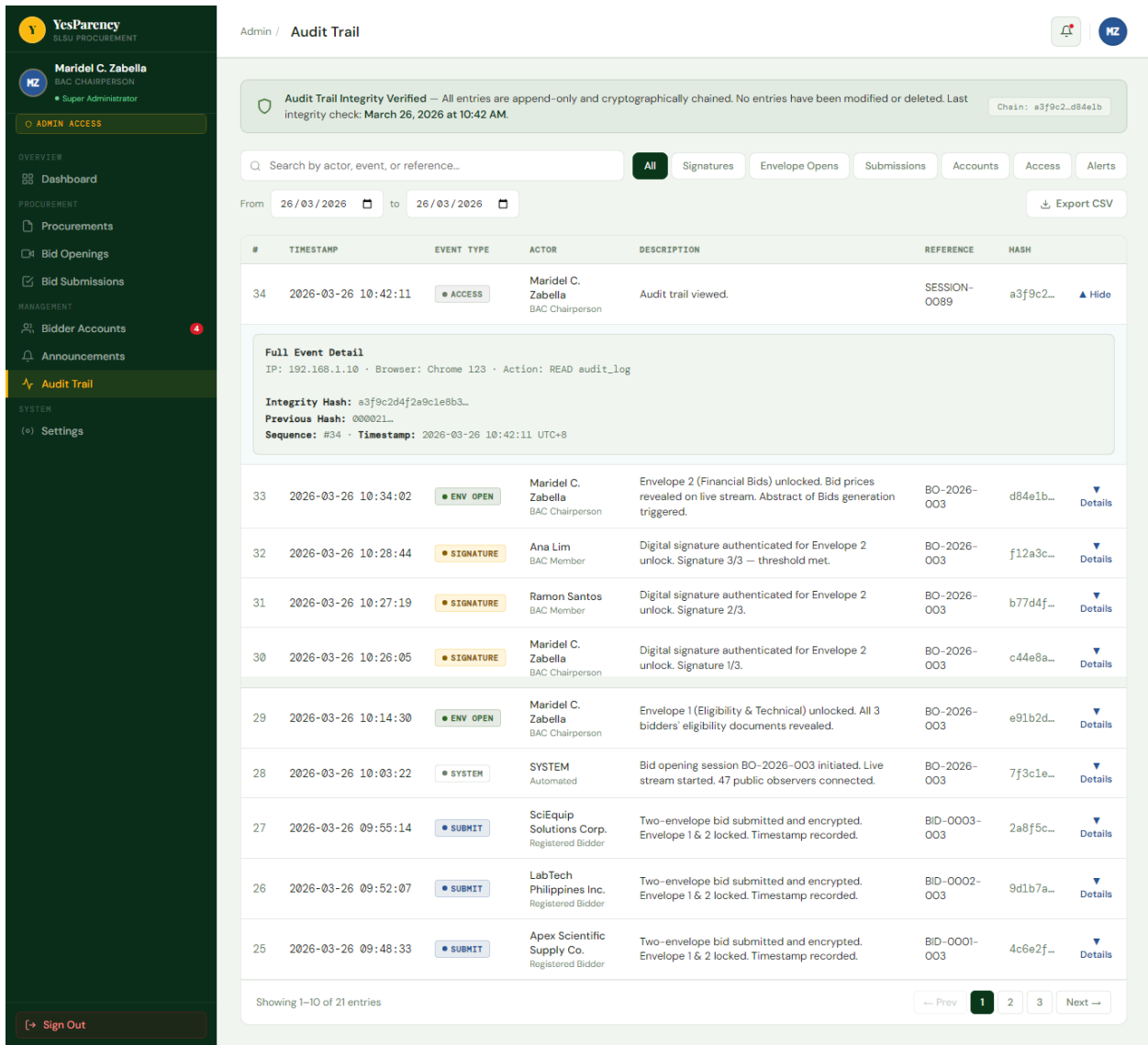


Figure 71. YesParency's Audit Trail Page (Admin View)

Figure 71 presents the Audit Trail page, which provides a comprehensive log of all system activities recorded in chronological order. A verification banner at the top confirms that all entries are append-only and cryptographically chained, ensuring log integrity. Each entry displays a sequence number, timestamp, event type tag, actor, description, reference number, and

cryptographic hash. Clicking “Details” on an entry expands the full event information, including IP address, browser, action performed, integrity hash, previous hash, and sequence timestamp. Filter buttons allow sorting by event type, such as Signatures, Envelope Opens, Submissions, Accounts, Access, and Alerts, while a date range selector and Export CSV option are also available.

System Setting Page

The screenshot displays the 'System Settings' page in the YesParency application, specifically the 'General' tab. The interface features a dark green sidebar on the left with navigation links for Overview, Procurement, Management, and System. The main content area is white with a light green header. A warning banner at the top states that the page is accessible to Super Administrators only and that changes affect the entire system. The settings are organized into two main sections: 'Institution' and 'Notifications'. The 'Institution' section includes fields for 'Institution Name' (Southern Luzon State University), 'Procurement Office Name' (SLSU Procurement Office), 'Fiscal Year' (2026), and 'Official Email' (procurement@sisu.edu). The 'Notifications' section includes toggle switches for 'Email Notifications' and 'Document Expiry Reminders', both of which are currently turned on. There is also a numeric input field for 'Expiry Grace Period' set to 15 days. A 'Save Changes' button is located at the bottom right of the settings area.

YesParency
SLSU PROCUREMENT

Admin / System Settings

This page is accessible to Super Administrators only. Changes made here affect the entire YesParency system and are recorded in the audit trail.

General Signatures Admin Users Security System

Institution

Institution Name
Displayed on all public-facing pages and documents. Southern Luzon State University

Procurement Office Name
Used in official announcements and bid documents. SLSU Procurement Office

Fiscal Year
Current active fiscal year for procurement tracking. 2026

Official Email
Used for system notifications and bidder communications. procurement@sisu.edu

Notifications

Email Notifications
Send email alerts to bidders for deadlines, bid openings, and awards. ☒

Document Expiry Reminders
Notify bidders 30 days before their eligibility documents expire. ☒

Expiry Grace Period
Days after document expiry before account is automatically suspended. - 15 +

Save Changes

Sign Out

Figure 72. YesParency's System Settings Page – General Tab (Super Admin View)

Figure 72 presents the System Settings page, accessible only to Super Administrators, with all changes recorded in the audit trail. The General tab contains institution settings, including the institution name, procurement office name, fiscal year, and official email, as well as notification settings with toggles for email notifications and document expiry reminders, and a configurable expiry grace period before automatic account suspension. Additional tabs for Signatures, Admin Users, Security, and System are available for managing other system-wide configurations.

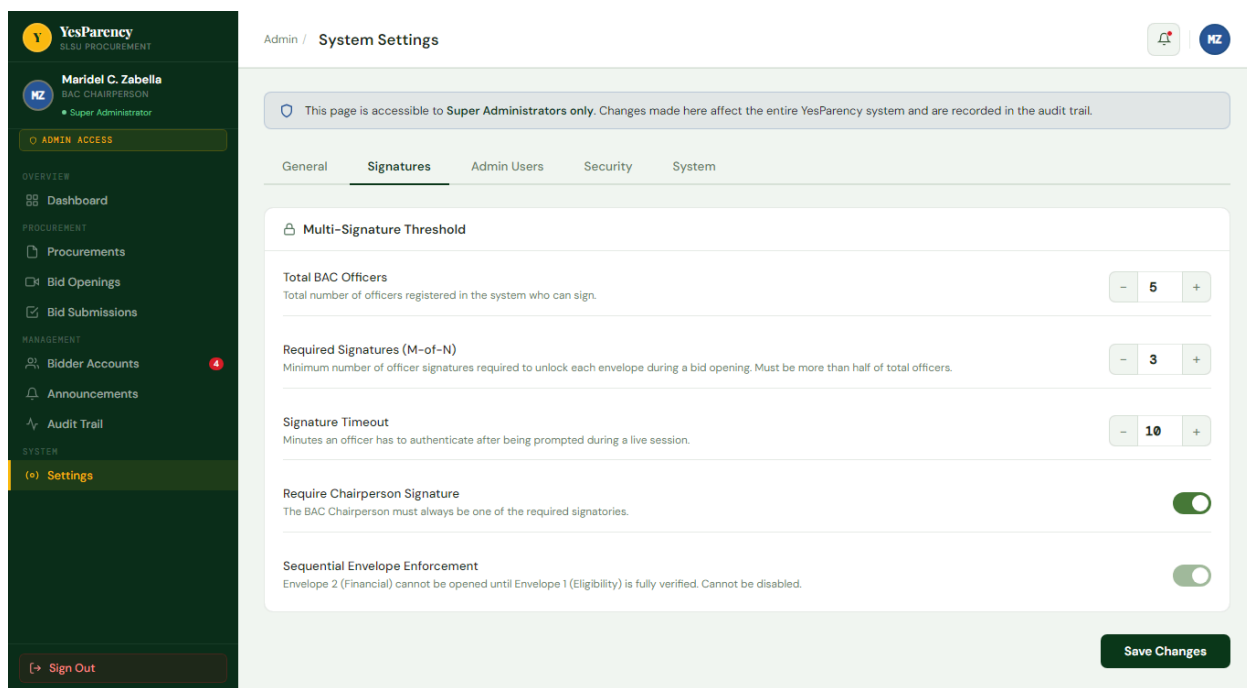


Figure 73. YesParency's System Settings Page – Signatures Tab (Super Admin View)

Figure 73 presents the Signatures tab of the System Settings page, which manages the digital signature configuration for bid openings. Settings include the total number of BAC officers registered, the required M-of-N signature threshold to unlock each envelope, and a

signature timeout period. Toggles are available for requiring the BAC Chairperson’s signature and enforcing sequential envelope opening, where Envelope 2 cannot be opened until Envelope 1 is fully verified.

The screenshot displays the 'System Settings' page in the YesParency system, specifically the 'Admin Users' tab. The page is accessible to Super Administrators only. The 'Admin Users' tab is selected, showing a list of 'Admin & Officer Accounts'. The table lists five users: Maridel C. Zaballa (Super Admin), Ramon Santos (BAC Member), Ana Lim (BAC Member), Pedro Garcia (BAC Secretariat), and Maria Cruz (TWG Observer). Each entry shows the officer's name, email, role, signing permission, last login, and status. The Super Admin can edit or revoke access for all other users. An 'Invite Admin' button is located at the top right of the table.

OFFICER	ROLE	CAN SIGN	LAST LOGIN	STATUS	
Maridel C. Zaballa m.zaballa@slsu.edu.ph	SUPER ADMIN	YES	Mar 26, 2026 - 8:45 AM	ACTIVE	Edit
Ramon Santos r.santos@slsu.edu.ph	BAC MEMBER	YES	Mar 26, 2026 - 9:02 AM	ACTIVE	Edit Revoke
Ana Lim a.lim@slsu.edu.ph	BAC MEMBER	YES	Mar 25, 2026 - 4:30 PM	ACTIVE	Edit Revoke
Pedro Garcia p.garcia@slsu.edu.ph	BAC SECRETARIAT	YES	Mar 24, 2026 - 2:15 PM	ACTIVE	Edit Revoke
Maria Cruz m.cruz@slsu.edu.ph	TWG OBSERVER	NO	Mar 20, 2026 - 11:00 AM	ACTIVE	Edit Revoke

Figure 74. YesParency's System Settings Page – Admin Users Tab (Super Admin View)

Figure 74 presents the Admin Users tab of the System Settings page, which displays all registered administrator and officer accounts in the system. Each entry shows the officer’s name, email, role, signing permission, last login, and account status. The Super Admin can edit or revoke access for each account, and an “Invite Admin” button at the top right allows adding new administrator or officer accounts to the system.

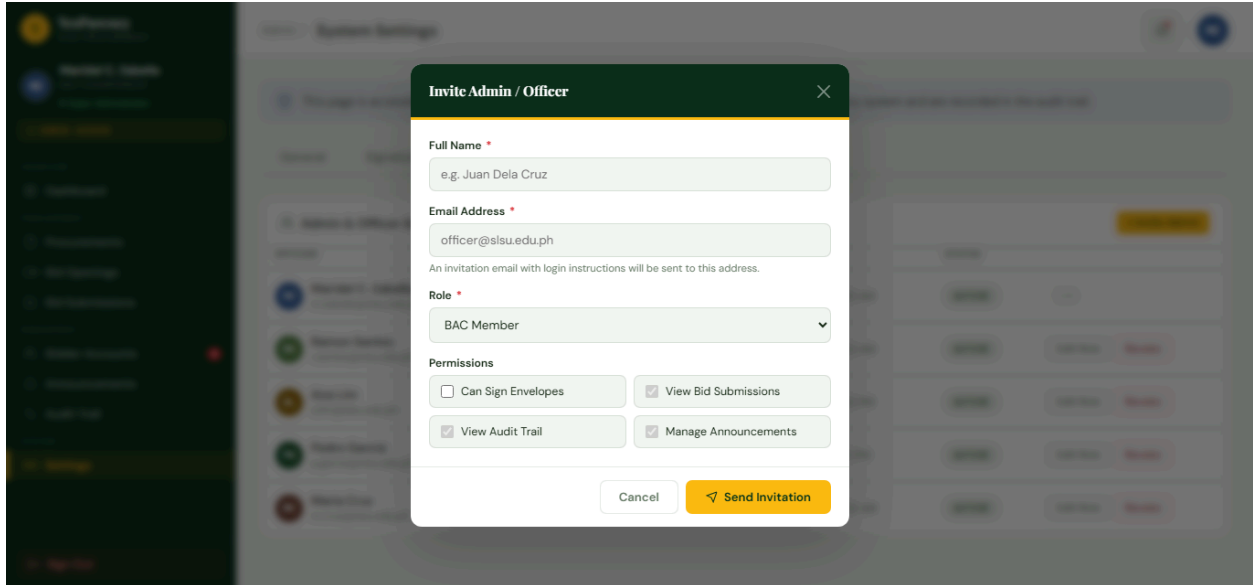


Figure 75. YesParenthood's Invite Admin/Officer Modal

Figure 75 presents the Invite Admin/Officer modal, which opens when the “Invite Admin” button is clicked. This modal allows the Super Admin to add new administrator or officer accounts by entering the full name, email address, and role of the new user. Permission checkboxes are provided to configure access rights, including the ability to sign envelopes, view bid submissions, view the audit trail, and manage announcements. A “Send Invitation” button submits the form and sends login instructions to the provided email address.

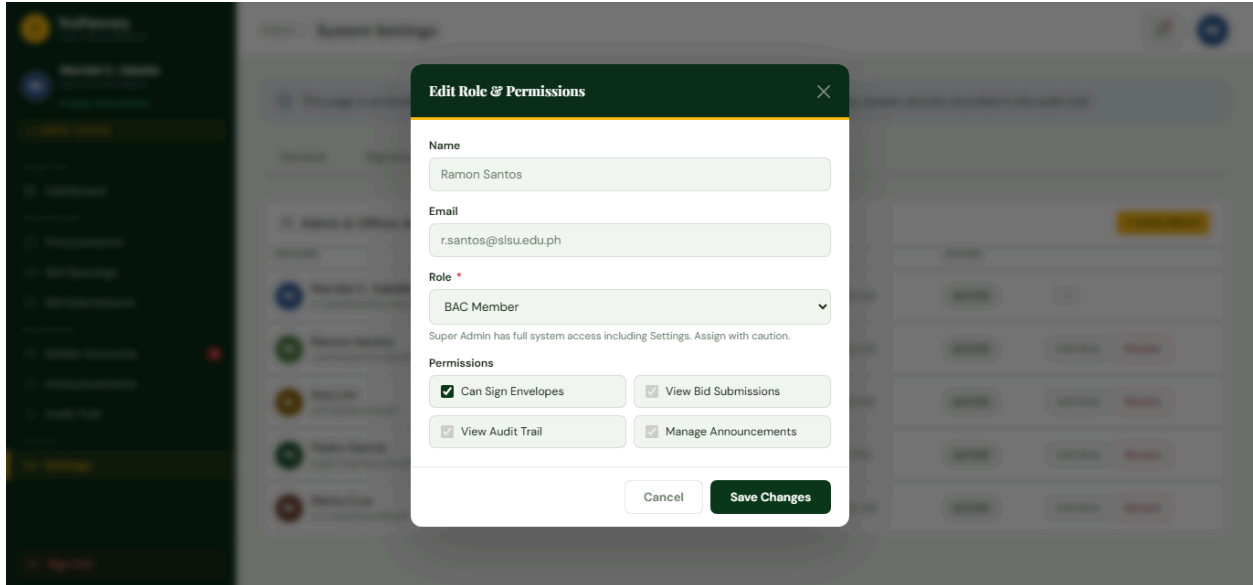


Figure 76. YesParency's Edit Role and Permissions Modal

Figure 76 presents the Edit Role and Permissions modal, which opens when the “Edit” button is clicked on an admin user entry. The modal follows the same layout as the Invite Admin form but is pre-populated with the selected officer’s existing details. The Super Admin can update the officer’s role and modify their permission settings, then save the changes using the “Save Changes” button.

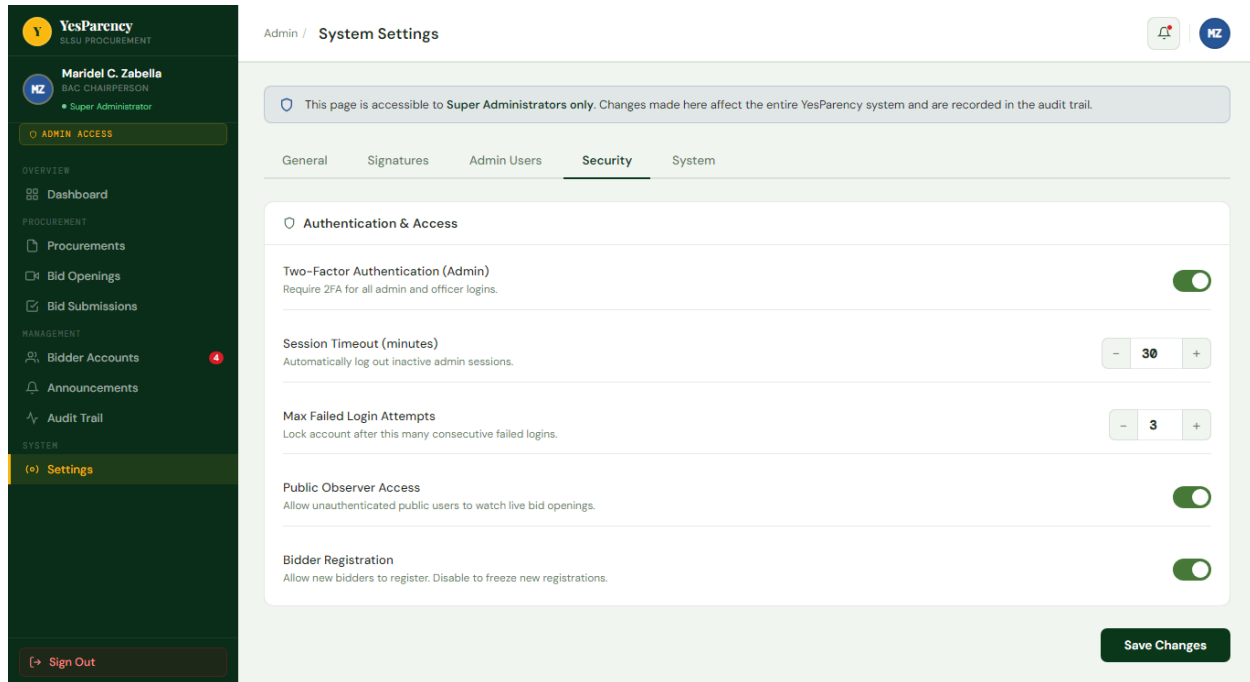


Figure 77. YesParency's System Settings Page – Security Tab (Super Admin View)

Figure 77 presents the Security tab of the System Settings page, which manages the system's authentication and access control settings. Configurable options include enabling Two-Factor Authentication for all admin and officer logins, setting the session timeout duration, defining the maximum number of failed login attempts before account lockout, toggling public observer access to live bid openings, and enabling or disabling new bidder registrations.

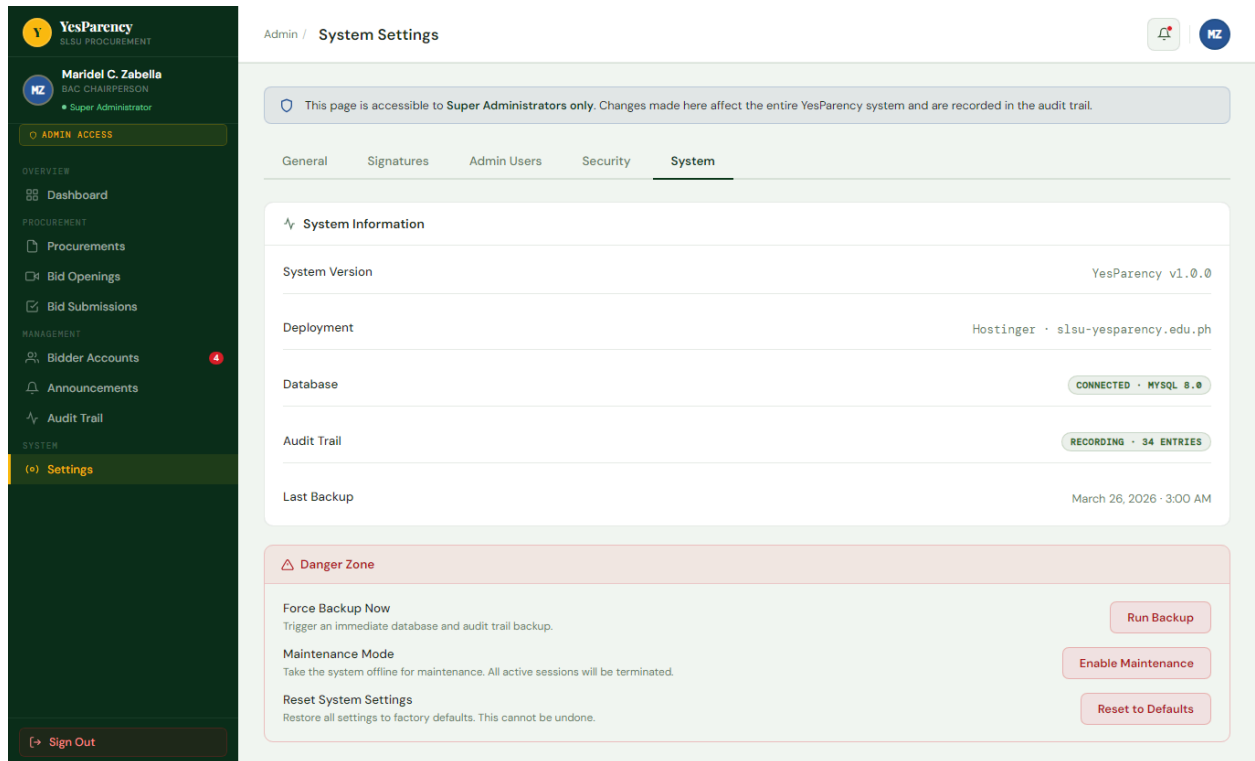


Figure 78. YesParenity's System Settings Page – System Tab (Super Admin View)

Figure 78 presents the System tab of the System Settings page, which displays technical information about the YesParenity system, including the system version, deployment details, database connection status, audit trail recording status, and the last backup timestamp. A Danger Zone section at the bottom provides options for running an immediate backup, enabling maintenance mode, and resetting all system settings to factory defaults, each accompanied by a warning that these actions have significant system-wide effects.

Implementation Plan

The implementation plan of YesParenCy will be guided by the principles of the Agile Scrum Methodology, which serves as the Software Development Life Cycle (SDLC) of the development team, to ensure the successful deployment and adoption of the system. The developers will first establish a thorough understanding of the target users for deployment, which include the procurement administrators, members of the Bids and Awards Committee (BAC), registered bidders, and public observers of Southern Luzon State University who are directly responsible for and involved in the institution's procurement activities.

The deployment strategy will be based on Agile Scrum, carried out iteratively through a series of sprints that allow for continuous feedback and improvement throughout the development and deployment process. For YesParenCy, the developers will begin by introducing the system to a controlled group of selected users, specifically the procurement administrators and BAC members of the Procurement Office, to gather initial feedback and implement necessary enhancements before a wider rollout. Following the refinement of the system based on this feedback, YesParenCy will then be made fully accessible to all registered bidders and public observers within the university's procurement ecosystem.

Implementation testing will be conducted iteratively at the end of each sprint, followed by a formal User Acceptance Testing (UAT) phase that involves the Procurement Office and BAC members to guarantee that the system meets institutional requirements and operates in accordance with the provisions of Republic Act No. 9184. The system will also undergo technical analysis to ensure that the database, operating system, server infrastructure, and hardware are all compatible and capable of supporting the system's core functionalities,

including bid document encryption, digital signature authentication, and live streaming of bid opening proceedings.

Procurement administrators, BAC members, and registered bidders will undergo formal training sessions supported by user manuals and system guides to ensure that all users are adequately prepared for the transition from the current manual bidding process to the fully digitized YesParenity platform. The implementation plan of YesParenity is specifically designed to ensure the secure, transparent, and efficient deployment of the system and to improve the procurement management operations of Southern Luzon State University in Lucban, Quezon.

Research Instrument

For the research instruments of this study, the researchers will use a structured questionnaire, interviews, and direct observation to evaluate the YesParenity system. The structured questionnaire will be used to collect data about the respondents' opinions and experiences while using the system.

A 5-point Likert scale will also be used to measure the respondents' level of agreement with the given statements. The scale is as follows:

5 – Strongly Agree

4 – Agree

3 – Neutral

2 – Disagree

1 – Strongly Disagree

This scale will help measure the system in terms of usability, efficiency, reliability, and overall user satisfaction during the document submission process.

Moreover, semi-structured interviews will also be conducted among selected respondents, including procurement officials and key users, in order to obtain more detailed information about their experiences with the system. This will allow them to share their insights, challenges, and recommendations regarding the improvement for the system. Direct observation will also be done while the system is being used. The researchers will observe how users interact with the system, especially during document submission, to identify any difficulties or issues encountered.

By using these instruments, questionnaires, interviews, and observation the study will gather both numerical and descriptive data to evaluate the system's performance and user experience.

Respondents of the Study

The respondents of this study were selected through purposive sampling to ensure that only relevant and knowledgeable participants were included in the evaluation of the YesParenity system.

The researchers included a total of 30 respondents, comprising 5 procurement administrators, 5 members of the Bids and Awards Committee (BAC), 5 registered bidders, 5 IT experts, and 10 public observers. The procurement administrators and BAC members are directly responsible for managing and overseeing all bidding activities at Southern Luzon State

University in accordance with Republic Act No. 9184. The registered bidders represent the supplier side of the procurement process and are the primary users of the bid submission portal. The IT experts were included to provide a technical evaluation of the system based on ISO/IEC 25010 Software Quality Standards. Public observers, on the other hand, were selected to assess the transparency and accessibility features of the live bid opening broadcast. The study was conducted at the Procurement Office of Southern Luzon State University in Lucban, Quezon, to further investigate how YesParency would help the institution manage its procurement activities more securely, efficiently, and transparently.

Data Gathering Procedures

Data will be gathered using two methods. Surveys will be provided to the 30 respondents selected through purposive sampling, focusing on those with direct involvement in the procurement process at Southern Luzon State University or with relevant technical expertise in software evaluation. This ensures relevant and insightful feedback for evaluating the YesParency system. The evaluation process is divided into two parts.

The first is the Non-Expert Evaluation, where surveys will be conducted with a total of 25 respondents, including 5 procurement administrators, 5 members of the Bids and Awards Committee (BAC), 5 registered bidders, and 10 public observers. This evaluation focuses on the system's usability, functional suitability, compatibility, safety, and security as experienced by the primary users of the system in their respective roles within the procurement process.

The second is the Expert Evaluation, where system testing and surveys will be conducted with 5 IT experts to assess technical aspects such as performance efficiency, reliability,

maintainability, flexibility, and security. Interviews will also be conducted with the head of the Procurement Office, BAC members, and IT experts to gain a deeper understanding of institutional requirements, operational workflows, policy alignment with Republic Act No. 9184, and technical recommendations for system improvement. System testing will focus on identifying and resolving any security vulnerabilities, performance bottlenecks, and usability issues based on the insights provided by these stakeholders. This method guarantees comprehensive and detailed feedback from all relevant stakeholders involved in the procurement activities of Southern Luzon State University.

Statistical Treatment

Scale	Descriptive Equivalent	Mean Range
5	Strongly Agree	4.21 - 5.00
4	Agree	3.41 - 4.20
3	Neutral	2.61 - 3.40
2	Disagree	1.81 - 2.60
1	Strongly Disagree	1.00 - 1.80

Table 1. Likert Scale

The table above showed the Likert scale. The data gathered from the questionnaire will be analyzed using the weighted mean to determine the overall responses of the participants. The weighted mean will be used to measure and interpret the respondents' level of agreement for each statement in the survey, particularly in terms of usability, efficiency, reliability, and user

satisfaction of the system. Each response will be assigned a numerical value based on the 5-point Likert scale, which is commonly used to measure attitudes and perceptions in research (Koo & Yang, 2025).

Formula for Weighted Mean:

$$\text{WM} = \frac{5(\text{f5}) + 4(\text{f4}) + 3(\text{f3}) + 2(\text{f2}) + 1(\text{f1})}{\text{N}}$$

Where:

WM -Weighted Mean

f5 – frequency of respondents who selected “ Strongly Agree”

f4 – frequency of respondents who selected “ Agree”

f3 – frequency of respondents who selected “Neutral”

f2 – frequency of respondents who selected “ Disagree”

f1 – frequency of respondents who selected “ Strongly Disagree”

N- Number of Respondents

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